

Game Design as the Dynamics of Learning

GAME DESIGN

AS THE DYNAMICS OF LEARNING

On the universality of play, the neural systems that support it, and the design principles that emerge when games are understood as systems for regulating the rate at which humans reduce uncertainty.

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The central claim

This book advances a single, unifying thesis:

Games are systems that regulate the rate at which players reduce uncertainty.

From this, two corollaries follow:

- **Fun** is the subjective experience of reducing uncertainty at an optimal rate
- **Flow** is the cognitive state that emerges when this rate is stable over time

The purpose of the book is to demonstrate that this claim:

1. Explains why games exist across cultures
2. Explains why they are engaging at the level of brain function
3. Explains why some games succeed while others fail
4. Provides a practical framework for designing better games

The argument proceeds in six parts, moving from anthropology to neuroscience to design. Each part builds on the last. The reader is asked to accept only what is demonstrated: that play is universal, that it is functional, that it operates through identifiable neural mechanisms, that these mechanisms can be formalised, and that this formalisation yields practical design principles. The goal is not persuasion through rhetoric, but convergence across domains. If anthropology, neuroscience, psychology, and game design all point to the same structure, then the explanation is unlikely to be accidental.

The book advances four novel theoretical contributions:

1. Flow as a third cognitive mode. Flow is neither System 1 (automatic, effortless processing) nor System 2 (deliberate, effortful reasoning) but a distinct configuration in which task-relevant executive control operates through well-trained procedural pathways while metacognitive overhead is suppressed. Dietrich's (2004) transient hypofrontality account, Weber and Huskey's synchronisation theory, and Harris et al.'s (2017) finding that objective mental effort peaks during flow while subjective effort is minimal all converge on this conclusion.

2. Games and language share frontal-basal ganglia circuits for hierarchical sequential structure. The procedural memory system rooted in the caudate nucleus, putamen, and Broca's area (BA 44/45) that computes grammatical rules also underpins game rule learning, strategic chunking, and expert intuition. Ullman's declarative/procedural model, Wan et al.'s (2011) demonstration of caudate activation in shogi experts, and formal parallels between game description languages and context-free grammars support this thesis.

3. Game quality correlates with the quality of the System 2-to-System 1 conversion pipeline. Every game is a machine for converting deliberate processing into automatic execution. The most acclaimed games introduce challenges that engage System 2 at the right rate, provide feedback that supports pattern extraction, and scale difficulty to match automaticity acquisition.

4. The learning gradient is the primary design metric. The rate at which a player reduces prediction error relative to their expected rate of progress is the hidden variable governing engagement. Wilson, Shenhav, Straccia, and Cohen's (2019) 85% rule for optimal learning provides quantitative grounding: gradient descent-based learning systems (including biological neural networks) learn exponentially faster at an optimal error rate of approximately 15.87%.

What follows

The chapters that follow build this argument step by step.

Games are often treated as trivial. They are not. They are one of the clearest windows we have into how the brain learns, predicts, and engages with the world. To understand games is to understand something fundamental about the human mind. This book is an attempt to make that understanding explicit.

PART I - PLAY IS UNIVERSAL

Chapter 1 - The Universality of Play

On the cross-species, cross-cultural persistence of play - the archaeological record of games - and why the existence of games is a phenomenon that requires explanation rather than a trivial consequence of culture.

1.1 The problem: games exist everywhere

There is a simple observation that, once stated plainly, becomes difficult to ignore: every known human culture plays games.

This is not a statement about modern entertainment industries or digital media. It is a statement about human behaviour across time and geography. Archaeological evidence reveals structured games in ancient Mesopotamia (the Royal Game of Ur), Egypt (Senet), India (Chaturanga), China (Go), Mesoamerica (ballgames), and medieval Europe (Chess). These systems differ in surface features - materials, rules, symbolic meaning - but share a deeper structural property: they are rule-bound environments in which agents act under uncertainty toward defined outcomes.

More strikingly, these systems arise independently. Chess is not a derivative of Go; the Royal Game of Ur is not a precursor to Mesoamerican ballgames. The recurrence of games across cultures is not explained by diffusion alone. It reflects repeated invention.

The implication is immediate. If a behaviour appears:

- across geographically isolated populations
- across historical epochs
- in structurally similar forms

then it is unlikely to be arbitrary. It is more plausibly the expression of an underlying constraint in human cognition.

Games, in this sense, are not merely cultural artifacts. They are solutions to a problem we have not yet fully specified.

1.2 Play is older than culture

The anthropological universality of games becomes more compelling when placed in a broader biological context: play predates humanity.

All four great apes - chimpanzees, bonobos, gorillas, and orangutans - engage in structured play behaviours. These include:

- Rough-and-tumble play, involving mock aggression without escalation
- Role reversal, where dominant individuals temporarily adopt subordinate positions
- Self-handicapping, where stronger individuals deliberately limit their advantage
- Teasing and provocation, including behaviours that appear to anticipate reactions

A landmark 2024 study by Laumer, Winkler, Rossano, and Cartmill (*Proceedings of the Royal Society B*) documented spontaneous playful teasing in all four great ape species, involving intentionally provocative, one-sided behaviours that include elements of surprise. This suggests that the cognitive prerequisites for humour - expectation, violation, and resolution - were present at least 13 million years ago in the hominoid lineage. Laughter-like vocalisations show homologous acoustic structures across great apes, reinforcing the evolutionary continuity of play-related affect.

These behaviours are not random. They are rule-governed in an implicit sense. Play fighting, for example, follows constraints:

- Biting is inhibited

- Signals (play faces, vocalisations) indicate non-serious intent
- Violations of these constraints terminate the interaction

In other words, even in non-human animals, play exhibits the core features of games:

- Bounded interaction spaces
- Implicit rules
- Uncertain outcomes
- Repeated engagement for its own sake

Jane Goodall's foundational observations at Gombe, published in *The Chimpanzees of Gombe* (1986), documented play across all age classes of wild chimpanzees. A subsequent analysis of 33 years of Gombe data by Lonsdorf and colleagues (2017) demonstrated that infant chimpanzees who spent more time in social play achieved motor and social milestones - independent travel, first grooming of non-maternal kin, first mating attempts - at significantly earlier ages. This was the first study in great apes linking play directly to developmental outcomes.

The conclusion is difficult to avoid. What we call "games" in human culture are not inventions ex nihilo. They are formal elaborations of a behavioural system that predates culture itself.

1.3 From play to games: formalisation and abstraction

If play is the biological substrate, games are its cultural crystallisation.

The transition from play to games involves three transformations.

First, constraint becomes explicit. In animal play, rules are implicit and enforced through social signalling. In human games, rules become externalised and formalised. "Do not bite" becomes "no fouls." "Take turns" becomes turn-based systems. Spatial boundaries become boards, fields, or arenas. This externalisation

allows rules to be transmitted across individuals, stabilised across generations, and modified deliberately.

Second, uncertainty becomes structured. Play contains uncertainty, but games engineer it. Dice introduce stochastic outcomes. Hidden information creates informational asymmetry. Skill-based systems create performance uncertainty. The result is not mere unpredictability, but controlled unpredictability; uncertainty that is neither trivial nor overwhelming.

Third, interaction becomes symbolic. Games detach from immediate physical reality and operate in abstract spaces. Stones on a board represent territory (Go). Pieces represent hierarchical power (Chess). Cards represent probabilistic distributions (poker). This abstraction allows games to scale in complexity far beyond what is possible in raw play behaviour.

The archaeological record documents this transition with striking clarity. **Senet**, the most iconic ancient Egyptian game, appears in fragmentary form in First Dynasty burials at Abu Rawash around 3100 BCE. Played on a grid of 30 squares with casting sticks determining movement, it began as secular entertainment but by the New Kingdom (c. 1550-1077 BCE) had acquired profound religious significance, appearing in Chapter 17 of the Book of the Dead as a representation of the soul's journey through the netherworld. Peter Piccione's 1990 University of Chicago dissertation traced this evolution in detail. Four senet boards were found in Tutankhamun's tomb; the game was played across all social classes, from elaborate inlaid boards to grids scratched into stone floors.

The **Royal Game of Ur**, discovered by Leonard Woolley during excavation of the Royal Cemetery at Ur between 1922 and 1934, dates to approximately 2600-2400 BCE. Irving Finkel of the British Museum reconstructed its basic rules from a cuneiform tablet written by the scribe Itti-Marduk-balāṭu in 177-176 BCE; a race game combining strategy and luck, broadly similar to backgammon. The earliest possible game board, found at Ain Ghazal in modern

Jordan, dates to approximately 5870 BCE.

1.4 Independent invention and convergent structure

One of the most important features of games is not their diversity, but their convergence.

Across cultures, we repeatedly observe similar structural solutions:

- Grid-based territorial games (Go, various indigenous strategy games)
- Race games involving probabilistic movement (Senet, Royal Game of Ur, Pachisi)
- Abstract strategy games with hierarchical pieces (Chess and its relatives)
- Physical competition games with bounded arenas (ball sports across continents)

These are not arbitrary categories. They represent distinct ways of structuring uncertainty and action: spatial control, temporal progression, hierarchical interaction, and motor skill execution. The recurrence of these forms suggests that there are only a limited number of ways to construct engaging systems under the constraints of human cognition. Cultures discover these solutions independently because they are locally optimal in a shared design space.

Roberts, Arth, and Bush's foundational cross-cultural study "Games in Culture" (1959, *American Anthropologist*) established the standard tripartite classification: games of physical skill, games of strategy, and games of chance. Analysing 50 societies, they found that games of strategy correlated with **social stratification**, while games of chance correlated with beliefs about supernatural controllability. Games, they argued, function as "expressive models" of real-world activities: physical skill models hunting and combat, strategy models political competition, and chance models the unpredictability of nature.

Certain children's games appear to be cross-cultural universals: hide-and-seek, tag and chase games, wrestling, and catch appear across societies with no historical connection, from Aboriginal Australia to the Arctic to sub-Saharan Africa to South America. Evolutionary psychologists interpret this convergence as evidence of deep biological function; these games simulate predator avoidance, terrain navigation, and emotional regulation under stress.

1.5 The invariants of games

Despite surface variation, games share a set of invariant properties:

1. **Rules.** Games define constraints on action. These constraints create the possibility space within which behaviour occurs.
2. **Agency.** Players can act within the system. Outcomes are not purely observational.
3. **Uncertainty.** The outcome is not fully determined in advance. This may arise from randomness, hidden information, or skill.
4. **Feedback.** Actions produce discernible consequences. The system responds to the player.
5. **Autotelic engagement.** The activity is performed for its own sake, not merely for external reward.

These properties are not definitional conveniences. They are necessary conditions. Remove any one of them and the system ceases to function as a game:

- Remove uncertainty and the system becomes trivial
- Remove agency and it becomes spectacle
- Remove feedback and it becomes opaque
- Remove rules and it becomes unstructured play

The persistence of these invariants across cultures reinforces the central claim: games are not arbitrary constructions. They are

constrained systems shaped by the architecture of the human mind.

1.6 The inadequacy of cultural explanations

A common response to the universality of games is to treat them as cultural byproducts: humans enjoy competition, humans enjoy storytelling, humans enjoy social interaction. Games, on this view, are composites of these tendencies.

This explanation is insufficient for two reasons.

First, it is descriptive rather than explanatory. It lists associated features without identifying the mechanism that makes games compelling.

Second, it fails to explain why games take the specific forms they do. Competition, for example, does not require rule-bound systems with structured uncertainty. Social interaction does not require abstract symbolic spaces. The particular architecture of games - rules, uncertainty, feedback - remains unexplained.

To explain games, we must move beyond surface motivations and ask a deeper question:

What kind of system does the brain find intrinsically engaging, and why?

1.7 Games as a problem in cognitive science

The existence of games poses a problem that sits at the intersection of anthropology, neuroscience, and computation.

We have established three facts:

1. Play is evolutionarily ancient, present across mammalian species and especially developed in primates.
2. Games are culturally universal, appearing independently across human societies.

3. Games share invariant structural properties, suggesting common underlying constraints.

These facts demand a unifying explanation.

The hypothesis that will guide the remainder of this book is the following:

Games are structured environments that exploit fundamental properties of the brain's learning and uncertainty-processing systems.

This is not yet a theory. It is a direction. To make it precise, we must understand how the brain processes uncertainty, how it updates its internal models of the world, and why certain patterns of interaction feel intrinsically rewarding. Only then can we explain not just why games exist, but why some games succeed where others fail.

1.8 Transition

Anthropology establishes that games are universal. Evolutionary biology suggests that play is functional. But neither tells us how games work at the level of mechanism.

To answer that, we must descend a level; from behaviour to biology.

The next chapter examines the evolutionary function of play and the neural systems that support it, beginning with a claim that will become central:

Play is not a luxury. It is a biological system for training the organism to act under uncertainty.

Chapter 2 - The Evolutionary Function of Play

On play as a biological system for training organisms to act under uncertainty - the neural circuits that generate it - and why its persistence across species implies function rather than frivolity.

2.1 Play is not optional

If a behaviour is metabolically expensive, developmentally prolonged, and evolutionarily conserved, it is unlikely to be incidental.

Play meets all three criteria.

- It consumes time and energy without immediate survival payoff
- It appears most intensely during development, when resources are most constrained
- It is present across mammalian species, with especially elaborate forms in primates

From an evolutionary perspective, this is paradoxical. Natural selection is not in the business of maintaining costly behaviours without function. The existence of play therefore demands explanation.

The most parsimonious hypothesis is not that play is leisure, but that it is work of a different kind:

Play is a training system.

The question is: training for what?

2.2 The structure of play behaviour

Across species, play exhibits a remarkably consistent structure.

Consider rough-and-tumble play in juvenile mammals. Individuals engage in mock aggression: they pursue, grapple, and attempt to dominate. Yet critical constraints are enforced: bites are inhibited, vulnerable positions are tolerated, and encounters terminate before escalation. The behaviour contains real motor patterns - chasing, striking, balancing - but stripped of lethal consequences.

Similarly, in object play, animals manipulate objects in ways that resemble foraging or tool use. Actions are repeated, varied, and recombined. Outcomes are explored rather than optimised. And in social play, individuals test boundaries: they provoke, retreat, re-engage. They learn the contingencies of interaction.

Across all forms, the same pattern appears:

- Action under uncertainty
- Immediate feedback
- Low cost of failure
- Repetition with variation

This is not random behaviour. It is a structured exploration of possibility space.

2.3 The PLAY system

Jaak Panksepp's work on affective neuroscience provides the clearest biological account of play.

He identified a primary emotional system - alongside SEEKING, FEAR, RAGE, CARE, LUST, and PANIC/GRIEF - dedicated specifically to play: the **PLAY system**. His landmark book *Affective Neuroscience* (Oxford University Press, 1998) and its successor *The Archaeology of Mind* (2012, with Lucy Biven) mapped these systems using electrical brain stimulation, pharmacological challenges, and

lesion studies. Critically, all seven systems arise from **subcortical brain regions** homologous across mammals. They do not require the neocortex: decorticated rats still play, and hydranencephalic children - born without a cerebral cortex - still laugh and smile (Merker, 2007).

Key properties of the PLAY system:

- It is subcortical, originating in ancient brain structures
- It is intrinsically motivated; play is pursued for its own sake
- It is socially modulated, especially in mammals
- It exhibits homeostatic rebound: deprivation increases subsequent play intensity

Panksepp's most celebrated experiments involved tickling rats. In publications from 2000 to 2003 (*Behavioural Brain Research, Physiology & Behavior*), he demonstrated that rats emit **50-kHz ultrasonic vocalizations** during rough-and-tumble play and when tickled by human hands at the nape of the neck; the same area targeted during conspecific play. These frequency-modulated calls, proposed as a homologue of human laughter, are positively correlated with the rewarding value of tickle stimulation, can induce approach behaviour in other rats when played back (Wöhr & Schwarting, 2007), and are distinct from the 22-kHz aversive calls emitted during distress. Rats selectively bred for high rates of 50-kHz vocalizations showed increased playfulness (Burgdorf et al., 2005, *Behavior Genetics*).

That play is a primary drive - not learned, not secondary - is supported by several converging lines of evidence. Play is tightly homeostatic: its amount can be titrated by varying social isolation duration, with longer deprivation producing rebound increases. Rats develop conditioned place preferences for play-associated environments and perform operant responses to obtain play opportunities. Neonatal decortication does not abolish play (Panksepp et al., 1994), confirming its subcortical origin. Conversely, frontal lobe lesions increase playfulness (Panksepp et

al., 2003), suggesting the maturing cortex normally inhibits subcortical play urges.

The neurochemical basis involves four interacting systems:

- **Dopamine** drives approach and engagement (motivational "wanting")
- **Opioids** mediate hedonic pleasure (hedonic "liking")
- **Endocannabinoids** enhance positive affect and flexibility
- **Oxytocin** supports social bonding

The neural architecture involves a distributed circuit centring on subcortical structures. Sivy and Panksepp (2011, *Neuroscience & Biobehavioral Reviews*) proposed a primary-process executive circuit including the **parafascicular area of the thalamus** (PFA), the **dorsal and ventral striatum**, and the **frontal cortex**, with contributions from the amygdala, periaqueductal grey, and ascending dopamine systems. The PFA appears to be a critical integration node: electrolytic lesions of the PFA reduced play pinning by 73% while having minimal effects on other sensory-motor processes (Sivy & Panksepp, 1985, *Behavioral Neuroscience*).

The PLAY system is not a cultural overlay. It is a biological drive, as fundamental as hunger or fear.

2.4 Neuroplasticity and skill acquisition

If play is a training system, it must produce measurable changes in the brain.

It does.

Experimental studies of play deprivation provide a critical insight. Animals deprived of play during development show:

- Reduced synaptic density in prefrontal cortex
- Impaired cognitive flexibility
- Deficits in social competence

- Increased anxiety-like behaviour

Bijlsma et al. (2022) demonstrated that play-deprived rats showed reduced inhibitory synapses in prefrontal cortex and impaired cognitive flexibility. The prefrontal cortex is not required to generate play but is profoundly shaped by it. Gordon et al. (2003) showed BDNF elevation after play; BDNF is the primary growth factor supporting synaptic plasticity throughout the brain.

Conversely, play enhances:

- Brain-derived neurotrophic factor (BDNF) levels
- Synaptic plasticity
- Motor coordination
- Social prediction ability

These effects are not subtle. They indicate that play is directly involved in shaping the neural architecture required for adaptive behaviour. Stuart Brown, in his clinical work, found that 26 young male murderers had no memories of normal play; a striking if preliminary correlation between play deprivation and catastrophic failures of social calibration.

The implication is clear:

Play is not rehearsal of specific behaviours; it is training the capacity to learn, adapt, and respond to novelty.

2.5 Safe failure and error tolerance

A defining feature of play is that it allows failure without catastrophic consequence.

This is not incidental. It is the central design feature.

In non-play contexts, failure may result in injury or death; errors are costly and must be minimised. In play, failure is expected, errors are tolerated, and exploration is encouraged. This creates a unique learning environment where the organism can test hypotheses,

explore edge cases, and experience near-failure states repeatedly.

From a learning perspective, this is ideal. Systems that learn through error correction require error signals, opportunity for adjustment, and repeated exposure. Play provides all three. The cost structure of play is what makes it such an effective training system: prediction errors are frequent and informative, but the consequences of getting things wrong are minimal. This maps directly onto how games function. Death in *Dark Souls* costs a few minutes of progress and some virtual currency. Death in real combat costs everything. The prediction error is structurally similar; the cost is incomparably different.

2.6 Uncertainty as the core variable

The common thread across all forms of play is uncertainty.

- In physical play: uncertainty of movement outcomes
- In social play: uncertainty of others' responses
- In object play: uncertainty of environmental interaction

Play is not simply activity. It is activity structured around unknowns.

Crucially, the uncertainty is:

- **Non-trivial:** outcomes are not predetermined
- **Reducible:** patterns can be learned over time

This places play in a specific regime; not pure randomness (which cannot be learned), not full determinism (which requires no learning), but environments where prediction is possible but not yet achieved. This is the regime in which learning systems operate most effectively.

2.7 Social calibration and boundary testing

In social species, play serves an additional function: calibration of interaction.

Through play, individuals learn how far they can push others, how others respond to provocation, how to signal intent, and how to repair violations. The mechanisms are subtle: a play face signals non-serious intent - overly aggressive behaviour leads to disengagement - successful interaction requires continuous adjustment. This is effectively real-time modelling of other agents. The organism is learning prediction of others' actions, adjustment of its own behaviour, and maintenance of cooperative equilibrium; precisely the skills required for navigating complex social environments.

2.8 The paradox of inefficiency

From a narrow perspective, play appears inefficient. Time spent playing is time not spent foraging. Energy is expended without immediate reward. Risks are taken without necessity.

Yet evolution has preserved it.

The resolution of this paradox lies in timescale. Play is inefficient in the short term but efficient in the long term. By investing in flexibility, adaptability, and learning capacity, the organism gains increased survival under novel conditions, improved social integration, and enhanced problem-solving ability. Play is therefore not a waste of resources. It is an investment in general competence.

Stephen Jay Gould described humans as a **neotenuous species**; retaining juvenile characteristics including curiosity, behavioural flexibility, and attraction to novelty into adulthood. Adult play persists in every human culture through sports, board games, music, dance, humour, and creative expression. Johan Huizinga's *Homo Ludens* (1938) argued that culture arises in the form of play and that civilisations lose their vitality as they lose their playfulness. Six evolutionary theories have been proposed for play's function:

Groos's practice theory, Spencer's surplus energy theory, Burghardt's developmental model, Spinka et al.'s training for the unexpected, Gray's self-education hypothesis, and Bateson's creativity engine. All converge on the same core insight: play builds the organism's capacity to handle novelty.

2.9 From play to games (revisited)

We can now reinterpret games in light of play's function.

If play is a biological system for training under uncertainty, then games are **artificial environments that replicate and amplify this training process**. Games take the core features of play - uncertainty, feedback, safe failure, repetition - and make them more structured, more controllable, and more scalable. They are, in effect, engineered play systems.

This reframes the design problem entirely. The question is no longer "How do we make games entertaining?" It becomes:

How do we construct environments that optimally engage the brain's learning systems?

2.10 Transition

We have established that play is functional: it is biologically grounded, it shapes neural development, and it trains organisms to act under uncertainty. We have also seen that games are formalised extensions of play.

The next step is to move from function to mechanism. If play trains the organism through interaction with uncertainty, then: what neural signals track that uncertainty? What makes certain interactions rewarding? Why do some patterns sustain engagement while others do not?

To answer these questions, we must examine the brain's reward and learning systems directly.

The next chapter begins with a claim that will anchor the rest of this book:

The brain does not reward outcomes. It rewards the reduction of uncertainty.

PART II - WHAT GAMES DO TO THE BRAIN

Chapter 3 - The Reward System and Prediction Error

On dopamine as a signal of prediction error - the difference between expected and actual outcomes - and why engagement is maximised not by reward itself, but by structured uncertainty.

3.1 The brain is a prediction machine

The foundational insight of modern neuroscience is that the brain is not a passive receiver of sensory input. It is an active generator of predictions. At every level of the neural hierarchy, from retinal ganglion cells to prefrontal cortex, the brain constructs models of what will happen next and compares those models against what actually occurs. The difference between prediction and reality - the prediction error - is the fundamental currency of neural computation.

This insight, formalised under the banner of predictive processing (Friston, 2010; Clark, 2013), reframes everything from perception to action to emotion. Perception is not the brain "seeing" the world; it is the brain generating a hypothesis about the world and checking that hypothesis against incoming sensory data. Action is not the brain "doing" something; it is the brain trying to make the world conform to its predictions. Emotion is not a reaction to events; it is a signal about how well the brain's predictions are faring.

For games, this framework is foundational. If the brain is fundamentally in the business of predicting and error-correcting, then a game is an environment specifically designed to generate structured, resolvable prediction errors at a controlled rate. The

question becomes: what neural systems process these errors, and what makes some error patterns feel rewarding while others feel frustrating?

3.2 Dopamine encodes reward prediction error

The answer begins in the midbrain, with a small cluster of neurons that produce the neurotransmitter dopamine.

Wolfram Schultz, Peter Dayan, and P. Read Montague established the modern framework in their landmark 1997 *Science* paper, "A Neural Substrate of Prediction and Reward." Recording from individual midbrain dopamine neurons in macaque monkeys, they identified three canonical response patterns:

- When a reward arrives **unexpectedly**, dopamine neurons fire a phasic burst above baseline. This is a positive prediction error: reality was better than expected.
- When a reward arrives **exactly as predicted**, neurons show no change in firing rate. This is zero prediction error: the model was correct.
- When a predicted reward **fails to arrive**, neurons suppress their firing below baseline. This is a negative prediction error: reality was worse than expected.

The signal is not about reward itself. It is about the **difference** between what was expected and what occurred. A monkey that has learned to expect juice after a tone shows no dopamine response when the juice arrives on schedule; the reward is fully predicted and therefore carries no new information. But if the juice is omitted after the tone, dopamine neurons actively suppress; the brain's model was wrong, and the suppression signal drives model updating.

This pattern implements the temporal difference (TD) learning algorithm formalised by Montague, Dayan, and Sejnowski (1996), where the error signal $\delta(t) = r(t) + \gamma V(t+1) - V(t)$. The elegance of

this signal is that it is both an evaluation (was this better or worse than expected?) and a teaching signal (update the model accordingly). As learning proceeds, the dopamine response **transfers forward in time**: the burst moves from the moment of reward to the earliest cue predicting reward. Once the association is fully learned, the cue triggers the burst, the reward itself produces no response, and omission of the reward produces suppression. Schultz's 1998 *Journal of Neurophysiology* review confirmed the generality of this pattern across reward types and experimental paradigms.

The implications for games are immediate and far-reaching.

3.3 What this means for games

A game that produces only expected outcomes generates no dopaminergic reward signal. The player may be performing well - executing flawlessly, winning every encounter, clearing every obstacle - but if nothing is surprising, the dopamine system has nothing to report. This is why fully mastered games feel flat even when the player is performing at a high level: performance without surprise produces no signal. The game has been "solved," and the prediction error that once fuelled engagement has been reduced to zero.

A game that produces outcomes **worse** than expected produces active suppression of dopamine. This is the neurochemical signature of the experience players describe as "unfair": the game violated their model in a way that could not be attributed to their own error. An enemy that kills the player through a wall, a jump that fails despite correct timing due to a hitbox glitch, a boss with an undodgeable attack; these produce negative prediction errors that signal "your model of how this system works is wrong," but the player cannot identify a correctable cause. The suppression is aversive, and repeated negative prediction errors without resolution drive disengagement.

Only a game that regularly produces outcomes **better than predicted, or different from predicted in ways that can be learned from**, sustains the neurochemical signature of engagement. The player who discovers that a grenade thrown behind an Elite forces it to dodge into their line of fire has experienced a positive prediction error: the outcome was better than their model predicted. The player who discovers that a particular card combination triggers an unexpected synergy has experienced the same thing. These moments of "it worked better than I thought" or "I didn't expect that, but now I understand why" are the raw material of engagement, and the dopamine system marks each one with a burst that says: update the model, and seek more experiences like this.

3.4 Variable rewards drive dopamine; fixed rewards do not

A finding of particular relevance to game design comes from Zald et al. (2004, *Journal of Neuroscience*), who used PET imaging to demonstrate that **unpredictable (variable ratio) monetary rewards** produced significant dopamine release in the medial caudate, whereas equivalent predictable (fixed ratio) rewards produced no measurable increase. The brain's reward system is specifically responsive to unpredictability; a fixed reward, no matter how large, becomes fully predicted and ceases to generate a dopamine signal.

This explains the pervasive effectiveness of variable reward systems in games. Critical hits that deal extra damage with some probability. Loot drops whose contents are unknown until opened. Procedurally generated environments that present familiar elements in novel configurations. All of these systems maintain prediction error by ensuring that the player cannot fully predict the outcome of any given action. The reward is not the item or the damage number; it is the **surprise** of not knowing what would happen.

The danger, of course, is that variable reward systems can sustain engagement without learning. Slot machines generate high prediction error with zero reducibility; the player cannot improve their model of a random number generator. The engagement is maintained through the dopamine system's responsiveness to uncertainty, but it is hollow engagement; "wanting" without genuine skill development. This is where the distinction between reducible and irreducible uncertainty becomes critical, as we will see shortly.

3.5 Uncertainty itself is rewarding: the Fiorillo signal

The Schultz framework explains why surprising outcomes feel good. But games are not primarily about receiving rewards. They are about operating under uncertainty; the outcome has not yet been determined, the player does not yet know whether they will succeed, and the system is in a state of unresolved suspense.

Fiorillo, Tobler, and Schultz (2003, *Science*) demonstrated that uncertainty itself generates a distinctive dopamine signal. Recording from the same midbrain neurons, they identified a **sustained, ramping activity** during the delay period between a reward-predicting cue and potential reward delivery. This ramp was distinct from the phasic burst that signals prediction error. It built gradually over the delay period. And it peaked at $P = 0.5$ - maximum uncertainty - decreasing symmetrically toward both $P = 0$ (reward certain to be absent) and $P = 1.0$ (reward certain to arrive). The function followed the mathematical variance of a Bernoulli distribution: $p(1 - p)$, an inverted U with its apex at maximum unpredictability.

When magnitude variance was manipulated at fixed $P = 0.5$, the ramp increased with the spread between possible outcomes, further confirming that the signal tracks uncertainty rather than expected value. The finding has been debated; Niv, Duff, and Dayan (2005) proposed an alternative interpretation involving backpropagating TD errors. But the functional implication is the same regardless of

mechanism: **the dopamine system is maximally active under maximum reducible uncertainty.** The condition game designers call "optimal challenge" or "balanced difficulty" is the condition under which the neurochemical substrate of engagement is most potent.

This finding resolves a puzzle that difficulty-based models of game design cannot. Two situations can have identical difficulty but produce completely different experiences. A coin flip and a close chess match both involve $P = 0.5$ uncertainty. But the chess match is engaging and the coin flip is not. The difference is that chess uncertainty is **reducible through learning**; the player's predictions can improve. Coin flip uncertainty is irreducible; no amount of study will improve the player's accuracy beyond 50%. The dopamine system, properly understood, does not simply reward uncertainty. It rewards the **opportunity to reduce** uncertainty through action.

3.6 The brain treats information as reward

A complementary finding comes from Bromberg-Martin and Hikosaka (2009, *Neuron*), who demonstrated that macaques prefer informative cues predicting reward size even when the information cannot change the outcome. Given a choice between a cue that reveals the size of an upcoming reward (but does not alter it) and a cue that reveals nothing, monkeys overwhelmingly preferred the informative option. The same dopamine neurons that signal expected reward also signalled the **expectation of information**. Blanchard, Hayden, and Bromberg-Martin (2015) showed that monkeys would sacrifice actual reward to gain advance information.

The brain treats uncertainty reduction as intrinsically valuable; not as a means to an end, but as an end in itself. This is the neural foundation for curiosity, and it explains one of the most distinctive features of games: exploration feels rewarding before any extrinsic reward appears. A player who opens a door to discover a new area

has not received any in-game currency or experience points. They have received **information**; a reduction in uncertainty about the game world. And this is processed by the same dopaminergic circuitry that handles food, water, and social reward.

3.7 Distributional reward coding

Dabney et al. (2020, *Nature*) extended the Schultz framework significantly by demonstrating that different dopamine neurons simultaneously encode different **quantiles** of the reward distribution, with asymmetric sensitivity to positive versus negative errors. Some neurons are optimistic (more responsive to better-than-expected outcomes), others are pessimistic (more responsive to worse-than-expected outcomes), and together they maintain a full probability distribution over possible outcomes rather than just an expected value.

This distributional coding has direct relevance to game design. It explains why games with variable outcomes - critical hits, loot drops, procedural generation - feel different from games with deterministic ones. The brain is not simply tracking the average outcome; it is modelling the **shape** of the uncertainty. A game where every hit deals exactly 10 damage and a game where hits deal between 1 and 19 damage (averaging 10) produce identical expected damage but different distributional prediction errors. The variable game produces a richer error landscape, with some neurons consistently surprised on the upside and others consistently surprised on the downside, sustaining a broader range of neural responses and, subjectively, a more engaging experience.

3.8 Games and the dopamine system: direct evidence

The theoretical framework connecting dopamine to game engagement is supported by direct empirical measurement.

Koepp et al. (1998, *Nature*) conducted the foundational study, using [11C]raclopride PET to show that playing a tank-navigation video game produced a ~13% reduction in radiotracer binding in the ventral striatum; direct evidence of endogenous dopamine release during gameplay. Performance scores correlated with the magnitude of dopamine displacement, establishing that the mesolimbic pathway tracks game success parametrically.

Subsequent fMRI work has consistently replicated striatal engagement. Hoeft et al. (2008, *Journal of Psychiatric Research*) found that a simple space-infringement game activated the nucleus accumbens, orbitofrontal cortex, and amygdala. Klasen et al. (2012, *Social Cognitive and Affective Neuroscience*; 2020, *Brain Structure and Function*) demonstrated during free play of a first-person shooter that game success events activated the caudate nucleus, nucleus accumbens, and putamen, with a dissociation between ventral striatal activation for non-violent success and dorsal striatal activation for violent success.

A critical finding is that **active agency amplifies reward signals**. Kätsyri et al. (2013, *Frontiers in Human Neuroscience*) showed that striatal reward responses were substantially stronger during active gameplay than during passive observation of identical game footage, with anterior putamen showing the sharpest differentiation for win events during active play. Watching someone else play a game produces a weaker dopamine signal than playing the same game yourself. This is why games are more engaging than movies, and why "let's play" videos, while entertaining, do not replicate the neurochemical intensity of actual gameplay: agency is necessary for robust striatal dopamine release.

3.9 "Wanting" without "liking": the dark side of game reward

Berridge and Robinson's incentive salience theory offers a crucial refinement that any complete account of game engagement must

address. Across foundational reviews spanning three decades (Robinson & Berridge, 1993, *Brain Research Reviews*; Berridge & Robinson, 2016, *American Psychologist*; Robinson & Berridge, 2025, *Annual Review of Psychology*), they established a dissociation that is consequential for game design: **dopamine mediates "wanting" (incentive salience)** through large, robust mesolimbic projections, while **hedonic "liking" depends on much smaller opioid and endocannabinoid hotspots** in the nucleus accumbens shell and ventral pallidum.

The dissociation means that the dopamine system's responsiveness to variable rewards can be sensitised through repeated exposure, escalating "wanting" (craving, compulsive engagement) while "liking" (actual hedonic enjoyment) remains flat or declines. This is the hallmark pattern of behavioural addiction, and it maps directly onto the commonly reported experience of gamers who feel compelled to continue playing despite diminished enjoyment. Singer et al. (2012) and Mascia et al. (2018, *Neuropsychopharmacology*) demonstrated that chronic variable-ratio reinforcement produces dopamine system sensitisation.

The wanting/liking dissociation is the neurochemical basis for the distinction between games that sustain genuine engagement through learning (where the reward is uncertainty reduction and the "liking" tracks the "wanting") and games that sustain compulsive engagement through variable reinforcement schedules (where the "wanting" escalates while "liking" stagnates). The unified model this book proposes addresses the former; the latter is a pathological exploitation of the same circuitry.

3.10 Transition

The dopamine prediction error framework provides the first pillar of the unified model: the brain generates a specific neurochemical signal when predictions fail, and this signal is the substrate of engagement. Variable, reducible uncertainty maximises this signal.

Agency amplifies it. And the brain treats information itself as intrinsically rewarding.

But prediction error alone is not enough. The dopamine system explains why surprising outcomes feel good. It does not explain why the brain actively seeks out situations of uncertainty rather than simply responding to them when they arise. For that, we need to understand curiosity; the drive that makes players explore, experiment, and pursue information even when no extrinsic reward is offered.

Chapter 4 - Curiosity and Exploration

On the distinction between intrinsic and extrinsic motivation - the brain's drive to resolve uncertainty - and why exploration feels rewarding before any external payoff arrives.

4.1 Curiosity as a drive state

Curiosity is not a luxury. It is a biological drive state, comparable in neural architecture to hunger and thirst, that motivates organisms to seek information and reduce uncertainty about their environment. The difference between curiosity and the prediction error system described in the previous chapter is the difference between reactive and proactive engagement: prediction error is what happens when expectations are violated; curiosity is what drives the organism to seek situations where expectations might be violated.

Gruber, Gelman, and Ranganath (2014, *Neuron*) provided the key empirical link between curiosity and reward circuitry. During fMRI, they presented trivia questions calibrated to induce varying levels of curiosity. High-curiosity states enhanced memory not only for the target trivia answers but for entirely incidental face stimuli encountered during the curiosity period; faces shown between the question and the answer were better remembered when the question was one the participant was curious about, even though the faces had nothing to do with the trivia content. This enhancement was predicted by **anticipatory activity in the midbrain (SN/VTA) and nucleus accumbens**, with functional connectivity between midbrain and hippocampus mediating the effect.

The implication is striking: curiosity does not merely prepare the brain to learn a specific piece of information. It opens a window of enhanced learning across the board, priming the hippocampal memory system through dopaminergic modulation. In game design terms, this means that a player in a state of curiosity; wondering what is behind the next door, what a new mechanic does, how two systems interact; is in a neurochemically enhanced state for learning of all kinds, including learning that has nothing to do with the original source of curiosity.

Gruber and Ranganath's (2019, *Trends in Cognitive Sciences*) PACE framework formalised this: curiosity is triggered by significant prediction errors, undergoes appraisal (is this error interesting or threatening?), and, if appraised as safe and potentially informative, enhances encoding through dopaminergic modulation of the hippocampus. The appraisal stage is critical for games: the same prediction error can produce curiosity or anxiety depending on whether the player perceives the environment as safe for exploration. This is why horror games must carefully calibrate threat; too much real danger converts curiosity into fear, collapsing the exploratory learning gradient.

4.2 The Goldilocks effect

Curiosity is not uniformly distributed across all levels of uncertainty. Kidd and Hayden (2015, *Neuron*) reviewed evidence for what they termed the "Goldilocks effect": organisms preferentially attend to stimuli of intermediate complexity, neither too predictable (boring) nor too surprising (confusing). Infants look longer at visual sequences of intermediate probability. Adults rate intermediate-difficulty trivia questions as most curiosity-inducing. The caudate nucleus and inferior frontal gyrus (Kang et al., 2009, *Psychological Science*) show maximal activation for information gaps that are large enough to be interesting but small enough to seem resolvable.

Gottlieb, Oudeyer, Lopes, and Baranes (2013, *Trends in Cognitive Sciences*) and Gottlieb and Oudeyer (2018, *Nature Reviews Neuroscience*) integrated this into a framework where exploration is guided by **expected information gain**: the brain estimates how much model improvement a particular experience is likely to yield, and directs attention toward experiences that promise the greatest reduction in uncertainty. This is not mere novelty-seeking; it is strategic uncertainty-reduction, guided by the brain's current model of what it does and does not understand.

For game design, this means that the most curiosity-inducing elements are not the most novel ones (which may be incomprehensible) or the most familiar ones (which are boring) but those that sit at the boundary of the player's current understanding; familiar enough to be approachable, unfamiliar enough to promise new learning. This is the curiosity analogue of the flow channel, and it explains why good level design introduces new elements gradually rather than all at once.

4.3 Intrinsic versus extrinsic motivation

Ryan and Deci's self-determination theory (SDT) provides the most influential framework for understanding why people engage voluntarily in activities. SDT identifies three fundamental psychological needs: **autonomy** (the sense of being the origin of one's own behaviour), **competence** (the sense of being effective in one's interactions), and **relatedness** (the sense of connection to others). When activities satisfy these needs, they are intrinsically motivating; people engage in them for their own sake, without requiring external incentives.

Rigby and Ryan (2011) applied SDT to games, demonstrating that player motivation tracks satisfaction of these three needs. Games that provide choices satisfy autonomy. Games that provide appropriate challenge and feedback satisfy competence. Multiplayer games that enable meaningful social interaction satisfy relatedness.

The framework correctly predicts broad motivational patterns: players prefer games that give them choices, challenge them appropriately, and connect them to others.

But SDT, like flow theory, is a static framework. It identifies what motivates people without describing the dynamic process that unfolds over time. A game can satisfy all three needs and still fail to engage if the moment-to-moment prediction error rate is wrong. SDT tells us that the player needs to feel autonomous, competent, and connected; it does not tell us what should happen in the next ten seconds of gameplay to sustain those feelings. That is the role of the learning gradient, which will be developed in later chapters.

4.4 The undermining effect

One of the most robust findings in motivation research is the **undermining effect**: extrinsic rewards can reduce intrinsic motivation. Deci (1971) demonstrated that paying participants to solve puzzles reduced their subsequent willingness to solve puzzles for free. The introduction of an external reason for the behaviour ("I'm doing this for money") displaces the internal reason ("I'm doing this because it's interesting"), and when the external reward is removed, the behaviour decreases below its pre-reward baseline.

In game design, this effect manifests as the difference between games that sustain engagement through genuine learning and games that sustain engagement through token accumulation. Explicit reward systems - XP bars, achievement notifications, daily login bonuses, progress percentages - provide extrinsic markers that can displace the intrinsic reward of uncertainty reduction. The player who is exploring a dungeon to discover what is inside it is intrinsically motivated; their curiosity drives engagement, and the reward is understanding. The player who is clearing a dungeon to fill a progress bar is extrinsically motivated; their engagement depends on the bar, and removing the bar removes the motivation.

The design implication is not that extrinsic rewards should never be used, but that they should be subordinate to intrinsic learning. Extrinsic rewards work best when they mark genuine achievement (you mastered this skill) rather than mere activity (you spent time here). They work worst when they become the primary reason for engagement, converting a curiosity-driven exploration into a token-collecting errand.

The contrast between *Breath of the Wild* and Ubisoft's marker-heavy open-world formula illustrates the point. *Breath of the Wild* invests enormous design effort in making the world visually legible: landmarks are visible from great distances, unusual structures signal hidden content, and terrain features guide the eye toward points of interest without explicit UI markers. The player's own perceptual system does the "quest marker" work, which means the exploration process itself engages active attention. Ubisoft's formulaic approach (map towers, objective markers, percentage completion trackers) reduces anxiety but simultaneously kills autotelic motivation by converting exploration into a to-do list. When every icon is visible on the map, exploration is no longer discovery; it is errand-running.

4.5 Exploration versus exploitation

The explore-exploit trade-off is a fundamental problem in adaptive behaviour, formalised in reinforcement learning as the multi-armed bandit problem: should the organism continue exploiting a known rewarding option, or explore an unknown option that might be better?

The locus coeruleus-norepinephrine (LC-NE) system mediates this trade-off in the brain. Aston-Jones and Cohen (2005) proposed a dual-mode model:

- **Phasic mode:** low baseline NE, task-evoked bursts. Attention is focused. The organism exploits what it knows. This is the mode associated with flow; sustained engagement with a

well-understood task.

- **Tonic mode:** high baseline NE, broad attentional sampling. The organism explores. Attention shifts frequently. This is the mode associated with curiosity, distraction, and the search for better options.

Games must manage both modes. A game that sustains phasic mode indefinitely will produce flow but eventually bore the player when the current task is exhausted. A game that sustains tonic mode indefinitely will produce restless exploration without the deep engagement that comes from sustained focus. The best games oscillate between the two: exploration phases (tonic mode) discover new challenges, which then demand sustained engagement (phasic mode) to master.

Breath of the Wild's structure exemplifies this oscillation. Traversal between shrines and points of interest is exploratory (tonic mode): the player scans the horizon, notices a distant landmark, decides to investigate. Shrine puzzles are focused (phasic mode): attention narrows to a single challenge, feedback is immediate, and the player enters a flow state. Combat encounters similarly punctuate exploration with episodes of focused engagement. The game's genius is that it never forces the player to stay in either mode for too long; the transitions are player-driven and feel natural.

4.6 How games structure curiosity

Games create curiosity through the systematic construction of **information gaps**; situations where the player knows enough to ask a question but not enough to answer it.

Fog of war hides portions of the map, creating spatial information gaps. Locked doors signal that something lies behind them, creating structural information gaps. Partially revealed loot tables let the player know that better items exist without revealing which ones, creating reward information gaps. Skill trees show abilities that are not yet unlocked, creating capability information gaps. Story hooks

introduce characters and conflicts without resolution, creating narrative information gaps.

Each of these systems works by the same mechanism: the player's model of the game world is incomplete, and the gap between what is known and what is unknown generates a prediction error that the curiosity system marks as worth pursuing. The gap must be large enough to be interesting (the Goldilocks effect) but bounded enough that the player believes resolution is achievable.

Outer Wilds represents the purest implementation of curiosity-driven game design. The game has no combat, no upgrades, no permanent progression. The only thing that changes between loops is what the player knows. Every piece of information in the game is available from the first minute; the "progression" is entirely epistemic. The learning gradient is maintained solely by the player's growing understanding of how the solar system works and what happened to the Nomai, and the game achieved widespread critical recognition as one of the decade's best designs despite having none of the conventional reward structures that most games rely on.

4.7 Transition

Curiosity provides the proactive complement to the reactive prediction error system. Together, they explain why games attract and sustain attention: prediction error rewards the player for encountering surprising outcomes, and curiosity drives the player to seek those encounters in the first place.

But attraction and reward are only half the story. The other half is what happens to the player's brain as they continue to engage: how skills are acquired, how knowledge is consolidated, and how the effortful processing of a novice transforms into the automatic fluency of an expert. This transformation - from conscious incompetence to unconscious competence - is the subject of the next chapter.

Chapter 5 - Learning Systems and Skill Acquisition

On the transition from effortful, declarative processing to automatic, procedural execution - how repeated interaction reorganises neural circuits - and the cortical-to-subcortical transfer that defines mastery.

5.1 Two memory systems

The brain does not have a single learning system. It has at least two, operating in parallel, with different computational properties and different neural substrates.

Michael Ullman's Declarative/Procedural (DP) model, first articulated in *Nature Reviews Neuroscience* (Ullman, 2001a) and expanded in *Cognition* (Ullman, 2004) and the *Neurobiology of Language* handbook (Ullman, 2016), provides the clearest framework. The **declarative memory system**, rooted in temporal-lobe structures centred on the hippocampus, handles facts and episodes: what happened, where things are, what the rules say. It is conscious, explicit, and fast to acquire but slow to retrieve under time pressure. The **procedural memory system**, rooted in frontal cortex and the basal ganglia (specifically the caudate nucleus, anterior putamen, and Broca's area), handles skills and habits: how to do things, how to apply rules automatically, how to execute complex sequences without conscious supervision. It is unconscious, implicit, and slow to acquire but fast to execute once learned.

This dissociation maps precisely onto game cognition. Game facts; this enemy has 200 HP, that weapon deals fire damage, the boss

attacks every three seconds; are declarative. Game skills; dodge-rolling the boss's third attack, executing a combo, reading an opponent's positioning in a fighting game; are procedural. The transition from knowing facts about a game to being able to play it well is the transition from hippocampal to striatal processing.

Critically, the DP model asserts that the procedural system is **domain-general**. Ullman (2004) describes it as supporting "the learning and processing of motor and cognitive skills, especially those involving sequences"; including navigation, motor sequences, rules, categories, and habits. Ullman (2016) further specifies that the procedural system "may be specialized for learning to predict (perhaps especially probabilistic outcomes), for example, the next item in a sequence or the output of a rule." This predictive function maps directly onto game cognition, where players must anticipate consequences of moves under rule constraints.

Clinical evidence supports the domain-generality claim. Parkinson's disease, which degrades dopaminergic projections to the striatum, produces parallel deficits in motor sequencing and grammatical processing. Ullman et al. (1997, *Journal of Cognitive Neuroscience*) showed that PD patients made disproportionately more errors with regular past-tense forms (requiring rule computation) than irregular forms (requiring lexical retrieval); the opposite pattern from Alzheimer's patients, whose hippocampal declarative system is degraded instead. If the same circuits underlie game rule application, PD patients should show parallel deficits in learning and applying game rules; a prediction ripe for empirical testing.

5.2 The three stages of motor learning

Fitts and Posner (1967) described three stages of motor skill acquisition that map cleanly onto the dual-system framework.

In the **cognitive stage**, the learner is aware of what they are trying to do but cannot do it smoothly. Movements are guided by verbal and declarative processes; the player tells themselves "press X to

dodge, then press Y to attack." Attention is fully consumed by the motor task, leaving no capacity for higher-order strategy. Errors are frequent, and performance is inconsistent. This stage is dominated by System 2: the prefrontal cortex, anterior cingulate cortex (ACC), and dorsolateral prefrontal cortex (dlPFC) are heavily engaged, maintaining task rules in working memory and monitoring for errors.

In the **associative stage**, performance becomes more fluid and reliable. Links form between actions and outcomes, and errors decrease. Neural activation shifts away from prefrontal regions toward increasing sensorimotor cortex and striatal involvement. The corticocerebellar loop begins to disengage as internal models improve (Doyon et al., 2002). The *Celeste* player who no longer thinks about wall-jumping but still occasionally misjudges dash angles is in this transitional phase.

In the **autonomous stage**, performance is accurate, consistent, and largely automatic. Activation concentrates in sensorimotor striatum (putamen), primary motor cortex, supplementary motor area, and cerebellar nuclei. dlPFC activation is markedly reduced. Critically, the player can now attend to higher-level information - game strategy, environmental cues, opponent behaviour - because motor execution no longer requires supervisory attention. The Guitar Hero expert sight-reading an unfamiliar song on Expert difficulty operates here.

Kim et al. (2015, *PLOS Biology*) confirmed that the early phase recruits frontal and parietal regions involved in attention, spatial working memory, and movement planning, while advanced performance shifts to sensorimotor and cerebellar circuits.

5.3 Cortical-to-subcortical transfer: the neural mechanism of mastery

The shift from System 2 to System 1 processing during skill acquisition is not metaphorical. It is a measurable cortical-to-subcortical transfer that has been mapped in detail across multiple research programmes.

Poldrack et al. (2005, *Journal of Neuroscience*) used fMRI during a serial reaction time task to index automaticity by elimination of dual-task interference. Before training, sequential performance activated broad frontal and striatal regions. After training, activation **decreased** in dlPFC, ventral premotor cortex, inferior frontal gyrus, and right caudate. Automaticity was characterised by reduced prefrontal engagement while dorsal premotor cortex and supplementary motor area maintained stable activation; the fundamental motor programming layer persists while the supervisory layer withdraws.

Lehéricy et al. (2005, *PNAS*) traced motor sequence learning over four weeks and found a shift *within* the basal ganglia itself: early learning activated **rostradorsal (associative) putamen** alongside dlPFC and premotor areas, while advanced performance shifted to **caudoventral (sensorimotor) putamen**. Error rates correlated positively with early-learning regions; reaction times correlated negatively with late-learning regions. The basal ganglia do not just receive transferred control from cortex; they internally reorganise from cognitive to sensorimotor circuits.

Haier et al. (1992) provided perhaps the most vivid demonstration. Using PET imaging, they showed that Tetris practice produced a **seven-fold performance improvement** alongside *decreased* cortical glucose metabolism. The brain got dramatically better at the task while using dramatically less energy. This is the neural signature of automaticity; and it explains the subjective experience of mastery: difficult things become easy not because they require less processing, but because the processing has migrated to circuits that are metabolically cheaper and computationally faster.

For game design, the cortical-to-subcortical transfer means that every game operates on a hidden clock. As the player practises, processing migrates from cortex (slow, effortful, attention-consuming) to basal ganglia and cerebellum (fast, automatic, attention-freeing). The game must introduce new challenges that re-engage cortical processing at approximately the same rate that existing challenges migrate to subcortical automaticity. If new challenges arrive too slowly, the player's cortex has nothing to do and boredom results. If they arrive too fast, the cortex is overwhelmed with unautomatised demands and frustration results. The optimal rate of challenge introduction is the rate that matches the player's cortical-to-subcortical transfer speed.

5.4 The chunking mechanism

Ann Graybiel's research provides the mechanistic framework for how the basal ganglia acquire automaticity. Through decades of work on rodent habit formation, she has shown that the striatum implements a **chunking mechanism**: complex action sequences are gradually consolidated into single, automated routines.

The key finding is a distinctive pattern of striatal neural activity called **task-bracketing**: neurons fire strongly at the initiation and termination of a learned sequence but go silent during execution. As a rat learns to navigate a maze, striatal neurons initially fire throughout the run. With practice, activity consolidates to the start and end points, with the middle of the sequence executed as a single automated chunk. The sequence has been "compiled" from a series of individual decisions into a single habitual action.

This is the neural implementation of what players experience when a complex combo in a fighting game stops feeling like a series of button presses and starts feeling like a single action. The quarter-circle-forward-plus-punch sequence that initially required conscious monitoring of each directional input becomes a single motor program executed as a unit. Chase and Simon's (1973,

Cognitive Psychology) chunking experiments in chess revealed the same principle at the cognitive level: masters could recall positions of approximately 16 pieces after a five-second glance; beginners recalled about 4. But with randomly placed pieces, masters performed no better than beginners. Expertise was not superior memory; it was a library of approximately 50,000-100,000 domain-specific patterns stored in long-term memory, each connected to plausible responses.

5.5 Expert cognition: the endpoint of the transition

The most compelling evidence for the dual-system framework in games comes from studies of expert performance.

Wan et al. (2011, *Science*) compared 11 professional and 17 amateur shogi (Japanese chess) players using fMRI. When generating the best next move within one second (forcing intuitive processing), professionals showed specific activation of the **caudate nucleus**; a basal ganglia structure that was completely silent in amateurs during the same task. When given eight seconds for deliberate search, the caudate remained silent even in professionals. A separate professional-specific activation appeared in the **precuneus** during board perception, and precuneus-caudate activity covaried, suggesting a circuit from pattern recognition to intuitive action selection. Wan et al.'s (2012, *Journal of Neuroscience*) follow-up trained novices for 15 weeks and found caudate activation developed in parallel with intuitive performance; confirming this is learned, not innate.

This is not a gradual difference between experts and novices. It is a qualitative shift in which brain system handles the task. The progression from complete novice (System 2 for every decision) through intermediate (basic tactics become System 1 pattern recognition) to grandmaster (a library of ~50,000-100,000 chunks enabling rapid positional evaluation) is the clearest illustration of the dual-process trajectory across years of practice.

As Kahneman himself framed it, citing Simon: "Intuition is nothing more and nothing less than recognition." The expert's System 1 has absorbed what once required their System 2.

5.6 Competing memory systems

Foerde, Knowlton, and Poldrack (2006, *PNAS*) added a critical finding: a secondary task during learning shifts reliance from declarative (hippocampal) memory to habit (striatal) learning. The two memory systems compete. When the hippocampal system is occupied by a concurrent task, the striatal system takes over, and the resulting learning is more habitual and less flexible.

This has direct implications for games that demand multitasking. A real-time strategy game that requires simultaneous base management and combat forces striatal procedural learning pathways, potentially accelerating automaticity but at the cost of explicit strategic understanding. The player may develop good habits without knowing why they work. A turn-based strategy game that allows time for reflection engages the hippocampal system, producing more flexible but slower learning.

5.7 Games and language share procedural circuits

One of the novel theoretical contributions of this book is the claim that games and language share frontal-basal ganglia circuits evolved for hierarchical sequential structure. The evidence is substantial.

The procedural memory system that computes grammatical rules - the caudate nucleus, putamen, and Broca's area (BA 44/45) - also underpins game rule learning, strategic chunking, and expert intuition. Both games and language exhibit **discrete infinity**: finite rules generating unbounded combinatorial spaces. A finite grammar with recursive rules generates infinite sentences; a finite game rule

set generates astronomically large game trees (chess: $\sim 10^{120}$ nodes; Go: $\sim 10^{360}$). Both face the same computational challenge of navigating enormous spaces through hierarchical chunking and heuristic search.

Thibault, Py, Gervasi, and colleagues (2021, *Science*) demonstrated **common neurofunctional substrates in the basal ganglia** for both tool use (hierarchical motor planning) and language syntax. Training on one function improved the other, demonstrating bidirectional transfer. The implied evolutionary trajectory runs: hierarchical action planning (present in primates) → tool use (recruiting IFG and basal ganglia) → language (exapting the same circuits for symbolic combination) → games (extending hierarchical rule-governed sequential behaviour into the cultural domain).

Riggins (2020, IEEE Conference on Games) explicitly defined a grammar-like formalism for games, treating game systems as formal structures analogous to grammars. Browne (2016) developed a context-free grammar for the Ludii general game system that describes games as trees of "ludemes" (game atoms), directly paralleling how sentences are described as trees of syntactic constituents. The formal parallels between game rules and formal grammars are not metaphors; they are computationally precise.

5.8 Games as conversion engines

Every game is, at its core, a machine for converting System 2 deliberation into System 1 automaticity. The quality of this conversion; introducing challenges that engage System 2 at the right rate, providing feedback that supports pattern extraction, and scaling difficulty to match automaticity acquisition; is a primary determinant of player experience.

Games that manage the conversion well produce flow, mastery, and the distinctive pleasure that Koster identified as "learning." Games that mismanage it produce boredom (System 1 saturated with nothing left to learn) or frustration (System 2 overwhelmed beyond

its conversion capacity).

The most enduring games create **nested automaticity cycles at multiple timescales**: moment-to-moment motor learning (seconds), encounter-level pattern acquisition (minutes), system-level strategic understanding (hours), and meta-level domain expertise (weeks to years); each operating as its own System 2→System 1 pipeline, feeding into the next. The brain does not play games with one system or the other. It plays with both, and the art of game design is choreographing their interaction.

5.9 Transition

The neural mechanisms of reward, curiosity, and skill acquisition converge on a single picture: the brain is an uncertainty-reduction machine that finds the process of reducing uncertainty intrinsically rewarding. Dopamine signals prediction error. Curiosity drives the search for informative experiences. And the procedural memory system converts effortful learning into automatic skill, freeing cognitive resources for the next layer of challenge.

Games exploit all three systems simultaneously by providing structured environments where uncertainty is calibrated, feedback is immediate, and skill acquisition proceeds at a measurable rate. The next section examines how this maps onto the subjective experience of play; the phenomenology that players actually report.

PART III - THE PHENOMENOLOGY OF GAMES

Chapter 6 - Flow

On the concept of flow; its defining characteristics, its limitations as an explanatory framework, and its relationship to dual-process cognition.

6.1 Csikszentmihalyi's model

Mihaly Csikszentmihalyi's concept of flow, developed across four decades of research beginning with *Beyond Boredom and Anxiety* (1975) and formalised in *Flow: The Psychology of Optimal Experience* (1990), remains the most widely referenced psychological construct in game design. Flow describes a state of complete absorption in an activity, characterised by eight dimensions:

1. **Clear goals:** the player knows what they are trying to do
2. **Immediate feedback:** the player knows how well they are doing
3. **Challenge-skill balance:** the demands of the task match the player's ability
4. **Merged action and awareness:** the player acts without conscious self-monitoring
5. **Loss of self-consciousness:** the inner critic goes quiet
6. **Altered sense of time:** hours pass in what feels like minutes
7. **Sense of control:** the player feels capable of handling the situation
8. **Autotelic experience:** the activity is intrinsically rewarding

The meta-analytic flow-performance correlation is $r = .31$ (Alameda et al., 2022, *Cortex*), confirming that flow is associated with

objectively better performance, not merely a pleasant delusion. Csikszentmihalyi's signal contribution was identifying this state as a coherent psychological phenomenon and locating it in the zone where challenge approximately equals skill; too much challenge produces anxiety, too little produces boredom, and the narrow band between them is where optimal experience lives.

The flow construct has been applied to games by Jenova Chen (2007), whose MFA thesis "Flow in Games" directly influenced the design of *fIOW* and *Journey*, and by numerous researchers using validated instruments (Flow State Scale, Dispositional Flow Scale, Experience Sampling Method) to measure flow during gameplay. The consensus finding is that games are unusually reliable flow inducers, probably because they satisfy the preconditions - clear goals, immediate feedback, calibrated challenge - more consistently than most natural activities.

6.2 Flow as a third cognitive mode

Despite its utility, Csikszentmihalyi's model describes flow at the psychological level without specifying the neural mechanisms that produce it. To understand what flow actually is at the level of brain function, we need to place it within the dual-process framework established in Chapter 5.

The question is: is flow a System 1 state, a System 2 state, or something else entirely?

Three competing models have been proposed.

Model 1: Flow as System 1 dominance. Arne Dietrich's **transient hypofrontality hypothesis** (2003, 2004, *Consciousness and Cognition*) proposes that flow requires temporary suppression of prefrontal analytical and meta-conscious capacities. Because the brain has finite metabolic resources, intense activation of motor and sensory processing systems during peak performance produces a concomitant decrease in prefrontal activity. The explicit System 2

system partially deactivates while the implicit System 1 system runs without interference. Dietrich's 2004 paper specifically frames flow as "a period during which a highly practised skill in the implicit system's knowledge base is implemented without interference from the explicit system."

Model 2: Flow as control-reward synchronisation. Weber and Huskey proposed synchronisation theory, which identifies flow with a specific pattern of functional connectivity: **dIPFC-NAc (dorsolateral prefrontal cortex to nucleus accumbens) synchronisation** within a modular brain-network topology characterised by the lowest global efficiency. In this model, flow is not the absence of prefrontal control but the precise synchronisation of control signals with reward signals, creating a feedback loop where task engagement and reward reinforce each other without conscious monitoring.

Model 3: Flow as optimised dual-process integration. Harris et al. (2017) challenged the transient hypofrontality account by demonstrating that **objective mental effort peaks during flow while subjective effort is minimal**. The brain is working hard - harder than in non-flow states - but the work does not feel effortful. This suggests that flow is not the absence of System 2 but its optimal operation through well-trained procedural pathways. Task-relevant executive control remains fully engaged; what is suppressed is not prefrontal activity per se but the metacognitive overhead - self-monitoring, self-criticism, temporal awareness - that normally accompanies System 2 operation.

The evidence best supports Model 3, which aligns flow with the cortical-to-subcortical transfer described in Chapter 5. Flow is neither pure System 1 (which would be automatic but disengaged; consider the experience of driving a familiar route while your mind wanders) nor pure System 2 (which would be effortful and self-aware; consider the experience of solving a difficult maths problem). It is a **third configuration**: task-relevant executive control operating through well-trained procedural pathways,

producing high performance at high demand without the subjective experience of effort.

This is why flow requires that the player's automaticity level precisely matches the game's demands. If the challenge exceeds automatised capability, System 2 reactivates in its full self-monitoring mode (frustration, conscious problem-solving). If automatised capability exceeds the challenge, neither system is fully engaged (boredom). The flow channel is literally the moving boundary between System 1 and System 2 processing, and the game's difficulty curve must track the player's automaticity acquisition rate to maintain it.

6.3 The locus coeruleus-norepinephrine system as the shared mechanism

The neurochemical system most consistently implicated in flow across all three models is the locus coeruleus-norepinephrine (LC-NE) system. Van der Linden, Tops, and Bakker (2021, *Frontiers in Psychology*) proposed the LC-NE system as the mediator of flow states, building on Aston-Jones and Cohen's (2005) dual-mode model.

In **phasic mode** (low baseline NE, task-evoked bursts), the LC-NE system facilitates focused attention. Norepinephrine is released in response to task-relevant stimuli, sharpening signal-to-noise ratios in cortical processing and maintaining engagement with the current task. Exploratory attention-shifting is suppressed. This is the mode associated with exploitation, focused performance, and flow.

In **tonic mode** (high baseline NE, broad attentional sampling), the LC-NE system promotes exploration. Baseline norepinephrine is elevated, reducing the signal-to-noise ratio and allowing attention to wander. The organism scans the environment for better options. This is the mode associated with distraction, curiosity, and task-switching.

Flow corresponds to sustained phasic mode: attention is locked onto the task, and the task is generating prediction errors at a rate that sustains this configuration. The phasic NE bursts reinforce attention to task-relevant events (enemies dodging, platforms appearing, patterns resolving), while the low tonic baseline prevents attention from drifting to task-irrelevant information (checking the clock, thinking about dinner, noticing background noise).

The neurochemical picture likely involves dopaminergic reward prediction error synchronising with noradrenergic arousal regulation; dopamine signals "this is worth learning from" while norepinephrine signals "stay focused on this." When both systems are operating in concert, the result is the absorbed, effortless engagement that Csikszentmihalyi described.

6.4 Cognitive ease versus effortless attention

A critical distinction that has produced confusion in the flow literature is the difference between **cognitive ease** and **effortless attention**.

Kahneman's cognitive ease is genuinely low demand. The task is simple, the answers are obvious, and the brain is coasting. System 2 is barely engaged because it is not needed. This is the experience of reading a familiar text, walking a familiar route, or playing a game on its easiest setting. It is pleasant in a mild way but not absorbing.

Csikszentmihalyi's effortless attention is something completely different. The task is demanding; objectively more demanding than a typical non-flow task. But the effort does not feel effortful because processing has migrated to procedural pathways that operate without metacognitive overhead. The player is working hard; Harris et al. (2017) confirmed this with objective measures; but the work feels natural, fluid, and automatic.

Games that produce cognitive ease (too-easy tasks) do not produce flow. The player is not absorbed; they are coasting. Games that

produce effortless attention (optimally challenging tasks processed through automatised pathways) do produce flow. The difference is whether System 2 is absent because it is not needed, or whether System 2's computational work is being performed through System 1's hardware.

6.5 Flow is fragile

The sustained phasic LC-NE mode that supports flow is self-reinforcing but fragile. As long as prediction errors arrive at the right rate; frequent enough to sustain phasic NE bursts, not so frequent as to overwhelm processing; the system maintains itself. The player stays locked in.

But any disruption can break the loop:

- A **difficulty spike** overwhelms the player's automatised capability, forcing System 2 back into full self-monitoring mode (frustration, conscious problem-solving)
- A **difficulty drop** eliminates prediction errors, causing the phasic mode to decay into tonic mode (boredom, attention-wandering)
- A **confusing design decision** produces prediction errors that cannot be resolved, blocking model updating (frustration, helplessness)
- A **loading screen** interrupts the temporal continuity of the action-feedback loop, allowing the phasic mode to decay
- A **cutscene** removes agency, eliminating the action component of the prediction error cycle
- An **unjust death** produces a negative prediction error attributable to the system rather than the player, breaking trust in the learnability of the system

Re-establishing flow after disruption requires recalibrating the learning rate from scratch. The player must re-enter the task, rebuild attentional focus, and re-engage the phasic LC-NE mode.

This takes time, and if disruptions are frequent, flow never establishes in the first place. This is why seamless design - minimal loading, integrated narrative, consistent rules, uninterrupted gameplay - correlates so strongly with critical acclaim. Every seam in the experience is a potential flow-breaker.

6.6 The limitations of flow theory for game design

Despite its influence, flow theory has a critical limitation when applied to game design: it is **static**.

Flow theory identifies challenge-skill balance as the key variable. At any given moment, the player is either in the flow channel (challenge $\approx$ skill), above it (anxiety), or below it (boredom). This is useful for diagnosis but insufficient for design, because it treats flow as a snapshot rather than a trajectory.

Consider two players with identical challenge-skill ratios. Both are in the flow channel at this exact moment. But one player is improving rapidly; each encounter teaches them something new, and their skill is rising. The other player is stagnating; they are performing adequately but learning nothing, and their skill is static. Both satisfy Csikszentmihalyi's challenge-skill balance criterion. But their experiences are qualitatively different. The first player is deeply engaged; the second is on the verge of boredom.

The difference is not where they are but **how they are moving**. Flow theory captures the position; it misses the velocity. What is missing is the temporal dimension: the rate at which the player is learning, improving, and reducing uncertainty about the game system. This is the variable that will be isolated in Part IV and formalised in the unified model of Chapter 12.

6.7 Transition

Flow theory provides a powerful description of optimal experience but an incomplete explanation of what produces it. Its core insight - that engagement requires a match between challenge and skill - is correct but insufficient. The match must be dynamic, not static. It must evolve over time as the player's skill increases. And the rate of that evolution - the learning rate - is the hidden variable that determines whether the experience is truly absorbing or merely adequate.

Before we can formalise this insight, we need to examine what successful games actually do - what design patterns recur across the most acclaimed titles - and what existing design theories have and have not captured.

Chapter 7 - What Great Games Share

On the recurring design patterns in critically acclaimed games; and why these patterns align with the conditions that support sustained engagement.

7.1 The common ground

The highest-rated games across the past two decades; *The Legend of Zelda: Breath of the Wild*, *The Last of Us*, *Elden Ring*, *Portal*, *Dark Souls*, *Celeste*, *Hades*, *Dota 2*, *Tetris*, *Outer Wilds*; span wildly different genres, aesthetics, and audiences. A puzzle game and a fighting game share almost no surface features. An idle game and a Soulslike appear to demand completely different explanatory models.

Yet analysis of the design features that recur across 95+ Metacritic games reveals a consistent set of structural properties. These are not stylistic choices or genre conventions. They are **conditions for sustained learning**, and their recurrence across genres is predicted by the neural mechanisms established in Part II.

7.2 Challenge-skill calibration

Every acclaimed game provides mechanisms for calibrating challenge to player skill, though the mechanisms vary.

Some games offer explicit difficulty selection: *Halo's* four tiers, *Celeste's* Assist Mode, *The Last of Us's* granular difficulty options. Others calibrate through player-directed exploration: *Breath of the Wild* and *Elden Ring* let the player choose where to go, implicitly

choosing their difficulty level. Others use hidden affordances: *Mario*'s "coyote time," aim assist in console shooters, generous hitboxes that make near-misses count as hits. And some build calibration into the structure itself: roguelikes like *Hades* offer persistent upgrades that reduce difficulty over repeated runs, ensuring that every player eventually reaches a manageable challenge level regardless of initial skill.

The common principle is that the game must maintain the player within the zone where prediction errors are frequent but resolvable. Too few errors produce boredom; too many produce overload. The specific mechanism matters less than the outcome: a learning gradient that stays positive across the widest possible range of player skill levels.

7.3 Tight feedback loops

Acclaimed games provide unusually clear and immediate feedback at the perceptual level. Steve Swink's *Game Feel* (2008) identified the experiential foundation: when driving a nail, you feel the nail through the hammer automatically, without conscious processing. Game feel is the equivalent perceptual extension; the screen becomes vision, speakers become hearing, and rumble motors become touch. The avatar becomes an extension of body and self through automatic proprioceptive transfer.

Swink's analysis places game feel firmly in System 1 territory; it operates at timescales below approximately 240ms, faster than conscious deliberation. Every frame of input lag, every ambiguous damage indicator, every unclear death screen degrades the System 1 feedback loop and slows the learning gradient. Acclaimed games are obsessive about feedback precision: *Mario*'s jump arc provides frame-level information about trajectory; *Halo*'s shield indicator, reticle colour change, and enemy flinch animations provide multi-layered combat feedback; *Celeste*'s death is instantaneous and respawn is immediate, eliminating the delay between error and

retry.

The LC-NE system provides the neural mechanism. Phasic norepinephrine bursts are triggered by salient events; task-relevant signals that demand processing. When feedback is immediate and precise, each action triggers a phasic burst that reinforces focused attention. When feedback is delayed or ambiguous, the phasic signal is weak, and the system drifts toward tonic mode (distraction, disengagement).

7.4 Intrinsic exploration and suppression of extrinsic markers

The most acclaimed open-world games suppress the extrinsic reward markers that less acclaimed titles rely on. *Breath of the Wild* removes quest markers, minimap objectives, and percentage completion trackers. *Elden Ring* provides no quest log and minimal map guidance. *Outer Wilds* has no upgrades, no experience points, and no permanent progression of any kind.

The design rationale, grounded in the curiosity neuroscience of Chapter 4, is that extrinsic markers activate phasic dopamine through anticipated-reward-delivery (the icon on the map predicts a reward at that location), producing intermittent reinforcement rather than sustained curiosity. Intrinsic exploration activates a tonic dopamine state of sustained curiosity; the reward is discovery itself, which is inherently unpredictable. The tonic state maps better onto the sustained, absorbed engagement that characterises flow; the phasic anticipated-reward pattern maps onto the intermittent reinforcement schedule that characterises addiction.

7.5 Systems that recombine rather than accumulate

Acclaimed games tend to introduce a limited set of mechanics that interact in rich, combinatorial ways, rather than accumulating a

large number of independent mechanics. *Halo's* Golden Triangle (guns, grenades, melee) generates more tactical depth through three-way interaction than a game with twenty independent abilities. *Breath of the Wild's* physics system (fire, electricity, magnetism, wind, temperature) produces emergent solutions that no designer anticipated. *Portal's* single mechanic (linked portals) is extended through spatial reasoning rather than through the addition of new portal types.

The prediction error logic is clear: combinatorial systems produce **more** unique prediction errors from **fewer** elements, because the interaction space grows multiplicatively rather than additively. And because the elements are familiar (the player already understands guns, grenades, and melee individually), the prediction errors are resolvable; the player can figure out why the new combination worked or failed because they already understand the components.

7.6 Readable enemies and environments

Acclaimed combat games invest heavily in making enemies and environments **readable**; legible in their behaviour, consistent in their rules, and expressive in their state. *Halo's* Covenant enemies display their internal state through animation and vocalisation. *Dark Souls* bosses telegraph their attacks with distinct wind-up animations. *Doom Eternal's* demons have weak points that glow, health states that are visually distinct, and stagger animations that communicate vulnerability.

The neural mechanism is feedback legibility: prediction errors are only useful for learning if the player can identify what went wrong. Readable enemies convert every encounter into an informative prediction error. Unreadable enemies (like *Halo's* Flood, which rush mindlessly with no state expression) produce prediction errors that cannot be resolved, and the learning gradient collapses.

7.7 The "one more run" structure

A distinctive feature of many acclaimed games, particularly roguelikes and challenging action games, is the **"one more run"** compulsion: the player who dies at 1am and immediately starts another attempt rather than going to bed. *Hades*, *Spelunky*, *Celeste*, and *Dark Souls* all produce this pattern reliably.

The unified model explains why. Death in these games generates a strong negative prediction error (the player's model failed) combined with a clear update signal (the player knows why they died and what to do differently). The updated model produces a prediction of improved performance on the next attempt, which generates anticipatory dopamine activity (the Fiorillo ramp). The player expects the next run to go better, and this expectation - this prediction of a positive prediction error - is itself rewarding. "One more run" is not compulsion; it is the rational response of a brain that has just updated its model and is eager to test the update.

7.8 Transition

The design patterns that recur across acclaimed games are not arbitrary. They are structural conditions for sustained prediction error generation and resolution: calibrated challenge, immediate feedback, curiosity-driven exploration, combinatorial depth, readable systems, and the anticipation of improvement. Each pattern maps onto the neural mechanisms established in Part II.

But existing game design theories capture these patterns only partially. The next chapter examines what current frameworks get right, where they fall short, and why a new theory is needed.

Chapter 8 - Existing Theories and Their Limits

On the strengths and limitations of six influential frameworks for understanding games; and why each captures an aspect of the phenomenon while failing to account for its dynamics.

8.1 The landscape

Game design theory has produced at least six frameworks of genuine explanatory power over the past three decades. Each identifies something real about how games work. None explains why some games sustain engagement for hundreds of hours while others - built on the same principles, satisfying the same criteria - fail to hold attention beyond a few sessions. This chapter takes each framework seriously on its own terms before identifying the specific gap that the unified model will fill.

8.2 MDA: structure without dynamics

Hunicke, LeBlanc, and Zubek's MDA framework (2004 AAAI Workshop on Challenges in Game AI) decomposes games into three layers. **Mechanics** are the base components: rules, player actions, algorithms, data structures. **Dynamics** are the run-time behaviour that emerges when mechanics interact with player input. **Aesthetics** are the emotional responses evoked in the player.

The framework identifies eight aesthetic categories: Sensation (game as sense-pleasure), Fantasy (game as make-believe), Narrative (game as drama), Challenge (game as obstacle course),

Fellowship (game as social framework), Discovery (game as uncharted territory), Expression (game as self-discovery), and Submission (game as pastime). Its most important insight is directional: designers control only mechanics; dynamics emerge from mechanics; aesthetics emerge from dynamics. The player encounters these layers in reverse order, perceiving aesthetics first, then dynamics, then inferring mechanics. This creates a fundamental design challenge: the designer cannot directly control player experience. They can only shape it indirectly through the rules and systems they build.

MDA is valuable. It correctly identifies the indirect, emergent relationship between what designers make and what players feel. It provides a shared vocabulary for discussing why two mechanically similar games can produce different emotional responses (they generate different dynamics from similar mechanics) and why two emotionally similar experiences can arise from different mechanics (multiple mechanical configurations can converge on the same dynamics).

But MDA has three critical limitations.

First, it is **descriptive rather than predictive**. The framework can decompose an existing game into its layers and classify its aesthetics. It cannot tell a designer which mechanics will produce which dynamics, or which dynamics will produce which aesthetics, before the game is built. As critics have noted, MDA "leaves the design process reliant on subjectivity and stakeholder knowledge"; it works better for post-hoc critique than forward-looking design.

Second, its **linear model is too simple**. The mechanics→dynamics→aesthetics pipeline assumes a unidirectional flow, but real games involve continuous feedback: playtesting reveals unintended dynamics, which prompt mechanical revision, which produces new dynamics. Complex games like *Europa Universalis III* or *Silent Hunter III* require documentation before players can engage; they do not conform to the linear progression from aesthetics perception to mechanics inference. The Redefining

MDA (RMDA) framework proposed in 2021 (*MDPI Information Journal*) attempted to address this bidirectionality.

Third, and most consequential for this book: MDA is **silent on time**. It describes the structural relationship between mechanics, dynamics, and aesthetics at a single moment. It says nothing about how that relationship changes as the player learns, improves, and exhausts the game's patterns. A game whose MDA decomposition is identical at hour one and hour fifty may produce completely different player experiences at those two points because the player's relationship to the system has changed. MDA captures what a game is. It does not capture what a game does to its player over time.

8.3 Meaningful play: outcomes without trajectories

Katie Salen and Eric Zimmerman's *Rules of Play: Game Design Fundamentals* (MIT Press, 2004) was the first comprehensive attempt to establish a theoretical framework for game design as a discipline. Their central concept is **meaningful play**, defined at two levels.

The **descriptive definition**: meaningful play resides in the relationship between action and outcome. The **evaluative definition**: meaningful play occurs when the relationship between actions and outcomes is both **discernible** (the player can perceive that their actions have effects) and **integrated** (those effects persist and contribute to the larger game context, not just the immediate moment).

This is a stronger framework than it initially appears. The discernibility criterion correctly predicts that games with opaque feedback will fail to engage; if the player cannot see the effects of their actions, the system becomes a black box and learning is impossible. The integration criterion correctly predicts that games with only immediate consequences will feel shallow; if nothing carries forward, each moment is disconnected from every other, and

the game has no arc. Together, discernibility and integration describe the conditions under which player actions feel **consequential** rather than arbitrary.

But meaningful play, like MDA, is a structural criterion. It asks whether the relationship between action and outcome has certain properties (discernible, integrated) without asking how that relationship evolves. A game can satisfy both criteria throughout its entire runtime and still fail to engage if the player's model of the action-outcome relationship stops improving. Consider a game where every action is discernible, every outcome is integrated, but the player has fully learned the mapping after two hours: the remaining content is structurally meaningful (actions still produce discernible, integrated outcomes) but experientially empty (no prediction errors remain to resolve). Meaningful play is necessary for engagement but not sufficient. What is missing is a criterion about the *trajectory* of the player's understanding.

Sidhu and Carter (2021, *Sage Publishers*) proposed "Pivotal Play" as an alternative - "appealing, memorable, and transformative play experiences" - arguing that Salen and Zimmerman's paradigm does not capture meaning that occurs outside the immediate gameplay context. This critique points toward the same gap: meaningful play describes structural properties of the game system without describing the dynamic, temporal experience of the player interacting with it.

8.4 Uncertainty: the right ingredient without the recipe

Greg Costikyan's *Uncertainty in Games* (MIT Press, 2013) makes a sharper claim than either MDA or meaningful play: games **require** uncertainty to hold interest. The struggle to master uncertainty is central to their appeal.

Costikyan identifies eleven forms of uncertainty, and the taxonomy is worth presenting because it is the most complete catalogue of the raw material that games use to generate engagement:

1. **Performative uncertainty:** can I physically execute this manoeuvre? (*Guitar Hero*, *Super Mario Bros.*)
2. **Solver's uncertainty:** can I find the solution? (*Portal*, adventure games)
3. **Player unpredictability:** how will other players act? (multiplayer, robust AI)
4. **Randomness:** what will fortune give me? (dice rolls, roguelikes, procedural generation)
5. **Analytic complexity:** what is the optimal move in this complex decision tree? (*Chess*)
6. **Hidden information:** what is being deliberately withheld? (poker, fog of war)
7. **Narrative anticipation:** what happens next in the story?
8. **Development anticipation:** what new content will appear as I progress?
9. **Schedule uncertainty:** what has changed since my last session? (idle games, live services)
10. **Uncertainty of perception:** can I filter the important data from the noise?
11. **Semiotic uncertainty:** what does my playing this game mean?

This taxonomy is genuinely useful. It explains why different games feel different despite all being "uncertain": *Chess* deploys analytic complexity with zero randomness; *Poker* combines hidden information with player unpredictability; *Dark Souls* generates primarily performative uncertainty. Costikyan's framework provides a vocabulary for identifying *which kind* of uncertainty a game deploys, not just how much.

But the taxonomy, powerful as it is, does not explain why some deployments of uncertainty produce sustained engagement and others do not. A slot machine generates randomness (type 4) at high intensity and frequency. A chess match generates analytic complexity (type 5) at comparable intensity. Both are uncertain. One sustains engagement for decades of serious study; the other sustains engagement only through the exploitation of variable-ratio reinforcement schedules. Costikyan correctly identifies that uncertainty is the essential ingredient. He does not explain the difference between uncertainty that sustains genuine learning and uncertainty that sustains compulsive behaviour. What is missing is a criterion about **reducibility**: uncertainty that the player can progressively resolve through skill development produces fundamentally different engagement than uncertainty that remains irreducible regardless of the player's actions.

8.5 Self-determination theory: motivation without mechanism

Ryan and Deci's self-determination theory, applied to games by Rigby and Ryan (2011) and in the foundational study by Ryan, Rigby, and Przybylski (2006, *Motivation and Emotion*; four studies, 2,685+ citations), identifies three basic psychological needs: **autonomy** (the sense of being the origin of one's own behaviour), **competence** (the sense of being effective), and **relatedness** (the sense of connection to others). When games satisfy these needs, players are intrinsically motivated to engage.

SDT has genuine predictive power at the macro level. It correctly predicts that games offering meaningful choices will be preferred over games that railroad the player (autonomy). It correctly predicts that games with appropriate challenge and clear feedback will be preferred over games that are too easy or too opaque (competence). It correctly predicts that multiplayer games with cooperative or competitive social structures will sustain engagement longer than

isolated single-player experiences (relatedness).

But SDT has been subjected to serious critique in the games research literature, and the criticisms are damaging.

A comprehensive review in *ACM Transactions on Computer-Human Interaction* found that the bulk of SDT game research consists of "shallow, perfunctory applications of the theory" rather than rigorous engagement. Even works claiming substantial engagement "contain prevalent misconceptions about SDT's fundamental concepts." The theory is used as "vague sources of hypotheses or post hoc explanations" rather than a predictive framework. Core aspects of SDT's broader framework (Basic Psychological Need Theory, Organismic Integration Theory) are largely ignored. And there is a "broad unwillingness to contest SDT tenets when study results are inconsistent with the theory"; SDT functions as an unquestioned paradigm.

More substantively, SDT has three gaps that matter for this book.

First, **relatedness is undefined for single-player games**. SDT does not officially define how relatedness contributes to intrinsic motivation in single-player contexts. This is a significant theoretical hole given that some of the most engaging games ever made (*Dark Souls*, *Zelda*, *Portal*) are single-player experiences. Przybylski, Rigby, and Ryan (2010, *Review of General Psychology*) extended the model to show that violence per se did not drive motivation - need satisfaction did - but the single-player relatedness gap remains.

Second, the three needs are **non-independent** in ways the framework does not specify. Autonomy must accompany competence for people to see their behaviours as self-determined by intrinsic motivation. Simply adopting game elements designed to support these needs does not always guarantee the desired results. The needs interact in complex, non-additive ways that SDT does not model.

Third, and most critically: SDT explains **why people start playing** but not **what sustains engagement moment to moment**. Two

games can satisfy autonomy, competence, and relatedness equally well and produce completely different levels of engagement because their moment-to-moment prediction error profiles are different. SDT tells you that the player needs to feel competent; it does not tell you what should happen in the next ten seconds of gameplay to sustain that feeling. SDT research in games typically measures aggregate engagement levels rather than explaining variation in moment-to-moment player engagement during gameplay. The theory lacks temporal granularity.

8.6 Flow: experience without trajectory

Csikszentmihalyi's flow theory was addressed at length in Chapter 6, but its limitations warrant restatement in the context of competing frameworks.

Flow theory's core claim is that optimal experience arises when challenge matches skill. Its eight dimensions (clear goals, immediate feedback, challenge-skill balance, merged action and awareness, loss of self-consciousness, altered time sense, sense of control, autotelic experience) describe a coherent phenomenological state that players recognise immediately. The theory is widely cited in game design education and has directly influenced commercial design (Jenova Chen's *fLOW* and *Journey*).

But the game design community has systematically misapplied flow by collapsing it into a single dimension: the challenge-skill balance diagram. This oversimplification conceals two serious problems.

First, a balance between **low** skill and **low** challenge does not produce flow; it produces apathy. Mere balance is insufficient. Both dimensions must reach a minimum threshold. The "flow channel" diagram, reproduced in hundreds of design presentations, obscures this by suggesting that any diagonal through the challenge-skill space is equivalent. It is not. A player whose skill is low and whose challenge is low is not in flow; they are disengaged. Flow requires not just balance but balance at a level of demand that engages the

organism's full processing capacity.

Second, and more damaging: the challenge-skill model is **static**. Flow has traditionally been used as a static construct, but flow literature increasingly suggests it should be understood as a dynamic psychological process. Flow theory was developed primarily for sustained activities like mountain climbing, surgery, and knowledge work - not for the rapid fluctuations in difficulty that characterise most games. Empirical work has shown that flow is not always optimised by challenge-skill balance; inferring flow from this condition alone is "not a safe bet." Skills and challenges function as independent cognitive factors, and the balance model cannot account for the rapid fluctuations in difficulty that characterise most game sessions.

The temporal problem is the most important. Two players can occupy the same position in the challenge-skill space; both are in the "flow channel" at this exact moment. But one player is improving rapidly; each encounter teaches them something new. The other is stagnating; they are performing adequately but learning nothing. Both satisfy Csikszentmihalyi's challenge-skill balance criterion. Their experiences are qualitatively different. Flow theory captures where the player is. It misses how fast they are moving and in what direction.

Cowley, Moutinho, Bateman, and Oliveira (2011, *Computers in Entertainment*, ACM) emphasised that learning principles and design interaction are equally or more important than flow-state induction, representing one of the earlier empirical challenges to flow's primacy in game design theory.

8.7 Koster: learning without rate

Raph Koster's *A Theory of Fun for Game Design* (2004; revised 2013) comes closest to the model this book proposes, and it deserves the most extended treatment.

Koster's central claim is that fun is the emotional response to learning patterns. "Games are just exceptionally tasty patterns to eat up." The mechanism is cognitive chunking: the brain divides information into usable groups, and the process of chunking is intrinsically rewarding. When a game's patterns are fully absorbed - when the player has "grokked" the system - the game becomes boring. There is nothing left to learn.

This is the most important single insight in game design theory. It correctly identifies learning, not reward or narrative or spectacle, as the core driver of engagement. It correctly predicts that games with deeper pattern spaces sustain engagement longer (chess outlasts tic-tac-toe). It correctly predicts the endpoint of engagement (pattern exhaustion, not content exhaustion). And it correctly connects game design to cognitive science rather than treating games as a purely aesthetic phenomenon.

In his 2024 GDC talk, "Revisiting Fun: 20 Years of A Theory of Fun," Koster reflected that he "wasn't excited about how narrow" his original formulation had been. He expressed a more relaxed perspective on how game designers can understand why systems that do not fit neatly into his theory still bring joy, suggesting an expansion beyond pure pattern-recognition. He explicitly connected his work to the predictive processing framework, highlighting Deterding, Andersen, Kiverstein, and Miller (2022, *Frontiers in Psychology*), which demonstrates that predictive processing provides a coherent formal cognitive framework explaining fun as "the dynamic process of reducing uncertainty surprisingly efficiently."

This is the right direction. But even with the predictive processing connection, Koster's theory does not formalise the variable that separates fun learning from frustrating learning. A student grinding through differential equations is learning patterns. A musician drilling scales for the hundredth time is learning patterns. These experiences are not fun in the way a good game is fun. The theory correctly identifies what is happening (pattern acquisition) without

specifying the conditions under which that process feels rewarding.

The missing variable is rate. Not whether the player is learning, but how fast - and how fast relative to their expectation. This is the gap between Koster's insight and the unified model: the first derivative of the player's model accuracy over time.

8.8 Daniel Cook's skill atoms: the closest predecessor

Daniel Cook, chief creative officer at Spry Fox, developed the most operationally precise design framework in the practitioner tradition, and it deserves recognition as the closest predecessor to the learning gradient concept.

Cook's **skill atom** is the minimal unit of game learning: a four-element feedback cycle comprising action (the player performs an action), simulation (the game updates), feedback (the player perceives the state change), and modelling (the player updates their mental model, informing the next decision). The learning arc within atoms follows a predictable trajectory: first pass is "vaguely interesting" but not understood; multiple iterations produce gradual refinement; an "aha" moment crystallises the mental model; and mastery follows. Cook explicitly locates fun at the crystallisation point: "This moment of understanding and mastery is at the heart of what we call Fun."

Skill chains extend the model: atoms link into directed graphs where each lower-level skill acts as a foundation for more complex ones, mirroring learning hierarchies in educational psychology. Skill chains can model virtually any game by breaking complex designs into dozens of simple atoms linked to form a clear map of progression.

Cook also distinguishes **loops** (structures exercised multiple times, delivering value through repeated execution) from **arcs** (single-execution structures that evoke past experiences). Games

consist of chemistry-like mixtures of both, nested and connected. The analytical question is always: "What repeats and what does not?"

Cook's framework is remarkably close to the learning gradient concept. The skill atom is effectively a micro-scale prediction error loop (action → outcome → error → model update). The skill chain maps the learning gradient's layered structure (micro → meso → macro). And the loop/arc distinction identifies the difference between renewable engagement (loops sustain prediction errors through repetition) and non-renewable engagement (arcs exhaust their prediction errors in a single pass).

What Cook does not formalise is the **rate** at which a player moves through a skill chain, or the relationship between that rate and their subjective experience. His framework describes the structure of learning in games with considerable precision but does not specify the dynamics; how fast the player should be progressing, what determines whether progress feels rewarding, and what happens when progress stalls. The skill atom describes the feedback loop. The learning gradient describes the velocity of that loop's operation.

8.9 The common limitation

Six frameworks. Each valuable. Each incomplete in the same way.

MDA describes structure without dynamics. Meaningful play describes outcomes without trajectories. Costikyan describes the essential ingredient (uncertainty) without the recipe (reducibility and rate). SDT describes motivation without moment-to-moment mechanism. Flow describes experience without temporal evolution. Koster describes the core process (learning) without the critical variable (rate). Cook describes the feedback loop without the velocity of its operation.

The shared gap is **time**. All six frameworks describe states, structures, or conditions that must be satisfied at a given moment.

None describes the **dynamic process** that unfolds as the player engages over minutes, hours, and days. None captures the **rate of change** that determines whether engagement is sustained, declining, or collapsing.

The next two chapters isolate this missing variable. Chapter 9 demonstrates that learning alone is insufficient to explain fun. Chapter 10 introduces the temporal dimension that transforms a static account into a dynamic one. Chapter 11 identifies the specific quantity - the first derivative of model accuracy over time - that governs the experiential landscape of play.

PART IV - THE MISSING VARIABLE

Chapter 9 - Fun Is Learning (But That Is Not Enough)

On Koster's insight that fun is learning; why it is correct but incomplete; and what it means for a theory to identify the right variable without specifying the right function.

9.1 Koster was right

The previous chapter surveyed six frameworks and found each wanting. Koster's theory of fun came closest to the mark, and this chapter begins by acknowledging why.

Fun is the emotional response to learning patterns. This is not a metaphor. It is a claim about mechanism, and it is supported by the neuroscience reviewed in Part II. The dopamine system rewards prediction error resolution (Schultz, Dayan, & Montague, 1997). The curiosity system drives the search for informative experiences (Gruber, Gelman, & Ranganath, 2014). The procedural memory system converts effortful learning into automatic skill (Poldrack et al., 2005; Lehericy et al., 2005). The brain is built to learn, and it finds the process intrinsically rewarding. Koster identified this at the level of design intuition before the neuroscience existed to confirm it.

His theory also correctly identifies the endpoint. When a game's patterns are fully absorbed - when the player has "grokked" the system - the game becomes boring. The prediction error rate has dropped to zero, and the dopamine system has nothing to signal. This explains why tic-tac-toe is interesting for approximately three games and chess is interesting for a lifetime: the pattern space of

tic-tac-toe is exhausted almost immediately, while the pattern space of chess exceeds any human's capacity to fully absorb.

Daniel Cook's skill atom framework operationalises Koster's insight at the design level: the minimal feedback loop of action → simulation → feedback → modelling describes the micro-structure of game learning with considerable precision. Cook explicitly locates fun at the moment of model crystallisation: "This moment of understanding and mastery is at the heart of what we call Fun."

So far, so good. The problem is that this is not enough.

9.2 Three learning scenarios

Consider three scenarios that share the same variable (learning is occurring) but produce completely different experiences.

Scenario 1: The steep gradient. A player encounters a new boss in *Dark Souls*. The first attempt is bewildering; attacks arrive from unexpected angles, the arena's geometry is unfamiliar, and death comes within seconds. The second attempt lasts thirty seconds longer because the player now recognises the opening attack and dodges it. The third attempt reaches the boss's second phase. The fifth attempt makes it to 50% health. By the tenth attempt, the player's model of the boss has crystallised sufficiently that victory feels imminent. Each death generates a large prediction error that is immediately resolvable; the player knows exactly what killed them and what to do differently. The learning rate is high, and the experience is deeply engaging.

Scenario 2: The stalled gradient. A student is studying organic chemistry. The material is new, the concepts are challenging, and the student has been reading the same chapter for three hours. They can define the terms; they can recite the mechanisms; but the intuitive understanding refuses to arrive. Learning is occurring (each pass through the material strengthens memory traces), but the rate is so slow that it is imperceptible from session to session.

The student knows they are not getting worse, but they cannot feel themselves getting better. The experience is tedious, draining, and unfun.

Scenario 3: The exhausted gradient. A player replays the opening world of *Super Mario Bros.* for the hundredth time. They can complete it with their eyes half-closed. Every enemy position is memorised, every jump arc is automatic, and every block location is known. Learning is no longer occurring because the player's model is complete. The experience is mildly pleasant (the motor execution is satisfying in a low-key way) but not engaging in the sense that matters. There is no absorption, no challenge, no sense of improvement.

Koster's theory correctly predicts that Scenario 3 (no learning) will be boring and that Scenario 1 (rapid learning) will be engaging. But it does not explain the difference between Scenario 1 and Scenario 2. Both involve active engagement with a system that contains learnable patterns. The student in Scenario 2 has not exhausted the patterns; there is much more to learn. But the experience is aversive rather than pleasurable.

The theory correctly identifies the variable (learning). It does not specify the function (the rate of learning relative to the learner's expectation).

9.3 The musician's paradox

The gap in Koster's theory is sharpest in cases of deliberate practice.

A guitarist practising scales is learning. Each repetition refines motor coordination, improves timing, and strengthens the cortical-to-subcortical transfer that will eventually make the scale automatic. By Koster's theory, this should be fun. Sometimes it is. But sometimes it is tedious, and the difference is not whether learning is occurring but how fast.

When the guitarist can hear their improvement - when each pass through the scale is noticeably cleaner, faster, and more accurate than the last - the practice is engaging. The rate of improvement is perceptible, and the experience has the quality of productive absorption. When the guitarist has been practising the same scale for an hour and can no longer detect any improvement - when each pass sounds identical to the last - the practice becomes tedious, even though learning is still occurring at the neural level. Hebbian plasticity continues to strengthen synaptic connections; the power law of practice ensures that each repetition produces some marginal improvement. But the improvement is below the threshold of conscious detection, and the subjective experience shifts from "I am getting better" to "nothing is happening."

The power law of practice (first described in the context of Bryan and Harter's 1899 study of telegraph operators, and subsequently confirmed across motor, perceptual, and cognitive domains) predicts this precisely: the logarithm of reaction time decreases linearly with the logarithm of practice trials. Initial learning is rapid and perceptible. Later learning is slow and imperceptible. The transition from perceptible to imperceptible improvement marks the transition from engagement to tedium, even though learning never actually stops.

Wilson, Shenhav, Straccia, and Cohen (2019, *Nature Communications*) provided the quantitative frame: for gradient-descent learning systems, the optimal error rate is approximately 15.87% (equivalently, approximately 85% accuracy). When the guitarist is making one mistake per seven notes, improvement is rapid and perceivable. When they are making one mistake per fifty notes, improvement is still occurring but at a rate too slow to register. The experience shifts from "I am getting better" to "nothing is happening" even though something is.

9.4 Productive failure and the role of struggle

Manu Kapur's productive failure research (2008, *Cognition and Instruction*; 2014, *Cognitive Science*) adds a crucial nuance. Students who struggle with problems before receiving instruction learn more than students who receive instruction first. Both approaches yield high levels of procedural knowledge. The critical difference: students who problem-solved first demonstrated significantly greater conceptual understanding and ability to transfer to novel problems. The struggle itself, even when it produced "suboptimal or even incorrect solutions," prepared the learner for deeper subsequent understanding.

This finding maps directly onto game design. A *Dark Souls* player who dies to a boss twenty times before defeating it has a deeper model of the boss's behaviour than a player who defeats it on the first attempt with an overpowered build. The deaths were not wasted time; they were productive failures that built the conceptual framework necessary for genuine mastery. Jesper Juul's *The Art of Failure* (MIT Press, 2013) captures this from the player's perspective: games exploit a "paradox of failure" in which humans have a basic desire to succeed yet voluntarily engage in activities where they are nearly certain to fail. The resolution is that the feeling of escaping failure - often through improving skills - is a central enjoyment of games.

Robert and Elizabeth Bjork's "desirable difficulties" framework complements this: conditions that slow apparent learning (spacing, interleaving, variation) actually accelerate long-term retention and transfer. The difficulty is desirable because it forces deeper processing; the struggle is productive because it builds more robust models. In game design terms, a desirable difficulty is one where prediction errors are slightly larger than comfortable but remain resolvable with effort. This describes exactly the experience of a well-designed difficulty curve: each new section feels slightly too hard at first, but the player's model catches up within a few attempts.

But productive failure works only when the failure is **informative**. When failure provides clear error signals that the player can use to update their model, each death generates genuine learning. When failure provides only "you died" with no causal information, each death generates frustration without learning. And failure with no consequence at all (unlimited retries, no stakes) is proven less effective because players can fail continuously without strategic engagement. The productive failure literature confirms what Koster's theory implies but does not specify: the quality of learning depends not on whether errors occur but on the rate at which errors convert into model improvements.

9.5 What the missing piece is

The missing piece is not learning itself but the **rate** of learning; and not the absolute rate, but the rate relative to the brain's expectation.

Koster identified the right variable: learning. Cook operationalised it: the skill atom feedback loop. But neither specified the function that determines whether the learning process feels rewarding or aversive. That function is the first temporal derivative of model accuracy: how fast the player's predictions are improving, evaluated against how fast the brain expects them to improve.

The next chapter introduces the temporal dimension that transforms this observation into a dynamic theory. Chapter 11 then identifies the precise quantity - converging from three independent theoretical traditions - that governs the experiential landscape of play.

Chapter 10 - The Temporal Dimension

On learning as a process characterised by change over time; skill acquisition as trajectory rather than position; and the empirical evidence that engagement follows predictable temporal patterns.

10.1 Position versus velocity

The core limitation of existing game design theories - every one of the six examined in Chapter 8 - is that they describe the player's position in a design space without describing their **velocity**: how fast and in what direction they are moving.

Csikszentmihalyi's flow channel describes a position: the player is in the zone where challenge approximately equals skill. But two players can occupy the same position with completely different velocities. One is improving rapidly; each encounter teaches something new, and their skill curve is steep. The other is stagnating; their skill curve is flat, and the challenge-skill match is maintained only because neither variable is changing. Both are in the flow channel. Their experiences are qualitatively different, and their futures are different: the first player will remain engaged; the second is one boring session away from quitting.

In physics, position is insufficient to predict the future of a system; you need velocity and acceleration. In game design, skill level is insufficient to predict engagement; you need the rate of skill change and, ideally, the rate of change of that rate. The unified model is, fundamentally, a velocity-based theory of engagement where existing theories are position-based.

10.2 The shape of engagement over time

Player engagement is not uniform across the lifecycle of a game. Empirical data on retention and completion reveals a consistent pattern.

Mobile games lose over 75% of new users within 24 hours. After one week, 85% have stopped playing. Only approximately 4% of mobile gamers remain active at 30 days. Industry benchmarks place Day 1 retention at approximately 40-50%, Day 7 at approximately 20%, and Day 30 at approximately 10%. The retention curve typically flattens around day 20-25, meaning Day 30 closely resembles Day 365; players who survive early attrition become long-term players.

On PC, the pattern is similar in shape if not magnitude. The average game completion rate on Steam is approximately 35%; roughly three out of four purchasers do not finish the main story. Specific examples are instructive: *Tomb Raider* (2013) at 42.4%, *BioShock Infinite* at 38.2%, *The Witcher 3* at 24.6%, *Red Dead Redemption 2* at 23.9%. *Spider-Man: Miles Morales* at 65% is the outlier, attributed to its relatively short runtime of 6-8 hours. The correlation between game length and completion rate is robust: shorter, tighter narratives show significantly higher completion.

These numbers are not merely industry statistics. They are the empirical signature of the learning gradient's temporal dynamics. The steep early drop-off corresponds to the overload phase: players whose error resolution rate cannot keep pace with the game's initial prediction error density quit. The gradual mid-game decline corresponds to the exhaustion phase: players whose learning gradient has flattened lose motivation. The completion-rate correlation with game length reflects the prediction that **content volume and learning depth are not the same thing**; a 60-hour game with 10 hours of unique prediction errors will lose players after 10 hours regardless of how much additional content remains.

10.3 The learning curve as trajectory

Skill acquisition follows a characteristic trajectory, and the shape of that trajectory determines the shape of engagement.

The power law of practice, first documented by Bryan and Harter in their 1899 study of Morse Code telegraph operators and subsequently confirmed across hundreds of studies, describes the overall shape: rapid initial improvement that gradually decelerates toward an asymptote. Initial learning is fast and perceptible; later learning is slow and often imperceptible. This explains why the first hour of a new game is frequently the most exciting and why late-game engagement is always at risk.

But learning curves are not smooth. Bryan and Harter's original study identified distinct **plateaus**: periods when subjects seemed unable to attain further improvement despite continued practice. The mechanism they identified remains relevant: stagnation occurred because learners had mastered lower-order elements (individual letters, words) but had not yet developed higher-order organisational skills (phrases, contextual meaning). The plateau represented the time required for the nervous system to reorganise and integrate disparate elements into a more efficient, automated system. A more recent piecewise model (2015, *PMC*) frames this as a sequence of **strategy shifts**: locally, gradual improvement follows a power law within a specific strategy; globally, progress involves a discrete sequence of strategy shifts, each better in the long term than the ones preceding it.

Each feature of the learning curve has implications for game design:

Plateaus are the most dangerous phase for engagement. The player is practising but not perceptibly improving. The learning gradient appears to be zero. In Bryan and Harter's terms, the player has mastered the current level of organisation but has not yet reorganised into the next level. Physical fitness plateaus typically last 2-3 weeks; complex procedural learning plateaus can extend to months. Effective game design recognises plateaus and introduces

new challenges, perspectives, or system combinations that prompt the reorganisation. *Dark Souls*'s introduction of new enemy types and environments across its mid-game serves this function: each new area forces the player to reorganise their combat model for a new context.

Breakthroughs are the most rewarding moments. The player suddenly "gets it"; a pattern clicks, a strategy crystallises, a skill snaps into place. In the piecewise model, this is a strategy shift: the player abandons a suboptimal approach for a fundamentally better one. The resulting jump in performance produces a spike in the learning gradient that generates intense positive affect. This is the "aha" moment that puzzle games are structured around and that boss-fight victories in Soulslike games deliver; a sudden resolution of accumulated prediction errors that produces Van de Cruys's amplified positive valence (the rate of error reduction far exceeds the brain's expectation).

Regressions are confusing and potentially frustrating. The player was performing well, and now they are performing worse. This often occurs during strategy shifts: the old model is being dismantled before the new one is complete. A player who has been playing *Dota 2* with a fixed hero pool and begins experimenting with new heroes will see their win rate decline temporarily as they build new models. Effective game design manages regressions by providing clear signals that the player is on the right track despite temporarily worse performance, or by creating safe spaces for experimentation (practice modes, low-stakes matches, optional challenges).

10.4 The lifecycle of game engagement

The learning curve framework maps onto a characteristic five-phase engagement lifecycle:

Phase 1: First contact. The player knows nothing. Prediction error density is at maximum. The risk is overload; the opportunity is the exhilarating novelty of a system that is entirely unknown. Design

requirement: controlled revelation, one system at a time.

Phase 2: Rapid learning. Basic systems are understood; the learning gradient is steep. New mechanics arrive frequently, and each encounter teaches something new. This is typically the most enjoyable phase. Design requirement: pacing new introductions to sustain the gradient without tipping into overload.

Phase 3: Deepening. The introduction rate of new mechanics decreases. Engagement depends on combinatorial depth: discovering new interactions between familiar elements. The player develops macro-level strategic frameworks. Design requirement: systems whose interactions produce emergent complexity exceeding the sum of their parts.

Phase 4: Mastery. The player's model is nearly complete. Remaining prediction errors are fine-grained. The learning gradient is shallow. Design requirement: depth sufficient to sustain refinement-level engagement for the player's remaining interest, or an acknowledgement that the game's learning content has been consumed.

Phase 5: Exhaustion and renewal. The gradient reaches zero. Engagement ends unless the game introduces new prediction error sources: procedural generation, human opponents, user-generated content, meta-game evolution, or expansion content.

10.5 Why static theories fail

The five-phase lifecycle explains why static theories cannot account for engagement. A game that satisfies MDA's structural criteria, produces Salen and Zimmerman's meaningful play, deploys Costikyan's uncertainty, satisfies SDT's three needs, and places the player in Csikszentmihalyi's flow channel can still fail to sustain engagement if the learning gradient collapses during Phase 3 or Phase 4.

The player's relationship to the game changes over time as their model improves. A theory that describes the game's structural properties without describing the trajectory of the player's understanding will correctly predict initial engagement (the structure is sound) but fail to predict sustained engagement (the gradient has flattened). This is why some games receive strong early reviews and strong early player counts but suffer steep engagement decline after the first week: the structural criteria were satisfied, but the temporal dynamics were not managed.

The missing dimension is rate: how fast the player's model is improving, and how fast that rate compares to the brain's expectation. The next chapter identifies this variable precisely and shows that three independent theoretical traditions converge on the same quantity as the hedonic signal governing the experiential landscape of play.

Chapter 11 - The Critical Variable: Learning Rate

On the isolation of learning rate as the hidden variable governing player experience; and the convergence between Van de Cruys's affective error dynamics, Schmidhuber's compression progress, and the predictive processing account of play.

11.1 The first derivative of prediction error

The critical insight that transforms "fun is learning" from a descriptive observation into a predictive theory comes from Sander Van de Cruys's 2017 contribution to *Philosophy and Predictive Processing*, "Affective Value in the Predictive Mind."

Van de Cruys proposed that **affective valence tracks the first temporal derivative of prediction error**; the rate at which errors are being reduced or increased over time. Positive valence corresponds to prediction errors that are **decreasing**: the world is becoming more predictable, the model is improving. Negative valence corresponds to prediction errors that are **increasing**: the world is becoming less predictable, the model is failing.

This is not merely a refinement of the reward prediction error framework. It is a shift in what the brain is tracking. The raw prediction error signal says "something unexpected happened." The derivative signal says "am I getting better or worse at predicting what happens?" The first is a snapshot; the second is a trajectory. And it is the trajectory that determines how the experience feels.

The formulation connects to a deeper biological principle. The reward value of water depends on thirst. The reward value of

warmth depends on cold. What matters for affective experience is not the absolute state of the organism but the **direction and speed of change** relative to homeostatic setpoints. Prediction error operates the same way:

- A **large error that is shrinking** feels good (the player is learning, the puzzle is yielding)
 - A **small error that is growing** feels bad (the player is losing ground, strategies are failing)
 - A **stable error** produces neutral affect (the plateau that plagues musicians and gamers alike)
-

11.2 The meta-level: expectations about the rate of learning

Van de Cruys's framework contains a further level that is critical for understanding games: the brain builds predictions not only about external events but about the **rate of error reduction itself**. When the rate of progress matches the brain's expectation, the experience is pleasant but unremarkable. When the rate of progress is **faster than expected**, the resulting positive affect is amplified.

Van de Cruys argues this is the processing signature of humour: a steep, sudden gradient of prediction error leads to a prediction of low rate of error reduction. If errors can in fact be reduced (through restructuring, through seeing the joke), the reduction rate will be much higher than expected, resulting in intensely positive affect. The punchline of a joke, the "aha" moment of a puzzle, and the moment a boss pattern clicks in *Dark Souls* all share this structure: the brain expected to be confused for longer, and the resolution arrived faster than predicted.

This meta-level explains why **difficulty is not the enemy of fun**. A difficult game that generates large prediction errors is not inherently aversive; it is aversive only if those errors resist resolution. A difficult game whose errors yield to study produces the

sharpest possible positive affect, because the brain expected the resolution to take longer than it did. This is the "aha" experience that *Dark Souls* players describe: the boss seemed impossible, and then suddenly the pattern clicked, and the resolution was faster than expected. The magnitude of the positive affect is proportional to the gap between the expected rate of resolution and the actual rate.

11.3 Schmidhuber's compression progress

Jürgen Schmidhuber arrived at an equivalent conclusion from a completely different direction. His compression progress theory (2006-2010; formalised in "Formal Theory of Creativity, Fun, and Intrinsic Motivation," *IEEE Transactions on Autonomous Mental Development*, 2010) proposes a single optimisation criterion: maximise **compression progress**.

Beauty is compressibility: a data stream is beautiful to the extent that it can be compressed into a shorter representation. **Interestingness is the first derivative of beauty**: the rate at which compressibility increases; the steepness of the learning curve. Curiosity is the drive to seek data that promises compression progress. Fun is the intrinsic reward signal generated by that progress. Boredom is the state in which no further compression progress is achievable (the data has been fully compressed).

Schmidhuber's formulation converges precisely with Van de Cruys's: both identify the same mathematical quantity - the rate of model improvement over time - as the hedonic signal. That two independent theoretical traditions, approaching the problem from opposite directions (predictive processing neuroscience and algorithmic information theory), converge on an equivalent formulation is strong evidence that the underlying insight is correct.

11.4 Andersen et al.'s predictive processing account of play

The final convergence comes from Andersen, Kiverstein, Miller, and Roepstorff (2023, *Psychological Review*), who proposed that play seeks "sweet spots" of relative complexity where prediction error is reduced faster than expected. Their analysis accommodates both idle games (continuous micro-uncertainty resolution through accumulation) and Soulslike games (massive uncertainty reduction upon eventual success after extended failure) within the same framework. This is a significant theoretical achievement: previously, idle games and *Dark Souls* seemed to require completely different explanatory models. Under the prediction-error-rate framework, they are instances of the same process operating at different timescales and magnitudes.

Deterding, Andersen, Kiverstein, and Miller (2022, *Frontiers in Psychology*) extended this framework to video games explicitly, finding the model explains "momentary jolts of positive affect whenever uncertainty is reduced faster than expected."

11.5 The hidden variable, specified

The critical variable governing player experience can now be stated precisely:

The learning rate (ΔL) is the first derivative of the player's model accuracy with respect to time, evaluated relative to the brain's expected rate of progress.

When ΔL exceeds expectations: fun, engagement, the "aha" moment. When ΔL matches expectations: sustained engagement, flow. When ΔL falls below expectations: boredom, grinding, disengagement. When ΔL is negative: frustration, helplessness, quitting.

This is the variable that Koster identified but did not formalise. It is the variable that Csikszentmihalyi's flow theory implies but does not measure. It is the variable that the dopamine prediction error system encodes. And it is the variable that game designers manipulate, often intuitively, through every decision about difficulty, feedback, pacing, and content.

The next chapter formalises this into the unified model and shows how it subsumes existing theories as special cases.

PART V - THE UNIFIED MODEL

Chapter 12 - The Unified Model: Fun, Learning, and the Dynamics of Flow

12.1 The problem no theory fully solves

Game design theory has made remarkable progress over the past three decades. We now have formal frameworks for rules and player interaction (Hunicke, LeBlanc, and Zubek's MDA model), for meaningful play as discernible and integrated outcomes (Salen and Zimmerman, 2003), for uncertainty as a prerequisite for engagement (Costikyan, 2013), for intrinsic motivation through autonomy, competence, and relatedness (Ryan and Deci's self-determination theory, applied to games by Rigby and Ryan, 2011), for flow as optimal experience arising from challenge-skill balance (Csikszentmihalyi, 1990; applied to games by Chen, 2007), and for fun as learning (Koster, 2004). Each of these claims is well-supported. Together, they still fail to explain a central phenomenon: why do some games sustain engagement for hundreds or thousands of hours while others, built on similar principles, fail to hold attention beyond a few sessions?

The issue is not that existing theories are wrong. It is that they are **incomplete**. They describe structures, relationships, inputs, motivations, and subjective states, but they do not describe the **dynamic process** that unfolds over time as a player interacts with a game. MDA tells you what a game is made of. Flow theory tells you what the optimal experience feels like. SDT tells you what motivates people in general. None of them tells you what is happening, moment to moment, inside the player's nervous system that makes one session feel transcendent and the next feel tedious, even when

the game has not changed.

This chapter proposes that the missing piece is **learning as a regulated process**, and that fun and flow emerge from how that process is managed. The argument synthesises four converging theoretical frameworks from outside game design; Schultz's reward prediction error, Van de Cruys's affective error dynamics, Schmidhuber's compression progress, and Andersen et al.'s predictive processing account of play; with the practitioner traditions of Koster, Swink, Cook, and Chen. The central claim can be stated simply:

Fun is the subjective experience of reducing uncertainty at an optimal rate. Flow is the cognitive state that emerges when this process is stable over time.

Everything that follows is an attempt to prove this claim is not a metaphor.

12.2 Revisiting Koster: what "fun is learning" actually means

Raph Koster's *A Theory of Fun for Game Design* (2004; revised 2013) remains the most influential single-sentence theory in the field: fun is the emotional response to learning patterns. The brain is "a voracious consumer of patterns," and games are "exceptionally tasty patterns to eat up." When a game's patterns are fully absorbed, the game becomes boring. The player has "grokked" it. There is nothing left to learn.

This insight explains a great deal. It explains why players enjoy mastering enemy behaviours, why puzzle solving is satisfying, why the first hour of a new game is often the most exciting, and why repetition becomes tedious once patterns are exhausted. It correctly predicts that games with deeper pattern spaces (chess, Go, *Dota 2*) sustain engagement longer than games with shallow ones (tic-tac-toe). It correctly identifies the relationship between novelty

and enjoyment. And Koster himself, in his 2014 GDC retrospective, connected the theory to dopamine: "It is a teaching signal to the brain. It gets dumped in you when there are unpredictable situations as well, in order to encourage you to solve them."

But the theory, as stated, leaves several questions unanswered. Why is some learning enjoyable and other learning frustrating? A student struggling with differential equations is learning, but the experience is rarely described as fun. Why does repetition sometimes feel rewarding (a musician practising scales, a basketball player shooting free throws) and sometimes tedious (grinding trivial enemies in an RPG)? Why can players feel deeply engaged even when they are not consciously aware of learning anything, as when navigating a familiar open world or replaying a beloved campaign?

The answer lies in a variable Koster identified but did not formalise: **the rate at which learning occurs**. Not whether the player is learning. Not what the player is learning. How fast, relative to what the brain expects. This is the hidden variable governing the entire experiential landscape of play.

12.3 The three core variables

Every moment of gameplay can be described as an interaction between three variables.

Pattern complexity (C) is the structure the player is attempting to understand. This is not the same as difficulty. A chess endgame with king and rook versus king is not difficult for a grandmaster, but the underlying pattern space of chess is extraordinarily complex. Complexity refers to how much structure must be internalised to predict and control outcomes. In *Dark Souls*, it includes the attack timing of an enemy, the spatial geometry of an arena, the properties of available weapons, and the interaction rules governing stamina, poise, and damage types. In *Dota 2*, it includes 120+ hero abilities, 200+ items, creep mechanics, vision rules, terrain elevation, and the behaviours of nine other human players. In *Portal*, it includes

the non-Euclidean spatial logic of linked surfaces. Pattern complexity determines the ceiling of possible learning.

Player skill (S) is the set of patterns the player has already learned. At low skill, actions are conscious and slow, errors are frequent, and systems feel opaque. This corresponds to what Fitts and Posner (1967) called the **cognitive stage** of motor learning: movements are guided by verbal and declarative processes, attention is fully consumed, and performance is inconsistent. At high skill, actions are automatic, predictions are accurate, and the system feels intuitive. This is the **autonomous stage**: execution occurs without conscious supervision, freeing attention for higher-order strategy. The transition between these stages corresponds to the shift from declarative to procedural memory systems documented by Ullman (2004, *Cognition*) and from cortical to subcortical processing documented by Poldrack et al. (2005, *Journal of Neuroscience*) and Lehéricy et al. (2005, *PNAS*). Player skill determines where the player currently stands in the learning space.

Learning rate (ΔL) is the rate at which the player reduces uncertainty about the system. This is the hidden variable. It is not directly visible to the player, and it is rarely discussed in design documentation, but it determines whether a player feels bored, frustrated, engaged, or deeply absorbed. When ΔL is zero (nothing new is being learned), the experience is boredom. When ΔL is negative (the player is getting worse, or the system is becoming less predictable faster than the player can adapt), the experience is frustration or helplessness. When ΔL is positive but slow, the experience is grinding. When ΔL is positive and matched to the brain's expected rate of progress, the experience is fun. And when that match is sustained over time, the experience is flow.

12.4 The neuroscience of prediction error: applying the framework

Chapter 3 established that the dopamine system encodes reward prediction error (Schultz, Dayan, & Montague, 1997), that uncertainty itself generates a sustained dopamine signal peaking at maximum unpredictability (Fiorillo, Tobler, & Schultz, 2003), that the brain treats information as intrinsically rewarding (Bromberg-Martin & Hikosaka, 2009), and that different dopamine neurons maintain a full probability distribution over possible outcomes rather than a single expected value (Dabney et al., 2020). Chapter 5 established that skill acquisition involves measurable cortical-to-subcortical transfer (Poldrack et al., 2005; Lehericy et al., 2005) and that expert performance activates qualitatively different brain circuits than novice performance (Wan et al., 2011).

These findings converge on a picture of the brain as a prediction-and-correction machine that finds the correction process intrinsically rewarding. The question for game design is: if prediction error is the neurochemical currency of engagement, what determines whether a given pattern of prediction errors produces sustained fun rather than frustration, boredom, or compulsion?

The answer is not in the errors themselves but in their **temporal dynamics**; the rate at which they are generated and resolved. This is the insight that transforms neuroscience into design theory.

12.5 From mechanism to dynamics

Chapter 3 established that uncertainty itself generates a sustained dopamine signal (Fiorillo et al., 2003), that the brain treats information as intrinsically rewarding (Bromberg-Martin & Hikosaka, 2009), and that reducible uncertainty (chess) produces fundamentally different engagement than irreducible uncertainty (slot machines). But reducibility is binary, and engagement is continuous. Two games with equally reducible uncertainty can produce wildly different levels of engagement. The prediction error framework explains why games are rewarding in general; it does not explain what determines whether a specific pattern of prediction

errors produces sustained fun, grinding tedium, or overwhelming frustration.

The resolution requires a shift from the errors themselves to their **temporal dynamics**: not what the prediction error is at any given moment, but how fast it is changing.

12.6 The critical insight applied: what the derivative means for games

Chapter 11 established the critical variable: affective valence tracks the first temporal derivative of prediction error (Van de Cruys, 2017), and interestingness is the first derivative of compressibility (Schmidhuber, 2010). Both frameworks, arriving from independent theoretical traditions, identify the same quantity as the hedonic signal: **the rate of model improvement over time**.

What does this mean concretely for game design?

It means that a large prediction error that is **shrinking** feels good. The player is learning; the puzzle is yielding; the boss pattern is crystallising. It means that a small prediction error that is **growing** feels bad. The player is losing ground; previously reliable strategies are failing. And it means that a stable error; one that neither grows nor shrinks; produces the plateau that plagues both musicians practising scales and gamers grinding experience points.

The framework also explains the **meta-level of emotional intensity** that characterises the best moments in games. The brain builds predictions not only about external events but about the rate of error reduction itself. When the rate of progress matches expectations, the experience is pleasant but unremarkable. When progress is **faster than expected**, the resulting affect is amplified. This is the processing signature of the "aha" moment: the brain expected to be confused for longer, and the resolution arrived faster than predicted. The punchline of a joke, the moment a *Portal* puzzle clicks, and the moment a *Dark Souls* boss pattern resolves all share

this structure. The magnitude of the positive affect is proportional to the gap between the expected rate of resolution and the actual rate.

This is why difficulty is not the enemy of fun. A difficult game that generates large prediction errors is not inherently aversive; it is aversive only if those errors resist resolution. A difficult game whose errors yield to study produces the **sharpest possible positive affect**, because the brain expected the resolution to take longer than it did.

12.7 The core claim, restated with teeth

We can now state the unified model with full neuroscientific grounding:

Fun is the subjective experience of reducing prediction error at a rate that matches or exceeds the brain's expected rate of progress. Flow is the cognitive state that emerges when this rate is sustained and stable over time.

The model has three components. First, games generate **prediction errors** through uncertainty, variability, and challenge. Every action a player takes produces an outcome that either confirms or violates their internal model. Second, players act to **reduce** those errors through practice, experimentation, and pattern extraction. This reduction is intrinsically rewarding because the dopamine system treats uncertainty resolution as valuable (Bromberg-Martin & Hikosaka, 2009). Third, the **rate** of this reduction determines affective valence (Van de Cruys, 2017; Schmidhuber, 2010) and the sustainability of engagement.

This differs from existing theories in a specific way. Csikszentmihalyi's flow model identifies challenge-skill balance as the key variable. But challenge-skill balance is a static snapshot; it tells you about the current moment, not the trajectory. Two players can have identical challenge-skill ratios and completely different experiences if one is improving and the other is stagnating. The

learning rate model adds the temporal dimension that flow theory lacks: it is not the balance itself that produces engagement but the **dynamics of the balance as it evolves**.

12.8 Prediction error as the engine of engagement: the loop

Games produce engagement through a continuous loop that maps directly onto the neural machinery described above:

1. The player forms an **expectation** about what will happen next (an enemy will attack from the left, a jump will clear the gap, a card combination will score well).
2. The game produces an **outcome** (the enemy flanks from the right, the gap was wider than expected, the combination triggers an unexpected synergy).
3. A **prediction error** occurs; the difference between expectation and reality.
4. The player **updates their internal model** to account for the new information (the enemy has a flanking behaviour, the gap requires a running start, that card combination is powerful).
5. The cycle repeats, with the updated model generating better predictions next time.

This loop is not unique to games. It operates in every domain of learning. What makes games unique is that they are **engineered to optimise this loop**. Unlike natural environments, where prediction errors arrive unpredictably and often in domains the learner would prefer to avoid (financial loss, social rejection, physical injury), games curate prediction errors to be frequent, manageable, domain-specific, and free from lasting consequence. Games are safe laboratories for a process the brain finds intrinsically rewarding.

12.9 The four states of player experience

From the model, four distinct experiential states emerge, corresponding to different learning rate regimes.

Boredom occurs when ΔL approaches zero. The player has fully learned the system. No new prediction errors are generated. The dopamine system has nothing to signal because outcomes are fully predicted. Examples: replaying an early Mario level after mastery, farming trivial enemies in an RPG.

Overload occurs when prediction errors arrive faster than the player can resolve them. The system is too complex or too opaque. Pattern extraction fails because the noise-to-signal ratio is too high. Examples: a new player entering a high-level *Dota* match, poorly tutorialised mechanics with no legible feedback.

Grinding occurs when ΔL is positive but far below the brain's expected rate. The player is active but not meaningfully improving. Repetition occurs without insight. Progress is measured in numbers (XP, currency, levels) rather than in understanding. Examples: repetitive resource farming, artificial progression systems that gate content behind time investment rather than skill development.

Flow occurs when ΔL is positive and matched to the brain's expected rate of progress. Prediction errors are frequent enough to demand processing but not so frequent as to overwhelm it. Feedback is immediate and legible. The player is continuously improving. The LC-NE system sustains phasic mode (Aston-Jones & Cohen, 2005), with task-evoked norepinephrine bursts facilitating focused attention while tonic levels remain moderate. The continuous model improvement generates the positive-valence signal that Van de Cruys's framework predicts. The absence of metacognitive interference (Dietrich's prefrontal suppression) prevents self-consciousness from interrupting the loop. Examples: learning boss patterns in *Dark Souls*, refining movement in *Celeste*, improving aim and positioning in *Halo*.

12.10 Rethinking difficulty

Traditional game design frames engagement as a balance between **challenge and skill**. This is imprecise in a way that matters.

Consider two situations with identical difficulty. In the first, a player faces a boss in *Dark Souls* whose attack patterns are consistent, telegraphed, and varied. Each death reveals new information: the overhead slam has a two-second wind-up - the sweep always follows the thrust - the charge attack tracks for 45 degrees and can be dodge-rolled to the right. In the second, a player faces a boss whose attacks are randomly selected from a uniform distribution with minimal wind-up and inconsistent timing. Both bosses kill the player at the same rate. Both produce the same measurable difficulty. But the first is widely beloved and the second would be universally hated.

The difference is **learnability**. The first boss generates prediction errors that systematically reduce with exposure. Each death teaches something specific, and the player's model of the boss's behaviour becomes measurably more accurate after each attempt. The second boss generates prediction errors that do not reduce because the underlying process is stochastic. No model can predict a random number generator. The errors are irresolvable.

What matters is not the magnitude of prediction errors (difficulty) but the rate at which they can be reduced (learnability). A game can be extremely difficult and extremely fun if its difficulty arises from patterns that yield to study. A game can be moderately difficult and extremely frustrating if its difficulty arises from randomness that resists modelling. Hidetaka Miyazaki, director of the *Dark Souls* series, has articulated exactly this principle: "It's not a matter of simply cranking up the difficulty; it's doing so fairly."

Wilson, Shenhav, Straccia, and Cohen (2019, *Nature Communications*) formalised this with the **85% rule for optimal learning**: for gradient descent-based learning systems (including biological neural networks), the optimal error rate is approximately **15.87%**; equivalently, an accuracy rate of approximately 85%. Training at this optimal difficulty produced exponentially faster

learning compared to training at suboptimal difficulty. The paper explicitly connects this to flow theory, Vygotsky's zone of proximal development, and the Goldilocks effect. The 85% figure provides a quantitative target for something game designers have always understood intuitively: the best games produce a success rate high enough that the player feels competent but low enough that genuine learning is occurring.

Robert Bjork's concept of **desirable difficulties** (1994; Bjork & Bjork, 2011) adds a further nuance. Certain conditions that slow apparent learning - spacing practice, interleaving different skills, varying conditions - actually accelerate long-term retention and transfer. A desirable difficulty in game design terms is one where prediction errors are slightly larger than comfortable but remain resolvable with effort. This describes exactly the experience of a well-designed difficulty curve: each new section feels slightly too hard at first, but the player's model catches up within a few attempts.

12.11 The learning gradient

We define the **learning gradient** as the rate at which a player improves their internal model of the game system. A well-designed game maintains a stable, positive learning gradient throughout its duration. Not flat (boredom). Not chaotic (overload). Not stagnant (grinding). But steadily advancing: a consistent reduction of prediction error across time, producing the sustained positive valence that Van de Cruys's framework predicts and that players experience as engagement.

The learning gradient is shaped by two forces. The first is the **rate at which the game introduces new prediction errors**. This is controlled by difficulty curves, new mechanic introductions, enemy variety, level design progression, and narrative revelations. The second is the **rate at which the player can resolve existing prediction errors**. This depends on the legibility of game feedback,

the consistency of game rules, the player's prior experience, and the availability of practice opportunities.

When the introduction rate exceeds the resolution rate, the gradient tips toward overload. When the resolution rate exceeds the introduction rate, the gradient tips toward boredom. The designer's task is to keep these two rates in approximate equilibrium; or more precisely, to ensure that the resolution rate stays slightly ahead of the introduction rate, producing a continuous sense of progress.

Learning in games occurs at multiple levels simultaneously:

- **Micro-learning (seconds):** Input timing, movement precision, reaction speed. Landing a jump in *Mario*. Hitting a headshot in *Halo*.
- **Meso-learning (minutes):** Encounter strategies, puzzle logic, tactical decision-making. Solving a room in *Zelda*. Clearing an encounter in *Dark Souls*.
- **Macro-learning (hours):** System mastery, build optimisation, strategic frameworks. Understanding the *Dota* meta. Mastering the *Celeste* movement vocabulary.

A strong game maintains a learning gradient at **all three levels simultaneously**. This is why Griesemer's nested loop architecture in *Halo* was so effective: by distributing prediction errors across three temporal scales (3-second motor loops, 30-second encounters, 3-minute combat spaces), the system had multiple redundant sources of learning gradient. If the motor loop was momentarily flat, the encounter loop might still be generating fresh tactical errors. Three nested learning gradients are more robust than one.

12.12 Play as epistemic niche construction

A crucial further insight: play is a form of **epistemic niche construction**. Organisms do not passively encounter optimal challenges; they create them. Children stack blocks to knock them down. They pretend to be monsters. They modify rules when games

get too easy or too hard. This maps directly to how players engage with game systems: seeking self-imposed challenges, ignoring dominant strategies, creating house rules, speedrunning, and modding. The prediction error landscape of a game is not fixed by the designer; it is co-constructed by the player's choices about how to engage.

Deterding, Andersen, Kiverstein, and Miller (2022, *Frontiers in Psychology*) extended this framework to video games explicitly, finding the model explains "momentary jolts of positive affect whenever uncertainty is reduced faster than expected." Their analysis accommodates both idle games (continuous micro-uncertainty resolution through accumulation) and Soulslike games (massive uncertainty reduction upon eventual success after extended failure sequences) within the same framework. This is a significant theoretical achievement: previously, idle games and *Dark Souls* seemed to require completely different explanatory models. Under the prediction-error-rate framework, they are instances of the same process operating at different timescales and magnitudes.

12.13 The roguelike structure as a compressed learning cycle

Roguelikes provide a natural experiment in learning gradient management. The genre's defining structure - short runs, permadeath, procedural generation, and iterative improvement - creates what may be the purest prediction error feedback loop in game design.

Each run follows a trajectory: encounter (high surprise, many prediction errors) → death (error signal, model fails) → reflection (update model with new information) → re-attempt (test updated model against novel procedural variation) → gradual mastery (prediction errors decrease across runs). The cycle is compressed into 15-45 minutes per run, compared to the hours-long arcs of conventional game campaigns. This compression means the learning

gradient is steep and the reinforcement cycle is tight.

Procedural generation is essential because it prevents rote memorisation from substituting for genuine model-building. If levels were identical each run, the player would eventually reduce all prediction errors to zero through memorisation, and the learning gradient would collapse. By varying the specific instantiation while preserving the underlying rules, roguelikes ensure that each run generates fresh prediction errors that test the player's general model rather than their specific recall.

Hades innovates further by integrating narrative scaffolding into the death cycle. Each death returns the player to the House of Hades, where new dialogue, character development, and story progression await. Greg Kasavin, Supergiant's creative director, described the design philosophy: "It was an explicit goal of our early development, to take the pain out of dying and having to restart." In prediction-error terms, *Hades* converts what would be a purely negative event (negative prediction error from death) into a mixed-valence event (negative combat error plus positive narrative surprise). The learning gradient is sustained not only through mechanical improvement but through a parallel narrative learning gradient; the player is simultaneously building a model of the combat system and a model of the story, and death advances the latter even as it resets the former.

12.14 Flow as a stable system

We can now describe flow not merely as a subjective feeling but as a **stable configuration of the player-game system**.

In this configuration, attention is fully engaged because prediction errors are frequent enough to demand processing but not so frequent as to overwhelm it. The LC-NE system sustains phasic mode (Aston-Jones & Cohen, 2005), with task-evoked norepinephrine bursts facilitating focused attention while tonic levels remain moderate. Feedback is immediate because the game's

output follows the player's input without perceptible delay. The temporal contiguity between action and outcome is essential for the dopamine RPE signal to assign credit correctly; delayed feedback impairs the brain's ability to identify which prediction was wrong. Actions map clearly to outcomes because the game's rules are consistent and legible. This does not mean simple; *Dota 2*'s rules are immensely complex. It means that outcomes follow from rules in a way the player can, in principle, learn to predict. Learning is continuous because new prediction errors are introduced at approximately the same rate as old ones are resolved. The learning gradient remains positive and stable.

This configuration is self-reinforcing but fragile. The sustained phasic LC-NE mode suppresses exploratory attention-shifting, keeping the player locked onto the task. The continuous model improvement generates the positive-valence signal that Van de Cruys's framework predicts. The absence of metacognitive interference (Dietrich's prefrontal suppression) prevents self-consciousness from interrupting the loop. But any disruption; a spike in difficulty, a confusing design decision, a broken mechanic, an unjust death, a loading screen; can knock the system out of its stable state, and re-establishing it requires recalibrating the learning rate from scratch.

12.15 Implications for design

This model changes the role of the designer. You are not designing mechanics, levels, rewards, or narratives in isolation. You are designing **a system that regulates the player's rate of learning**.

Every design decision should be evaluated against this criterion. Does this increase **meaningful prediction errors**? A new enemy type generates fresh errors; a reskinned version of an existing enemy does not. Does it make **feedback clearer**? A legible damage indicator helps the player identify which prediction failed; a confusing death screen obscures the connection between action and

outcome. Does it support **adaptation**? A difficulty system that lets the player self-select into their optimal challenge zone (like *Halo*'s four tiers or *Celeste*'s Assist Mode) serves different learning rates simultaneously. Does it sustain engagement **over time**? A game that front-loads all its prediction errors in the first hour and repeats them for the remaining ten will produce a declining learning gradient regardless of how polished the mechanics are.

The model also identifies a common design failure: confusing **content volume** with **learning depth**. A game with 100 hours of content but only 10 hours of unique prediction errors will sustain engagement for 10 hours. The remaining 90 hours will feel like grinding, because the player's model has been fully built and the additional content generates only trivially small errors. Conversely, a game with 10 hours of content but 10 hours of unique prediction errors will sustain engagement for its entire duration. Content volume is a proxy for learning depth, but the relationship is not linear.

12.16 Synthesis: unifying the theories

We can now place existing game design theories within the unified model as special cases.

Koster correctly identified that fun is learning. The unified model specifies what makes learning fun: the rate of error reduction must be positive and matched to the brain's expected rate of progress. Learning that is too slow (grinding) or too fast (trivial) fails to produce the rate-of-change signal that Van de Cruys's framework identifies as the hedonic variable.

Csikszentmihalyi correctly identified that flow requires challenge-skill balance. The unified model adds the temporal dimension: it is not the balance at any given moment that matters but the **dynamics** of the balance as it evolves. A stable balance with zero learning is not flow; it is stasis. Flow requires the balance to be continuously shifting in a direction that sustains prediction error

generation and resolution.

Schultz and the neuroscience of reward correctly identified that dopamine encodes prediction error. The unified model connects this to game design by specifying that the rate and resolvability of prediction errors, not their raw magnitude, determines whether the experience is rewarding.

Van de Cruys and Schmidhuber correctly identified that valence tracks the first derivative of prediction error. The unified model applies this to games: the subjective quality of a gaming experience is determined by the rate at which the player is reducing uncertainty about the game system, relative to their expected rate of progress.

Andersen et al. correctly identified that play seeks sweet spots of relative complexity. The unified model operationalises this for design: the designer's task is to maintain the player in the zone where prediction errors are being generated and resolved at the rate that maximises compression progress.

The missing link that connects all of these theories is the learning rate itself; the first derivative of the player's model accuracy over time. This is the hidden variable that each theory touches on without fully formalising, and that the unified model places at the centre of the analysis.

12.17 Boundary conditions: where the model is weakest

A model that claims to explain everything explains nothing. The learning gradient framework claims to identify the **primary** variable governing moment-to-moment gameplay engagement. It does not claim to be the only variable that matters, and intellectual honesty requires identifying the cases where its explanatory power is weakest.

Comfort-food replays. Players replay *Final Fantasy VII*, *Chrono Trigger*, and *Stardew Valley* with genuine enjoyment despite having fully internalised the game's patterns. The learning gradient is at zero; no new prediction errors are generated. The model predicts this should produce boredom, yet the experience is pleasurable.

The resolution is that replay engagement is driven by systems the model does not claim to subsume. Aesthetic pleasure (music, visual beauty, the satisfaction of a well-crafted world), emotional resonance (attachment to characters encountered during a formative period), and the hedonic value of motor automaticity itself (the satisfying feel of executing well-learned skills without cognitive effort) are genuine sources of positive affect that operate through different neural circuitry than the prediction error system. The model covers the primary driver of **first-play** engagement; the learning gradient explains why the player was absorbed the first time through. Replay engagement involves additional affective systems; opioidergic hedonic responses, nostalgia-mediated memory reconsolidation, aesthetic appreciation circuits; that are outside the model's scope. The model draws its boundary here: it explains why *Final Fantasy VII* was captivating the first time, not why it feels like home the fifteenth time.

Social belonging. MMO players log in for guild chat, raid nights with friends, and community events. Their engagement is sustained by relatedness (SDT's third need) rather than by a learning gradient. A player whose mechanical skill has plateaued and whose strategic model is complete may continue playing for years because the social bonds make the game their primary social environment.

The model acknowledges this directly: social belonging is a real motivator that operates through oxytocin, mentalising networks (Gallagher et al., 2002; Rilling et al., 2002), and the human need for connection. The learning gradient explains why the game's **mechanics** sustain engagement; it does not claim that mechanics are the only reason people play. Social games are often sustained by social bonds long after their mechanical learning gradients have

collapsed, and this is not a failure of the model but a delineation of its domain. The model's claim is about the game-as-system; the player's relationship to the game's rules, patterns, and challenges. Social motivations operate at a different level of analysis.

Aesthetic experience. Walking simulators (*Dear Esther*, *Firewatch*), meditative games (*Flower*, *Journey*), and ambient experiences (*Proteus*) sustain engagement with minimal challenge and minimal mechanical learning. The model predicts these should produce boredom, yet they produce absorbed, contemplative engagement.

The resolution requires broadening "prediction error" from the narrow sense (mechanical challenge) to the broad sense that predictive processing theory intends. In these games, the player IS building and refining a model, but the model is of the environment's aesthetic and narrative structure rather than its mechanical rules. The player explores *Firewatch*'s wilderness and builds a model of the landscape, the story, the relationship between the two characters. Each new vista, each dialogue exchange, each environmental detail generates a prediction error in the player's narrative-aesthetic model. The learning gradient is present; it is simply operating in a domain (emotional understanding, spatial appreciation, narrative comprehension) that looks different from the mechanical domain the model was primarily designed to address. The model accommodates aesthetic games if "prediction error" is understood at the level of generality that predictive processing theory operates at: any discrepancy between the brain's model and incoming sensory evidence, in any domain.

Meditative and zen states. Some players use games as tools for relaxation and mental decompression. They replay familiar levels, engage in repetitive farming, or play simple mobile games not for learning but for the calming effect of structured, low-demand activity. This is a case where the game functions as a different kind of tool than the model addresses; more like a fidget spinner than a learning environment. The model does not claim that every

interaction with a game is engagement in the technical sense the model defines; some interactions are relaxation, socialisation, or habit, and these are outside the model's primary explanatory target.

The honest boundary. The model's claim is specific: the learning gradient is the primary variable governing the quality of **gameplay engagement**; the absorbed, attentive, improving state that characterises the best moments of play. It is not the only reason people interact with games. Social connection, aesthetic appreciation, nostalgia, relaxation, habit, and community belonging are all real motivators that can sustain game-related behaviour independently of the learning gradient. The model explains why some games produce deeper and more sustained gameplay engagement than others; it does not explain every reason a person might choose to spend time with a game.

This boundary is a strength, not a weakness. A model that tried to explain social belonging AND mechanical engagement AND aesthetic appreciation AND nostalgia AND relaxation within a single framework would be too diffuse to make testable predictions. By specifying its domain precisely; the moment-to-moment quality of gameplay engagement as a function of the learning gradient; the model makes specific, falsifiable predictions about which design decisions will improve or degrade that engagement. That specificity is what makes it useful.

12.18 Final statement

We conclude:

Games are systems that generate and regulate prediction error. Players engage by reducing that error through action. When this process occurs at a rate that matches or exceeds the brain's expected rate of progress, it is experienced as fun. When that rate stabilises over time across multiple timescales, it becomes flow. When it collapses, engagement ends.

Fun is not a property of the game. It is not a property of the player. It is a property of the **dynamic relationship** between the game's pattern generation and the player's pattern absorption; a relationship that must be actively maintained through design decisions at every scale, from millisecond input responsiveness to hundred-hour content pacing. The designer who understands this relationship; who sees their work not as creating content but as engineering a learning gradient; has the most powerful lens available for predicting, diagnosing, and improving player engagement.

This is the foundation for everything that follows.

PART VI - GAME DESIGN AS CONTROL OF LEARNING

Chapter 13 - The Learning Gradient: Designing the Rate of Learning

13.1 From theory to control

In the previous chapter, we established the core idea:

Fun is the subjective experience of reducing uncertainty at an optimal rate.

This raises an immediate question: **how does a designer actually control that rate?**

If games are systems that regulate learning, then design is not about creating isolated mechanics. It is about shaping the **trajectory of understanding over time**. This trajectory is what we call the **learning gradient**.

13.2 What is the learning gradient?

The learning gradient is:

The rate at which a player improves their internal model of the game.

It is not difficulty, not complexity, and not progression speed, but rather the **speed and quality of insight**.

A steep gradient means rapid improvement and frequent "aha" moments. A shallow gradient means slow or stalled understanding. An unstable gradient means oscillation between confusion and triviality.

13.3 The shape of good learning

A well-designed learning gradient has three properties.

First, it is continuous. The player is always learning something. Even small refinements matter: tighter timing, better positioning, improved prediction. There should be no extended periods where the player's internal model is static. This does not mean every moment must introduce new mechanics; refinement of existing skills counts as learning. A *Celeste* player who shaves 0.3 seconds off a room clear by optimising their dash arc is learning, even though no new mechanic was introduced.

Second, it is legible. The player can understand why outcomes occur. Without legibility, errors do not produce learning; repetition becomes noise. This is the principle of **feedback clarity**: the player must be able to trace the causal chain from their action to the outcome. When a *Dark Souls* player dies to a boss, the death must communicate which attack killed them, when they should have dodged, and what pattern they failed to read. If the death communicates only "you are dead," no model update occurs and the gradient stalls.

Third, it is layered. New knowledge builds on prior knowledge. Good systems recombine existing mechanics, deepen understanding, and avoid replacing knowledge entirely. When *Portal* introduces fling mechanics, it does not replace the player's existing understanding of portal placement; it extends it. The new skill layer integrates with and builds upon the old one, producing a sense of growing competence rather than perpetual starting over.

13.4 The units of learning

Learning in games occurs at multiple levels simultaneously, and a well-designed game maintains a gradient at all of them:

- **Micro-learning (seconds):** Input timing, movement precision, reaction speed. Landing a jump in *Mario*. Hitting a headshot in *Halo*. Parrying an attack in *Sekiro*. These are processed through motor cortex and cerebellum, with automaticity developing through the cortico-striatal-cerebellar loop.
- **Meso-learning (minutes):** Encounter strategies, puzzle logic, tactical decision-making. Solving a shrine in *Zelda*. Clearing a room in *Doom Eternal*. Choosing which items to buy in a *Dota 2* mid-game transition. These engage prefrontal working memory and strategic planning circuits.
- **Macro-learning (hours):** System mastery, build optimisation, strategic frameworks. Understanding the *Dota* meta. Mastering the *Elden Ring* weapon upgrade system. Developing a coherent strategy in *Civilisation*. These involve the construction of abstract mental models that organise lower-level skills into coherent frameworks.

A strong game maintains a learning gradient at **all three levels simultaneously**. The nested structure is what prevents the gradient from flattening; it provides multiple simultaneous channels of uncertainty reduction, each operating at a different timescale and a different level of cognitive abstraction.

13.5 Designing for insight

The core unit of the learning gradient is **insight**. An insight occurs when a pattern becomes visible, a prediction becomes reliable, or a system becomes intuitive. Design must maximise:

Insight frequency × insight clarity

Every design element should be evaluated against this product. A new mechanic that generates frequent but ambiguous insights (the player keeps discovering things but cannot tell what caused them) is poorly designed. A mechanic that generates rare but crystal-clear insights (the player rarely learns something new, but each learning

moment is unmistakable) may be well-designed for a particular pacing target. The ideal is both: frequent insights that are immediately understandable.

13.6 The role of feedback

Learning depends on feedback, and feedback must satisfy three criteria.

Immediate. Delay destroys the connection between action and outcome. The dopamine RPE signal requires temporal contiguity to assign credit correctly. Steve Swink's *Game Feel* (2008) established that real-time control exists within the constraints of human perceptual timing: below approximately 240ms response loops, input feels responsive. Above that threshold, the connection between action and outcome degrades. Every millisecond of input lag is a tax on the learning gradient.

Precise. The player must know exactly what happened. A health bar that decreases tells the player they took damage. A health bar that flashes red at the point of impact and shows a directional damage indicator tells the player they took damage from a specific source at a specific moment. The second provides richer error signals and supports faster model updating.

Interpretable. The player must understand why it happened. This is the most commonly violated criterion. Many games provide immediate, precise feedback about outcomes without making the causal chain legible. A player who dies to an off-screen enemy received immediate feedback (death) with zero interpretability (no idea what killed them or how to prevent it). The prediction error is generated but cannot be resolved, and the learning gradient stalls.

13.7 Hidden assistance and gradient smoothing

Many great games secretly assist the player without their knowledge. "Coyote time" in platformers gives the player a brief grace period after leaving a ledge during which they can still jump. Aim assist in console shooters subtly guides the reticle toward targets. Input buffering in action games queues the next input during an animation so it executes at the earliest possible frame. These do not reduce difficulty. They **smooth the learning gradient** by preventing meaningless failure; deaths that teach the player nothing because they resulted from a frame-level timing error rather than a strategic mistake. The distinction is between failures that generate informative prediction errors (the player positioned badly, chose the wrong weapon, misread the enemy pattern) and failures that generate uninformative noise (the player pressed the button 30ms too late due to input lag). Smoothing eliminates the latter while preserving the former.

A 2024 CHI study (n=1,699) on "juice" (redundant, amplified sensory feedback) found that juicy feedback's effect on enjoyment was almost fully mediated by **competence and curiosity**, not by deliberative evaluation. But overloading amplified feedback "interferes with competence and effectance by occluding action-feedback links"; when juice overwhelms perceptual processing, it breaks the System 1 feedback loop. The lesson is that feedback amplification supports the learning gradient only when it increases signal clarity, not when it adds noise.

13.8 Avoiding gradient collapse

Three common design failures map directly onto gradient pathologies.

Plateau: no new patterns are introduced. The player's model is complete for the current level of engagement, but no new structure is offered. Result: boredom. The fix is not more content but more **structure**; new combinations of existing mechanics that generate novel prediction errors without requiring new systems.

Spike: sudden increase in complexity. The game introduces multiple new mechanics, enemy types, or systems simultaneously. The player's error resolution rate cannot keep pace with the error introduction rate. Result: overload and frustration. The fix is pacing; introducing one new element at a time and allowing consolidation before the next.

Noise: unclear or inconsistent feedback. The game's rules are inconsistent, or the connection between action and outcome is obscured. The player cannot extract patterns because the signal is buried in noise. Result: frustration and helplessness. The fix is not reducing difficulty but increasing **legibility**; making the rules clearer, the feedback more precise, the causal chains more visible.

13.9 Player-controlled gradient

The best games allow players to regulate their own learning gradient through mechanisms including optional challenges, exploration-based progression, build diversity, and nonlinear level selection.

Breath of the Wild is the paradigm case. Players choose where to go, what to fight, and when to engage. This allows them to maintain their own optimal gradient, avoid overload, and seek challenge when ready. The player who finds Hyrule Castle too difficult can leave and explore elsewhere, building skills and equipment that reduce the castle's prediction error density to a manageable level. The player who finds the early game too easy can head directly to difficult areas and self-select into a steeper gradient.

This creates agency (the player feels in control of their experience), stability (overload is self-correcting because the player can retreat), and personalisation (different players with different skill levels find different optimal paths through the same content).

13.10 The gradient over time

The learning gradient should evolve across the arc of the game:

- **Early game:** steep gradient, frequent insights, rapid onboarding. The player is learning basic systems and building foundational models. New mechanics should arrive quickly but one at a time.
 - **Mid game:** layered complexity, recombination, deeper systems. Existing mechanics interact in novel ways. The player's model grows more sophisticated. New prediction errors arise from combinations rather than introductions.
 - **Late game:** refinement, mastery, emergent play. The player's model is nearly complete for the game's core systems. Remaining prediction errors are fine-grained; precision timing, optimal routing, creative problem-solving. Engagement is sustained through the depth of the existing system rather than the breadth of new additions.
-

13.11 The end of learning

All games eventually face the exhaustion of the learning gradient. When patterns are fully internalised and no new structure remains, the game ceases to generate meaningful prediction errors and engagement declines. This is not a design flaw; it is an inevitability. Every finite system has a finite learning gradient.

Solutions include introducing new systems (DLC, expansion content), enabling emergence (combinatorial systems whose interaction space exceeds any player's capacity to fully explore), relying on social complexity (competition against other human beings, whose behaviour generates an inexhaustible prediction error landscape), and supporting player-generated content (modding, map editors, custom challenges).

13.12 The designer's task

The designer's role is now clear:

Design the slope, not just the system.

This means controlling complexity, structuring feedback, pacing new information, and enabling adaptation. A great game is not one that is fun. It is one that **continuously teaches at the right rate.**

Chapter 14 - Core Game Systems as Learning Structures

On how combat, movement, puzzles, and strategy each generate and resolve uncertainty through different mechanisms; and why genre distinctions map onto distinct prediction error profiles.

14.1 Systems, not genres

Game genres are marketing categories. From the perspective of the unified model, what matters is not whether a game is labelled an "action RPG" or a "puzzle platformer" but what kind of prediction errors its core systems generate, at what timescale, and through what neural circuitry. A game's core system determines the shape of its learning gradient: the type of uncertainty the player must resolve, the feedback that supports resolution, and the cognitive resources the resolution demands.

This chapter analyses five core system types - combat, movement, puzzles, strategy, and narrative - as distinct structures for generating and resolving prediction errors. Each maps onto different neural circuitry and different regions of the System 1/System 2 spectrum established in Part II.

14.2 Combat systems: prediction error through opponent modelling

Combat systems generate prediction errors through enemy behaviour, timing, and spatial positioning. The learning gradient is motor-heavy at the micro level (can I execute the dodge in time?),

tactical at the meso level (which enemies should I prioritise?), and strategic at the macro level (what build or loadout optimises my approach?).

The neural profile of combat engagement is dominated by the **cortical-to-subcortical transfer** described in Chapter 5. A novice player fighting a *Dark Souls* boss operates in the cognitive stage: dlPFC maintains attack patterns in working memory, ACC monitors for timing errors, and every action requires conscious planning. After twenty attempts, processing has migrated toward basal ganglia pattern recognition and cerebellar motor timing. The player has not merely "learned the boss"; their brain has physically reorganised which circuits handle the task (Poldrack et al., 2005; Lehericy et al., 2005).

Combat systems vary along two dimensions that determine their learning gradient profile:

Readability determines how efficiently prediction errors convert into model updates. *Halo's* Covenant enemies display their internal state through animation and vocalisation; every interaction is an informative prediction error. The Flood's mindless rush provides no readable state changes; prediction errors cannot be resolved. Miyazaki's *Dark Souls* bosses telegraph attacks with distinct wind-up animations, making difficulty arise from learnable patterns rather than irreducible randomness.

Combinatorial depth determines how long the learning gradient persists. A combat system with three enemy types in fixed configurations exhausts its prediction errors quickly. A system like *Halo's*, where three enemy types combine into dozens of compositionally distinct encounters, sustains the gradient far longer because the combinatorial space exceeds any individual's capacity to fully model it. *Doom Eternal* extends this further by requiring the player to match specific weapons to specific enemy weak points, creating a moment-to-moment weapon-selection decision that generates tactical prediction errors layered on top of the motor execution errors.

The genre-level mapping from the dual-process research confirms this: fighting games are the most System 1-dominant at expert level (sub-second decisions, pattern-based reads, motor automaticity), while action RPGs layer System 1 combat under System 2 strategic planning (Bediou et al., 2018; Konsolaki et al., 2024).

14.3 Movement systems: prediction error through physics

Movement systems generate prediction errors through spatial reasoning, momentum, timing, and physics. The learning gradient is dominated by the cortical-to-subcortical transfer of motor schemas; the player is building an internal model of how the game's physics work and automating that model through repetition.

Super Mario's jump is the paradigm case. The physics are not realistic; they include mid-air control, variable jump height based on button hold duration, and momentum that can be redirected after launch. Each of these properties generates prediction errors for a new player whose model is calibrated to real-world physics. The learning gradient is steep because the feedback is immediate (you either make the platform or you do not), precise (the gap between your landing point and the target is visible), and interpretable (the causal chain from button press to outcome is unambiguous).

Celeste extends the movement vocabulary to include dashes, wall-climbs, and momentum chains, each of which generates its own prediction error profile. Every room in *Celeste* teaches a specific movement concept; the level design is a structured curriculum in spatial physics. Maddy Thorson's design ensures that failure is immediate, respawn is instantaneous, and the learning loop (attempt → fail → adjust → retry) cycles in seconds rather than minutes. A 2024 study in the *International Journal of Human-Computer Studies* (n=10) found that players who enjoy challenging games like *Celeste* persist after failure because they make meaning from it, see purpose in it, and draw persistence from

the design itself; the game's architecture converts failure from punishment into information.

Steve Swink's *Game Feel* (2008) identified the perceptual foundation: game feel operates at timescales below ~240ms, in System 1 territory. Empirical research on input latency confirms the importance: Long and Gutwin (2018, CHI PLAY) developed predictive performance models showing latency thresholds as low as 50ms cause problems depending on game speed, and Liu et al. (2021, CHI) found even 25ms of additional latency negatively affected competitive FPS accuracy.

14.4 Puzzle systems: prediction error through logical structure

Puzzle systems generate prediction errors through logical structure, spatial reasoning, and rule discovery. Unlike combat and movement systems, where the learning gradient is continuous (each attempt refines motor execution by a small amount), puzzle systems produce **discrete learning events**; the "aha" moment when the solution becomes visible.

This discontinuous profile maps onto a different neural signature. Puzzle-solving is predominantly System 2: working memory maintains candidate solutions, the ACC monitors for contradictions, and the dlPFC evaluates logical relationships. The shift to System 1 is minimal because puzzles are typically solved once; there is no repetition to drive automaticity. Instead, the reward comes from the single moment of restructuring when the solution appears.

Van de Cruys's framework explains why this moment feels so good. Before the insight, prediction errors are large and apparently irresolvable; the puzzle seems impossible, and the brain predicts a low rate of error resolution. When the insight arrives, the prediction error collapses to zero in a single cognitive event. The rate of error reduction spikes far above the brain's expectation, producing the

amplified positive valence that Van de Cruys identifies as the processing signature of humour and the "aha" experience.

Portal is the paradigm case. Shute, Ventura, and Ke (2015, *Computers & Education*; n=77) conducted a randomised controlled experiment comparing *Portal 2* and *Lumosity*: *Portal 2* players showed significant advantages on problem solving, spatial skill, and persistence, while *Lumosity* players showed no gains on any measure. The game's single mechanic (linked portals) generates an enormous space of spatial reasoning puzzles, each requiring the player to restructure their spatial model of the environment. The learning gradient is sustained not through new mechanics but through increasing spatial complexity applied to the same mechanic.

The Witness extends this further with a meta-puzzle structure: the player must first learn the local rules of each puzzle family (what do the dots mean? the stars? the coloured blocks?), then apply those rules in novel configurations, then discover that the entire island is a puzzle whose solution requires integrating knowledge across all families. The learning gradient operates at three nested scales: individual puzzles (minutes), puzzle families (hours), and the meta-puzzle (the full game).

14.5 Strategy systems: prediction error through opponent modelling and long-term planning

Strategy systems generate prediction errors through resource management, opponent modelling, and long-term planning. The learning gradient operates at longer timescales and deeper abstraction levels than combat or movement systems.

In turn-based strategy (*Chess*, *Civilisation*, *XCOM*), the prediction errors are predominantly cognitive: the player's model of the game state, their prediction of the opponent's response, and their evaluation of long-term consequences. The learning gradient is

sustained by the combinatorial depth of the decision space; chess has approximately 10^{120} possible game positions, and no human player can exhaust its prediction error landscape in a lifetime.

In real-time strategy (*StarCraft*, *Age of Empires*), the prediction errors are split between cognitive strategy and motor execution. The player must simultaneously manage base construction, resource gathering, unit production, and combat; each generating its own prediction error stream. Foerde, Knowlton, and Poldrack (2006, *PNAS*) showed that multitasking during learning shifts reliance from hippocampal declarative memory to striatal habit learning, which may explain why RTS mastery feels qualitatively different from turn-based mastery: RTS players develop automatic habits (build orders, hotkey sequences) that free cognitive resources for strategic decision-making.

Competitive multiplayer strategy adds the deepest layer: the opponent is another human whose behaviour generates an **inexhaustible prediction error landscape**. Zhu, Mathewson, and Hsu (2012, *PNAS*) demonstrated two neurally dissociable learning signals in competitive games: reinforcement prediction errors (tracked by bilateral putamen) and belief prediction errors about opponents' strategies (tracked by dmPFC/TPJ). The strategic learning gradient in competitive games persists indefinitely because the opponent adapts, meta-games evolve, and the prediction error landscape shifts faster than any individual can fully model.

14.6 Narrative systems: prediction error through expectation violation

Narrative systems generate prediction errors through story structure: plot twists, character revelations, moral dilemmas, and emotional surprises. Unlike the other system types, narrative prediction errors are typically unrepeatable; once a twist is known, the prediction error it generated cannot be regenerated.

This creates a distinctive learning gradient profile: narrative engagement is front-loaded and non-renewable. The first playthrough of *The Last of Us* generates intense prediction errors through character development and plot revelation. The second playthrough generates far fewer, because the narrative model is already complete. Narrative games compensate through branching structures (multiple paths create multiple prediction error sequences), emergent storytelling (procedurally generated events that even the designer cannot predict), and environmental narrative (world-building details that reward close observation with model-expanding discoveries).

Anderson, Karzmark, and Wardrip-Fruin (2019, FDG; n=39) conducted the first empirical test of Bogost's procedural rhetoric, finding that players accurately identified rhetorical arguments in persuasive games and that *September 12th* significantly shifted anti-war attitudes ($t(38) = 3.73, p = .001$). Narrative systems, when integrated with mechanical systems, can generate prediction errors that operate simultaneously at the level of story (what will happen next?) and the level of meaning (what does this game system imply about the world?).

14.7 The spectrum and its hybrids

The most acclaimed games typically operate at **multiple points on this spectrum simultaneously**. *Hades* combines fighting-game System 1 combat with strategy-game System 2 build planning and narrative-game prediction errors through its death-cycle storytelling. *Elden Ring* layers Soulslike combat, open-world exploration, RPG character building, and environmental narrative into a system where multiple learning gradients operate concurrently. *Dota 2* fuses real-time motor execution, tactical encounter planning, strategic resource management, and social opponent modelling into five simultaneous prediction error streams.

The design principle is redundancy: by generating prediction errors across multiple system types at multiple timescales, the probability that all gradients simultaneously reach zero is kept very low. When motor execution is automatised, tactical decisions continue to generate errors. When tactical patterns become familiar, strategic depth sustains the gradient. When even strategy is mastered, the opponent adapts.

Chapter 15 - The Design Laws: Turning Theory into Practice

On the five principles that emerge directly from the learning gradient model; their manifestation across genres; and how they function as diagnostic tools for identifying and fixing engagement failures.

15.1 From model to method

The unified model tells us that fun equals optimal-rate learning and that flow equals stable learning over time. But a model is not enough. Design requires **actionable principles that can be applied to real systems**. This chapter presents five design laws derived directly from the learning gradient, illustrates each with examples across genres, and shows how violations of each law produce specific, diagnosable engagement failures.

These laws are not rules to follow mechanically. They are lenses for evaluating design decisions against a single criterion: does this system support the player's learning process?

15.2 Law 1: Maintain the learning gradient

The player must always be learning something.

If learning stops, engagement collapses. If learning is too fast, overload occurs. The gradient must remain positive and sustainable across the player's entire engagement with the game.

This does not mean every moment must introduce new mechanics. Refinement of existing skills counts as learning. A *Celeste* player

who shaves 0.3 seconds off a room clear by optimising their dash arc is learning, even though no new mechanic was introduced. A *Halo* player whose grenade placement improves from "near the enemy" to "where the enemy will be in 1.5 seconds" is learning. The gradient must be positive; it need not be steep.

Manifestation across genres:

In *Halo*, the gradient is maintained through combinatorial encounter design. Each combat space recombines enemy types, weapons, and terrain in novel configurations. The game introduces no new mechanics after the first few hours, but the combinations of existing mechanics continue to generate fresh prediction errors throughout the campaign because the combinatorial space is vastly larger than the number of individual elements.

In *Celeste*, the gradient is maintained through level design that sequences movement concepts with precise pacing. Each screen introduces one spatial challenge, allows the player to master it through rapid death-and-retry cycles, and then combines it with previously mastered concepts in subsequent screens. The mechanical vocabulary is fixed by the second chapter; everything after that is combinatorial application.

In *Civilisation VI*, the gradient is maintained through the intersection of multiple strategic systems (technology, civics, diplomacy, warfare, religion, trade) whose interactions produce emergent strategic situations that no individual system generates alone. A player who has mastered each system individually still faces novel prediction errors when the systems interact in unexpected ways (a religious victory creates diplomatic tension that triggers a surprise war during a technology race).

In *Stardew Valley*, the gradient is maintained through seasonal revelation. Each in-game season introduces new crops, new fish, new events, and new relationship opportunities. The player cannot exhaust the game's content in a single season because the system withholds information until the appropriate time. The gradient is

paced by the calendar rather than by the player's skill, which creates a different temporal profile: anticipatory curiosity (what will happen next season?) rather than mastery-driven engagement (can I execute this challenge?).

Diagnostic: When a player reports boredom, the first question is: what is the player currently learning? If the answer is "nothing" - if their model of the game is complete and no new prediction errors are being generated - then Law 1 has been violated. The fix is not more content (which may generate no new errors if it reuses existing patterns) but more **structure**: new combinations, new contexts, new applications of existing mechanics.

15.3 Law 2: Hide the learning

Players should feel like they are playing, not studying.

Explicit instruction breaks immersion and slows engagement. Implicit learning feels natural and sustains flow. The brain's procedural memory system acquires patterns through exposure and practice without requiring conscious awareness; games should target this system rather than the declarative system that processes tutorials and tooltips.

Manifestation across genres:

Zelda: Breath of the Wild teaches its physics chemistry system entirely through environmental interaction. The player discovers that metal conducts electricity not because a tutorial says so, but because they hold a metal weapon during a thunderstorm and get struck by lightning. The prediction error is vivid, the feedback is immediate, and the learning is permanent. No tutorial could produce the same retention because the discovery pathway engages both the error signal (surprising outcome) and the curiosity reward (I figured this out myself).

Super Mario Bros. World 1-1 teaches running, jumping, blocks, power-ups, enemies, and pits through spatial design alone. Shigeru Miyamoto's level design places a Goomba in the player's path early enough that collision is likely, teaching the death mechanic. It places a block at the right height to encourage jumping, teaching the block-breaking mechanic. It places a mushroom inside a block positioned so the player will naturally hit it, teaching the power-up mechanic. No text appears on screen.

Dark Souls teaches through lethal consequence. The Undead Burg's first encounter with a hollow soldier teaches blocking and attacking. The first encounter with a firebomb-throwing hollow teaches spatial awareness. The first mimic (a chest that attacks when opened) teaches the player to verify before grabbing. Every lesson is delivered through gameplay experience rather than instruction, and the lessons are retained because they are emotionally salient; death is a powerful mnemonic.

Portal teaches through chamber design, as detailed in Chapter 23: each room is a controlled experiment that introduces one concept, provides the materials for the player to discover that concept through interaction, and requires application of the concept to progress. The tutorial is the game; the game is the tutorial. They are indistinguishable.

Diagnostic: When a player skips tutorials, ignores tooltips, or expresses frustration with "hand-holding," the game is likely violating Law 2 by attempting to teach declaratively what should be taught procedurally. The fix is to redesign the teaching as gameplay: create situations where the correct behaviour is the natural response to the game's stimuli, rather than the prescribed response to an instruction.

15.4 Law 3: Player-regulated challenge

The player must be able to control their own difficulty.

Fixed difficulty risks boredom for skilled players and overload for beginners. Player-controlled systems stabilise the learning gradient by allowing each player to find their own optimal challenge zone. The mechanism of control varies widely across genres, but the principle is universal: the player must have some influence over the rate at which prediction errors arrive.

Manifestation across genres:

Halo's four difficulty tiers (Easy, Normal, Heroic, Legendary) are the most transparent implementation. The player explicitly selects their gradient steepness before beginning and commits to it. Bungie was explicit about the intended experience: "Normal is for beginners who are embarrassed to pick Easy. Heroic is what we intended." The transparency preserves the sense of control that Csikszentmihalyi identified as a prerequisite for flow.

Celeste's Assist Mode provides granular sub-controls: the player can individually adjust game speed, dash count, and invincibility. This deconstructs "difficulty" into its component prediction error sources, allowing the player to reduce motor-execution errors (slower speed, more dashes) while preserving spatial-reasoning errors (the puzzles are unchanged). The design acknowledges that "difficulty" is not a single variable but a composite of multiple prediction error types, and players may want to control each independently.

Breath of the Wild and *Elden Ring* implement difficulty regulation through geography. The player can go anywhere, but some areas are much harder than others. A player who finds the current region too difficult can leave and explore elsewhere, building skills and equipment in lower-difficulty areas before returning. The difficulty is not set by a menu; it is set by the player's choice of destination. This preserves immersion (no menus break the fiction) while providing the same gradient-matching function as explicit difficulty tiers.

Dota 2's MMR system implements difficulty regulation through matchmaking. The system automatically adjusts the difficulty of each match by selecting opponents of approximately equal skill. The player's difficulty is their opponent's skill, and the system calibrates this continuously. The player has no explicit control over this process, but their MMR rises or falls in response to their performance, maintaining the challenge-skill balance automatically.

Roguelikes implement difficulty regulation through persistence mechanics. *Hades* offers permanent upgrades (Mirror of Night, weapon aspects) that reduce difficulty over repeated runs. A player who cannot clear a boss with base equipment may succeed after investing persistent resources in damage and survivability upgrades. The game adjusts to the player through accumulated progress rather than through an explicit setting.

Diagnostic: When a player reports that a game is "too hard" or "too easy" and there is no mechanism for adjustment, Law 3 has been violated. The fix depends on genre: explicit difficulty settings for linear games, geographic difficulty variation for open-world games, matchmaking for competitive games, or persistence mechanics for roguelikes.

15.5 Law 4: Immediate, legible feedback

Learning requires clear and immediate error signals.

Without feedback, prediction errors cannot be resolved, learning stalls, and the gradient collapses. Feedback must be immediate (millisecond-scale response to input), precise (the player knows exactly what happened), and interpretable (the player understands why it happened). These three criteria are independently necessary: a system can be immediate but imprecise (the player died but does not know what killed them), precise but uninterpretable (the player knows they took 47 damage from "environmental hazard" but does not know which environmental hazard), or interpretable but delayed

(the player understands what went wrong but only after a loading screen and respawn sequence).

Manifestation across genres:

Mario's jump exemplifies perfect micro-level feedback. The response to the button press is instantaneous (zero perceptible input lag). The physics are predictable (the same input always produces the same arc). The outcome is immediately visible (you either land on the platform or you do not). Long and Gutwin (2018, CHI PLAY) demonstrated that latency thresholds as low as 50ms degrade performance depending on game speed; *Mario's* sub-frame response time is a necessary condition for the motor-level learning gradient that its platforming demands.

Halo's Covenant enemies exemplify perfect meso-level feedback, as detailed in Chapter 24. Grunts panic and flee when their leader dies. Elites stagger when their shields break. Jackals turn in surprise when flanked. Each reaction is readable confirmation that the player's action had the intended effect. The Flood's absence of readable state changes demonstrates the converse: without legible feedback, the learning gradient collapses even when the difficulty is appropriate.

Dota 2's damage numbers, status effect icons, and ability cooldown indicators exemplify information-dense feedback in complex systems. The game provides enough information for the player to reconstruct the causal chain of any encounter, but the information is presented in layers: essential combat feedback (health bars, damage numbers) is pre-attentive, while detailed tactical information (exact cooldown timers, buff durations) requires deliberate attention. This layered approach ensures that novice players can extract the most important error signals without being overwhelmed by the full information density.

Dark Souls's death screen exemplifies feedback through consequence. The screen goes dark, "YOU DIED" appears, and the player respawns at the last bonfire. There is no kill-cam, no damage

breakdown, no suggestion of what to do differently. The feedback is the death itself: which attack killed you (visible in the final animation before the screen fades) and where you were standing when it happened (visible from the respawn position relative to the death location). The austerity of the feedback forces the player to extract the lesson from their own memory of the encounter, which engages deeper processing than a post-death analysis screen would.

Diagnostic: When a player reports that deaths feel "random" or "unfair" - when they cannot explain why they failed - Law 4 has been violated. The fix is not reducing difficulty but increasing **legibility**: making the causal chain from player action to outcome visible, consistent, and traceable. The player must be able to answer "what killed me and what should I have done differently?" after every failure.

15.6 Law 5: Never fully solve the system

The game must stay slightly ahead of the player.

If the system is fully understood, the learning gradient reaches zero and engagement ends. The game must maintain a horizon of unresolved prediction errors that recedes as the player advances.

This law creates an apparent tension with Law 4 (legible feedback): if the system is fully legible, won't the player eventually learn everything? The resolution is that **legibility and depth are not the same thing**. A system can be perfectly legible (every outcome is traceable to its cause) and infinitely deep (the combinatorial space of outcomes exceeds any individual's capacity to fully explore). Chess is the paradigm: every move's consequence is deterministic and visible, yet the game has never been "solved" by a human player because the decision tree is too vast to fully compute.

Manifestation across genres:

Dota 2 exemplifies the infinitely deep system. 120+ heroes, 200+ items, five human teammates, five human opponents. The combinatorial space is astronomical, and human opponents ensure that the system's prediction error landscape shifts faster than any individual can fully model. The game can never be "solved" because the solution space includes the adaptive behaviour of nine other humans.

Breath of the Wild's physics chemistry system achieves depth through emergence. The rules are simple (fire burns wood, metal conducts electricity), but their interactions produce outcomes that neither the player nor the designer anticipated. A player who has spent 200 hours in Hyrule can still discover new interactions because the combinatorial space of physics objects, terrain features, weather conditions, and rune abilities exceeds what any single player can exhaustively explore.

Spelunky achieves unsolvability through procedural generation. Each run presents a new configuration of rooms, enemies, items, and traps. The underlying rules are learnable (spike traps kill on contact, arrow traps fire when a line-of-sight trigger is crossed), but the specific configurations are unpredictable. The player's general model improves continuously, but the specific instantiation always contains novel prediction errors.

Tetris achieves unsolvability through speed scaling and random piece generation. The rules are deterministic and fully known, but the combination of increasing speed and unpredictable piece sequences ensures that the player is always operating at the boundary of their motor automaticity. There is always a faster level.

Chess achieves unsolvability through sheer combinatorial depth. The rules are complete and have been known for centuries. The decision tree contains approximately 10^{120} possible game positions. No human has exhausted this space, and none will.

Diagnostic: When a player reports "grinding" - repetitive engagement without meaningful learning - Law 5 has likely been

violated. The system's prediction errors have been exhausted, but the game continues to demand engagement through numerical progression (XP, currency, levels) rather than genuine learning. The fix is introducing new structure (emergence, combination, social complexity) rather than new content (more of the same enemies in a different skin).

15.7 The interaction of laws

These laws are not independent. They form a reinforcing system:

- Learning requires feedback (Law 4 enables Law 1)
- Feedback enables gradient maintenance (Law 4 sustains Law 1)
- Gradient stability requires player control (Law 3 stabilises Law 1)
- Player control depends on hidden learning (Law 2 supports Law 3; a player who understands the teaching mechanism can game it, collapsing the gradient)
- Sustained engagement requires unsolved systems (Law 5 extends Law 1)

The most acclaimed games satisfy all five laws simultaneously. *Breath of the Wild* maintains the gradient through physics emergence (Law 1), teaches through environmental interaction (Law 2), lets the player choose their own path and difficulty (Law 3), provides immediate physics feedback (Law 4), and generates emergent interactions that no player can fully exhaust (Law 5). *Halo* maintains the gradient through combinatorial encounter design (Law 1), teaches through encounter structure rather than tutorials (Law 2), offers four explicit difficulty tiers (Law 3), provides multi-layered combat feedback (Law 4), and uses the nested loop architecture to prevent any single gradient from flattening (Law 5).

15.8 Diagnosing failure with the laws

When a game fails, one or more laws have been violated. The diagnostic framework:

- **Boredom:** Law 1 (no learning) and/or Law 5 (system exhausted). The gradient has reached zero.
- **Frustration:** Law 4 (unclear feedback). Prediction errors are present but irresolvable because the player cannot identify what went wrong.
- **Overload:** Law 3 (no control over difficulty) and/or Law 2 violation (explicit instruction overwhelming the player with declarative information rather than letting them learn procedurally).
- **Grinding:** Law 5 (system exhausted, engagement sustained through token accumulation rather than genuine learning).
- **Tutorial fatigue:** Law 2 (the game is teaching declaratively when it should be teaching procedurally).

Each diagnosis points to a specific corrective. The laws do not tell the designer what to build, but they tell the designer what to fix.

15.9 Final principle

All five laws reduce to one:

Design for sustained, interpretable learning.

If the player understands what is happening, can improve over time, encounters new structure regularly, and continues to find the system deeper than their current understanding, then the game will be engaging. The five laws are the operational decomposition of this single principle into actionable design criteria.

Chapter 16 - Exploration and Curiosity in Design

On the design principles that transform open worlds from checklists into curiosity engines; the structural differences between directed and self-directed learning gradients; and practical techniques for sustaining exploration-driven engagement.

16.1 The design problem of exploration

Chapter 4 established the neuroscience: curiosity is a dopaminergic drive state that transforms information gaps into intrinsic rewards (Gruber et al., 2014), organisms preferentially attend to stimuli of intermediate complexity (Kidd & Hayden, 2015), and extrinsic markers can undermine intrinsic motivation (Deci, Koestner, & Ryan, 1999). Chapter 25's analysis of *Breath of the Wild* demonstrates these principles in a single game.

This chapter addresses the **design problem** that sits between the neuroscience and the case study: how do you build a world that sustains curiosity-driven exploration across dozens of hours? What are the structural techniques, and how do they differ from the challenge-driven techniques that sustain combat or puzzle engagement?

16.2 Directed versus self-directed gradients

Combat and puzzle games impose a learning gradient: the designer controls which challenges appear, in what order, at what difficulty. The player's role is to resolve the prediction errors the designer has

placed in their path. This is a **directed gradient**; the designer is the author of the learning trajectory.

Exploration-driven games invert this relationship. The designer creates a world full of potential prediction errors, and the player chooses which ones to pursue. The learning trajectory is co-constructed: the designer provides the raw material (a world with consistent rules and discoverable secrets), and the player provides the direction (where to go, what to investigate, when to engage). This is a **self-directed gradient**, and it requires fundamentally different design techniques.

The directed gradient's failure mode is overload or boredom (the designer's pacing is wrong for this player). The self-directed gradient's failure mode is aimlessness (the player cannot identify which direction offers productive prediction errors) or exhaustion (the player has explored everything accessible and cannot find new territory).

16.3 Techniques for sustaining self-directed gradients

Successful exploration-driven games share a set of structural techniques that keep the self-directed gradient positive:

Visible horizons. The player must be able to see, from their current position, at least one point of interest that they have not yet investigated. *Breath of the Wild*'s triangle rule (peaks, towers, and unusual structures visible from great distances) and *Elden Ring*'s Erdtree (a constant landmark orienting the player in the world) both serve this function. The horizon provides a continuous source of spatial prediction errors: what is that? What is over there? How do I get to it?

Layered information density. The world should reveal different information at different distances. From far away, a landmark communicates "something is here." From medium distance, the

player can identify what kind of thing it is (a ruin, a camp, a shrine). From close range, the player discovers the specific content (what enemies guard it, what puzzle it contains, what reward it offers). This layering ensures that the prediction error resolves gradually rather than all at once, sustaining the curiosity signal across the approach.

Consistent environmental language. The player must learn to read the world. Consistent visual cues (a specific architectural style signals a specific type of challenge, a specific plant signals a specific resource) allow the player to build a model of the environment that generates productive predictions. When the player sees a familiar cue, they predict what they will find, and the prediction is tested upon arrival. *Elden Ring's* glowing skulls signal rune drops. *Breath of the Wild's* Sheikah structures signal shrines. *Dark Souls's* fog gates signal boss encounters. Each cue is a learned environmental rule that transforms visual scanning into model-based prediction.

Density calibration. Points of interest must be spaced so that the player always encounters something new before the curiosity signal decays. Too sparse, and the player traverses empty space with no prediction errors (boredom). Too dense, and the player is overwhelmed with options (overload). *Breath of the Wild's* designers described calibrating density so that the player always discovers something interesting within approximately 30-60 seconds of traversal in any direction. This calibration is the exploration equivalent of *Halo's* encounter pacing: it maintains the gradient by controlling the rate at which new prediction errors appear.

Reward variety. If every exploration target produces the same reward (a small amount of currency, a generic collectible), the prediction error resolves after the first discovery: "this is just another coin." Variety in reward type (mechanical upgrades, cosmetics, lore, shortcuts, new abilities, narrative revelations, environmental puzzles) preserves uncertainty about what each target contains. The prediction error "what will I find?" remains open because the answer is different each time.

Return value. Previously explored areas should change over time or reveal new information after the player has acquired new abilities or knowledge. *Metroidvania* design is built entirely on this principle: areas the player passed through early in the game contain paths accessible only with later abilities, creating a second (and third) exploration pass through familiar territory with fresh prediction errors. *Tears of the Kingdom*'s construction abilities transform previously explored terrain into engineering playgrounds, adding a new prediction error layer to locations the player thought they already understood.

16.4 Why markers fail: the mechanism

The previous chapters noted that quest markers undermine exploration. The design mechanism is worth specifying precisely because it illustrates how a well-intentioned feature can destroy a learning gradient.

A quest marker converts an exploration problem (where is this thing?) into a navigation problem (follow the line on the map). The exploration problem generates prediction errors about the world's structure: the player must build a spatial model, identify landmarks, read environmental cues, and make inferences about where interesting content might be. The navigation problem generates zero prediction errors: the line tells the player exactly where to go; the only remaining uncertainty is whether they can walk there without dying.

By eliminating the spatial prediction errors, markers eliminate the learning gradient that makes exploration engaging. The player is no longer building a model of the world; they are following instructions. The transition is from System 2 spatial reasoning (rewarding) to System 1 waypoint-following (automatic and disengaging). The exploration gradient collapses to a traversal grind.

The design lesson: every navigation aid should be evaluated against the question "does this help the player build their own model of the world, or does it replace the need for a model?" Compasses that point toward a general direction (north, toward a region) help the player orient without replacing spatial reasoning. Map markers that identify exact locations replace spatial reasoning entirely. The former supports the gradient; the latter destroys it.

16.5 Transition

The techniques described in this chapter provide the structural toolkit for exploration-driven engagement. The next chapter examines a different design challenge: how to manage the learning gradient across the temporal arc of a complete game, from the first minute to the last.

Chapter 17 - Pacing and Session Design

On the micro-structure of engagement within and across play sessions; why the first five minutes and the transition between sessions are the hardest design problems; and how pacing mechanics regulate the learning gradient at timescales below the lifecycle.

17.1 Below the lifecycle

Chapter 10 described the macro-scale lifecycle of engagement: first contact, rapid learning, deepening, mastery, exhaustion. That analysis operates at the timescale of hours to hundreds of hours. But engagement is also structured at shorter timescales: individual play sessions (30 minutes to 4 hours), encounter sequences (5-20 minutes), and moment-to-moment pacing (seconds to minutes). This chapter examines how the learning gradient is managed at these shorter timescales.

17.2 The first five minutes

The opening minutes of a game are the highest-risk moment in the entire experience. The player's model is empty, prediction errors arrive at maximum density, and the risk of overload is acute. Industry retention data confirms this: mobile games lose over 75% of new users within 24 hours, and the steepest drop occurs in the first session.

The design challenge is threading between two failure modes. **Too much information** produces overload: the player cannot identify which prediction errors to focus on and disengages. **Too little**

information produces confusion: the player does not understand what the game is or what they should do and disengages for different reasons.

Successful openings share a structure: **a single, clear action with immediate, vivid feedback**. *Mario* starts with running and jumping. *Portal* starts with walking through a portal. *Dark Souls* starts with rolling and hitting. The first action teaches the most fundamental mechanic, and the feedback confirms that the player's input matters. Everything else; story, systems, menus, options; is deferred.

The onboarding gradient should introduce one system per learning cycle. A learning cycle is the minimum sequence of: encounter the system → attempt to use it → receive feedback → adjust understanding. The length of a learning cycle varies by system complexity (10 seconds for "press A to jump," several minutes for "manage your inventory"), but the principle is constant: one new system per cycle, with consolidation time before the next introduction.

Breath of the Wild's Great Plateau solves this by gating four rune abilities behind four separate shrines, each teaching one concept in isolation. *Halo*'s first level introduces the control scheme, then adds weapons, then adds grenades, then adds vehicles across a sequence of encounters that each focus on one new element. The worst onboarding designs (*Dota 2*'s new player experience, many complex RPGs) present multiple interlocking systems simultaneously and expect the player to sort them out; this is the design equivalent of teaching someone to drive by putting them on a motorway.

17.3 Session rhythm

Within a single play session, the learning gradient benefits from a **rhythm of tension and release** that maps onto the LC-NE phasic/tonic oscillation.

Sustained phasic mode (intense focus) is metabolically expensive. Wiehler et al. (2022, *Current Biology*) demonstrated that prolonged cognitive control produces glutamate accumulation in the lateral prefrontal cortex, which may drive the subjective experience of mental fatigue. Games that demand constant high-intensity engagement without breaks produce exhaustion rather than flow.

Effective session rhythm alternates between:

- **High-intensity segments** (combat encounters, puzzle chambers, boss attempts) that demand focused attention and generate dense prediction errors
- **Low-intensity segments** (traversal, dialogue, exploration, inventory management) that allow cognitive recovery while maintaining engagement through lower-density prediction errors

Halo's campaign paces encounters with traversal and vehicle sections. *Dark Souls* paces boss fights with bonfire-to-boss runs that allow mental preparation. *Zelda* paces shrine puzzles with overworld exploration. *Portal* paces test chambers with transitional spaces that deliver narrative without mechanical challenge. Martin O'Donnell's use of silence between combat music cues serves the same function at the audio level: contrast that prevents habituation and allows autonomic recovery.

The optimal rhythm depends on the intensity of the high segments. More intense challenges (boss fights, competitive matches) require longer recovery periods. Lower-intensity challenges (standard encounters, simple puzzles) can be paced more densely. The design principle is that the player should feel ready to re-engage by the time the next high-intensity segment arrives; if they feel fatigued, the recovery period was too short.

17.4 Session boundaries and the return problem

The transition between play sessions is a design problem that most games ignore and few handle well. When a player stops playing and

returns hours or days later, they face a **re-entry problem**: their working model of the game's current state has partially decayed, and they must reconstruct context before they can resume productive engagement.

Games that handle this well provide **re-entry scaffolding**: recaps of recent events (*Witcher 3*'s loading screen summaries), persistent UI elements that show current objectives (quest logs, map markers), or environments designed so that the player's position communicates their current goal (standing outside a boss fog gate, at the entrance to a new area).

Games that handle this poorly expect the player to remember exactly where they were and what they were doing. A player who returns to a complex RPG after a two-week break and finds themselves in the middle of a dungeon with a full inventory and no recollection of which quest they were pursuing faces a re-entry gradient so steep that many players simply start over or quit.

The save system is a pacing tool, not merely a convenience feature. Save points at natural transition moments (between levels, after boss defeats, at the start of new areas) align session boundaries with gradient boundaries: the player stops at a point where one learning arc is complete and the next has not yet begun. Saving mid-challenge forces the player to resume in the middle of an unresolved gradient, which may produce confusion on return.

17.5 The endgame problem

The final hours of a game present the inverse of the opening problem. During onboarding, the gradient is too steep (too many errors, too fast). During the endgame, the gradient is approaching zero (the player's model is nearly complete, and few new prediction errors remain).

Games address this in several ways. **Escalation** introduces the most complex challenges at the end, testing the full range of skills the

player has developed (final boss fights, endgame dungeons, climactic encounters that combine every mechanic). **Narrative climax** provides a non-mechanical gradient: the player's model of the story is approaching resolution, and narrative prediction errors (how will it end?) sustain engagement through the mechanical gradient's decline. **New Game Plus** extends the gradient by modifying the game's parameters (harder enemies, different item placements, new abilities) so that the same content generates fresh prediction errors on a second pass.

The games that handle the endgame best are those where the mechanical gradient and the narrative gradient converge: the final challenge is both the hardest test of skill and the resolution of the story's central tension. *Portal*'s final encounter with GLaDOS. *Halo 3*'s Warthog run. *Dark Souls*'s final boss. Each is simultaneously a mechanical climax (the most demanding test of the skills the game has trained) and a narrative climax (the resolution of the story's central conflict). The learning gradient and the narrative gradient peak at the same moment, producing the most intense engagement of the entire experience.

17.6 Transition

Pacing and session design operate at the timescale below the lifecycle, managing the learning gradient across minutes and hours rather than across the full arc of the game. The next chapter examines what happens when this management fails: the specific pathologies that emerge when the gradient collapses, spikes, stalls, or oscillates.

Chapter 18 - Failure Modes

On boredom, frustration, grind, and anxiety as breakdowns in learning dynamics rather than isolated design flaws.

18.1 Failure is gradient failure

When a game stops being engaging, the cause is always a breakdown in the learning gradient. The specific phenomenology of the breakdown - whether the player feels bored, frustrated, anxious, or stuck in a grind - corresponds to a specific pathology of the prediction error dynamics.

This chapter provides a systematic taxonomy of engagement failures, each mapped to a specific gradient pathology and illustrated with concrete game examples.

18.2 Boredom: gradient at zero

Boredom occurs when the learning gradient reaches zero. The player has fully internalised the game's patterns, and no new prediction errors are being generated. Outcomes are fully predicted, and the dopamine system has nothing to signal (Schultz, 1997: zero prediction error produces no dopaminergic response).

Symptoms: the player feels that nothing is happening, that they are going through the motions, that the game has nothing left to teach. Time drags. Attention wanders. The LC-NE system drifts from phasic mode (focused engagement) to tonic mode (broad scanning for better options).

Common causes:

- Repetitive content with no variation (identical enemy encounters, copy-pasted side quests)
- Insufficient combinatorial depth (systems that do not interact in surprising ways)
- Player skill exceeding game challenge with no mechanism for self-regulation

Design diagnosis: Law 1 (maintain the learning gradient) and/or Law 5 (never fully solve the system) have been violated.

Examples: The late-game grind in many open-world RPGs, where clearing the 47th identical bandit camp generates no prediction errors that the first camp did not already provide. The final hours of *Assassin's Creed* games, where the map is covered in icons but none represents a genuinely new challenge.

18.3 Frustration: irresolvable errors

Frustration occurs when prediction errors are being generated but cannot be resolved. The player is failing, but they cannot identify why or what to do differently. The error signal is noise rather than information.

Symptoms: the player feels that the game is unfair, that deaths are random, that improvement is impossible. Negative valence is high because prediction errors are increasing rather than decreasing (Van de Cruys, 2017: negative affect tracks errors that are growing).

Common causes:

- Unclear feedback (the player cannot trace the causal chain from action to outcome)
- Inconsistent rules (the same action produces different results in different contexts)

- Stochastic difficulty (challenge arising from randomness rather than learnable patterns)
- Off-screen or undodgeable attacks (prediction errors that are irresolvable by design)

Design diagnosis: Law 4 (immediate, legible feedback) has been violated.

Examples: *Halo 2*'s Legendary Jackal Snipers, which kill the player from off-screen with hitscan weapons; the error signal is "I got one-shot by something I couldn't see," which teaches nothing actionable. Poorly designed camera systems in 3D platformers where the player cannot see the next platform. Boss attacks that have no telegraph and cannot be dodged.

Jesper Juul's *The Art of Failure* (2013, MIT Press) provides the theoretical framework: players enjoy feeling responsible for failure, and "fairness" is the perception that the player's actions, not external randomness, determined the outcome. Frustration arises when this attribution is impossible.

18.4 Grinding: gradient near zero with token progress

Grinding occurs when the learning gradient is near zero but the game continues to demand engagement through repetition. The player is active but not improving; prediction errors are trivially small and immediately resolvable, producing token progress (XP accumulation, currency farming, percentage completion) without genuine learning.

Symptoms: the player feels that they are "doing their time" rather than engaging with meaningful challenges. The activity is repetitive but not difficult. Progress is measured in numbers rather than understanding. The wanting/liking dissociation (Berridge & Robinson, 2016) may be evident: the player feels compelled to continue (dopaminergic "wanting" driven by variable-ratio

reinforcement) without enjoying the experience (opioidergic "liking" is flat or declining).

Common causes:

- Content padding (stretching 10 hours of unique prediction errors across 40 hours of gameplay)
- Artificial progression gates (requiring players to reach a numerical threshold before accessing new content, regardless of skill)
- Variable-ratio reinforcement schedules that sustain engagement without learning (loot grinding, gacha mechanics)

Design diagnosis: Law 5 (never fully solve the system) has been violated; the system has been exhausted, but the game pretends otherwise through numerical inflation.

Examples: MMO daily quests that require completing the same activities for incremental currency rewards. Mobile games that gate progress behind timers or premium currency rather than skill development. The middle sections of JRPGs where random encounters generate no new tactical challenges but are required to reach an adequate level for the next story boss.

18.5 Anxiety: unstable gradient

Anxiety occurs when prediction errors arrive unpredictably, alternating between trivial and overwhelming without a stable rhythm. The player cannot establish a consistent model-building process because the demands are chaotic.

Symptoms: the player feels uncertain about what the game expects, unable to prepare for what comes next, and unable to trust that their improving model will generalise to future challenges. The LC-NE system oscillates between phasic and tonic mode without settling into either.

Common causes:

- Sudden difficulty spikes (a boss fight orders of magnitude harder than the preceding content)
- Inconsistent pacing (long stretches of trivial challenge punctuated by punishing encounters)
- Genre confusion (the game shifts between system types without warning, e.g. a platformer that suddenly becomes a bullet-hell shooter)

Design diagnosis: Law 3 (player-regulated challenge) has been violated; the player has no mechanism to control the rate of prediction error delivery.

Examples: Games that alternate between cutscenes and combat without establishing a rhythm. Difficulty curves that oscillate wildly between trivial and punishing. *Halo: Reach*'s hybrid health system, which reintroduced the goal-displacement problem of health packs after Halo CE had solved it with regenerating shields.

Andersen et al. (2020, *Psychological Science*; n=110) empirically demonstrated the principle in a non-game context: enjoyment of a haunted house showed an inverted-U relationship with fear, with heart rate data confirming that "just-right" deviations from physiological baseline maximised enjoyment, while excessive deviations (too scary) and insufficient deviations (not scary enough) both reduced it.

18.6 Overload: gradient exceeding processing capacity

Overload occurs when prediction errors arrive faster than the player can process them. The system is too complex, too fast, or too opaque for the player's current skill level.

Symptoms: the player feels overwhelmed, confused, and unable to extract any pattern from the chaos. Cognitive load exceeds working memory capacity. The learning gradient is technically steep (there is much to learn) but the player cannot access it because the signal is

buried in noise.

Common causes:

- Insufficient onboarding (too many systems introduced simultaneously)
- Expert-level content without adequate scaffolding
- UI clutter that competes for attention with gameplay-relevant information

Design diagnosis: Law 2 (hide the learning) and Law 3 (player-regulated challenge) have been violated.

Examples: The new-player experience in *Dota 2* or *Path of Exile*, where dozens of interacting systems are presented simultaneously with minimal guidance. Flight simulators that present every cockpit control at once rather than introducing them sequentially.

18.7 The diagnostic framework

Every engagement failure reduces to a gradient pathology:

- **Boredom** = gradient at zero (no errors to resolve)
- **Frustration** = errors present but irresolvable (feedback is noise)
- **Grinding** = gradient near zero, masked by token progress
- **Anxiety** = gradient unstable (errors arrive unpredictably)
- **Overload** = gradient exceeds processing capacity (too many errors to process)

The designer's diagnostic task is to identify which pathology is present and apply the corresponding corrective: introduce new structure (boredom), improve feedback legibility (frustration), replace padding with depth (grinding), stabilise pacing (anxiety), or improve onboarding and scaffolding (overload).

Chapter 19 - Cross-Genre Analysis

On how the learning gradient manifests differently across genres; why content volume is not learning depth; and the design trade-offs that determine where on the gradient spectrum a game can operate.

19.1 The gradient spectrum

The case studies in Part VII demonstrate the learning gradient in action across individual games. This chapter steps back to examine how the gradient's properties vary systematically across genres, identifying the design trade-offs that shape different gradient profiles.

19.2 Content volume versus learning depth

The model identifies a design failure that content-based analyses consistently miss: confusing **content volume** with **learning depth**.

A game with 100 hours of content but only 10 hours of unique prediction errors will sustain engagement for 10 hours. The remaining 90 hours will feel like grinding, because the player's model has been fully built and the additional content generates only trivially small errors. Conversely, a game with 10 hours of content but 10 hours of unique prediction errors will sustain engagement for its entire duration.

This explains the recurring pattern in critical reception where short, focused games (*Portal* at 3 hours, *Celeste* at 8 hours, *Outer Wilds* at 20 hours) receive higher per-hour critical acclaim than sprawling 100-hour RPGs. It is not that short games are inherently better; it is

that their **prediction error density** (unique informative errors per hour) is higher. Every hour contains genuine learning. Long games that maintain high prediction error density (*Dota 2*, *Dark Souls* across multiple playthroughs, *Breath of the Wild*) sustain engagement for their full duration. Long games that front-load their prediction errors and pad the rest (*Assassin's Creed*, many Ubisoft open-world games) sustain engagement only until the errors are exhausted.

The correlation is not with length but with the ratio of learning depth to content volume. When the ratio approaches 1:1 (every hour of content contains an hour of unique learning), the game sustains engagement throughout. When the ratio drops (10 hours of learning spread across 60 hours of content), the back half becomes a grind regardless of production quality.

19.3 Genre gradient profiles

Different genres generate characteristically different gradient profiles:

Action games (*Halo*, *Doom Eternal*, *Devil May Cry*) generate steep, continuous, motor-dominant gradients. The learning is primarily cortical-to-subcortical transfer: aiming, timing, positioning. The gradient is sustained by combinatorial encounter design and enemy variety. Typical learning depth: 15-30 hours for a single campaign, extended indefinitely by competitive multiplayer.

Puzzle games (*Portal*, *The Witness*, *Baba Is You*) generate discontinuous "aha" gradients with peaks separated by plateaus. The learning is primarily System 2 cognitive restructuring. The gradient is sustained by conceptual deepening rather than mechanical novelty. Typical learning depth: the full runtime, because every puzzle generates a unique prediction error by design.

Roguelikes (*Hades*, *Spelunky*, *Slay the Spire*) generate compressed, repeated gradients through the death-and-retry loop.

Each run is a mini-lifecycle (first contact → rapid learning → mastery attempt → death). Procedural generation ensures that the general model improves across runs while specific configurations remain novel. Typical learning depth: 50-200 hours before the general model stabilises.

Open-world exploration (*Breath of the Wild*, *Elden Ring*, *Outer Wilds*) generate self-directed, curiosity-driven gradients. The learning is spatial (building a world model) and systemic (discovering rule interactions). The gradient is sustained by combinatorial physics, environmental density, and the player's freedom to choose their own challenge level. Typical learning depth: 40-100 hours, bounded by world size and system complexity.

Competitive multiplayer (*Dota 2*, *Counter-Strike*, *Chess*) generate the deepest gradients because human opponents provide an inexhaustible prediction error source. The learning is layered (mechanical, tactical, strategic, social, meta-game) with each layer operating at a different timescale. Typical learning depth: effectively infinite; professional players continue improving after 10,000+ hours.

Narrative games (*Disco Elysium*, *Planescape: Torment*, *Firewatch*) generate front-loaded, non-renewable gradients. The learning is model-building about characters, plot, and thematic meaning. The gradient is bounded by the story's length and collapses to zero upon completion, with limited replay value unless the game offers branching paths or hidden content. Typical learning depth: the full runtime of a single playthrough.

19.4 The DDA trade-off

Dynamic difficulty adjustment attempts to regulate the learning gradient algorithmically: if the player is succeeding too easily (gradient flattening), increase difficulty; if the player is failing too often (gradient approaching overload), decrease difficulty. Mortazavi, Moradi, and Vahabie (2024, *Multimedia Tools and*

Applications) reviewed the literature and found that most studies showed significant DDA effects on enjoyment, flow, and immersion.

But DDA introduces a trade-off that the learning gradient framework makes explicit. Baldwin et al. (2017, CHI PLAY) found that player-oriented DDA preserved sense of control while system-oriented DDA reduced self-consciousness. The trade-off: transparent DDA (the player knows difficulty is being adjusted) preserves agency but may undermine the sense of genuine achievement. Hidden DDA (the player does not know) preserves the feeling of achievement but undermines agency if discovered, and Csikszentmihalyi identified sense of control as a prerequisite for flow.

The model suggests that **player-directed difficulty regulation** (geographic freedom in *Elden Ring*, difficulty settings in *Halo*, build diversity in *Dark Souls*) is generally superior to algorithmic DDA because it preserves both agency and achievement. The player chose this challenge; they know the system did not adjust for them; and their success is therefore attributable to their own improvement. DDA is most useful in contexts where player-directed regulation is impractical: educational games for children, rehabilitation games for patients, and casual games where explicit difficulty settings would be stigmatising.

19.5 Transition

The cross-genre analysis confirms that the learning gradient framework applies across every major game genre, though the specific gradient profile; its shape, its timescale, its dominant prediction error type; varies characteristically. The case studies that follow demonstrate these principles in depth through individual games.

PART VII - CASE STUDIES

Chapter 20 - Dark Souls: The Conversion Chamber

On boss fights as System 2-to-System 1 conversion chambers; death as the purest error signal in game design; and why "fair difficulty" is a commitment to learnable prediction errors.

20.1 Why Dark Souls matters for the model

If *Halo* demonstrates how to maintain a learning gradient through nested loops and combinatorial encounter design, *Dark Souls* demonstrates something more fundamental: that **difficulty itself is not the variable that determines engagement**. What determines engagement is the **learnability** of the difficulty; the rate at which the player can convert prediction errors into model improvements.

Dark Souls is the hardest widely loved game in modern design. It kills the player frequently, offers minimal guidance, and provides no difficulty options. By any conventional difficulty metric, it should produce frustration and disengagement. Instead, it produces some of the deepest and most sustained engagement in the medium. The unified model explains why: every source of difficulty in *Dark Souls* generates prediction errors that are **resolvable through improved play**. The game is not hard for the sake of being hard. It is hard in ways that teach.

20.2 Boss fights as conversion chambers

Each boss in *Dark Souls* functions as a dedicated **automaticity conversion chamber**. Boss attack patterns are complex enough to

prevent full automation; five or more distinct attacks deployed in variable sequences; but consistent enough that pattern recognition develops across attempts. Death resets the encounter, forcing repeated exposure that drives cortical-to-subcortical transfer.

The first encounter with a boss is pure System 2. The player's dlPFC maintains a verbal catalogue of observed attacks ("the overhead slam has a two-second wind-up; the sweep follows the thrust; the charge tracks for 45 degrees"). The ACC monitors for timing errors. Working memory is saturated. Every dodge requires conscious deliberation.

After ten attempts, the cognitive load has decreased. Some attacks have been automatised; the dodge-roll for the overhead slam is now reflexive rather than deliberate. The player's attention, freed from motor execution, shifts to higher-order patterns: which attacks follow which, what the boss does at different health thresholds, where the safe windows for counter-attacks are.

After twenty attempts, the motor layer is largely automatic. The player responds to attack telegraphs without conscious processing; the basal ganglia handle what the prefrontal cortex once managed. The remaining prediction errors are strategic: timing heal windows, managing stamina, finding optimal damage windows. The boss has served its function as a conversion chamber, transforming System 2 knowledge into System 1 skill.

Specific bosses illustrate different gradient profiles within this general pattern.

The Asylum Demon, the first boss encountered, is a tutorial disguised as an impossible challenge. The player's initial encounter is scripted to fail; they are dropped into an arena with a massive demon while carrying a broken sword. The prediction error is maximal and binary: this enemy is too strong, I must flee. The actual lesson is environmental: a door to the left provides an escape route, and the game is teaching the player that retreat is a valid strategy. When the player returns with proper equipment, the Asylum

Demon's attack patterns are the simplest in the game; slow overhead slams with long wind-ups. The conversion chamber operates in miniature: the player learns to read wind-up animations, time dodge-rolls, and find punishment windows within a single encounter that takes five minutes to master.

The Bell Gargoyles introduce the combinatorial complexity that defines the Soulslike learning gradient. The first gargoyle is a manageable one-on-one fight. At 50% health, a second gargoyle joins. The prediction error spikes: the player's model of the fight (dodge the gargoyle's attacks, punish during recovery) is violated by the arrival of a second threat that demands simultaneous tracking. The learning gradient now has two concurrent streams: the motor layer (dodging two enemies' attacks) and the strategic layer (which gargoyle to focus, when to split attention, how to use the arena's geometry to separate them). Players who have automated the motor layer from the first gargoyle must now re-engage System 2 for the tactical problem of managing two threats.

Ornstein and Smough, widely considered the franchise's defining encounter, push this to its extreme. Two bosses with radically different movement profiles (Ornstein is fast and lunging; Smough is slow and sweeping) fight simultaneously, and killing one causes the other to absorb its power and gain new attacks. The encounter generates prediction errors across four distinct phases: the two-boss phase, Ornstein's powered-up solo phase, Smough's powered-up solo phase, and the strategic decision of which boss to kill first (which determines which powered-up solo phase the player faces). The learning gradient is sustained across dozens of attempts because each phase has its own error profile, and the player must build four partially independent models before the full encounter is mastered.

Seath the Scaleless demonstrates what happens when the conversion chamber fails. Seath's first encounter is scripted as an unwinnable fight; the player is killed and respawns in a prison. This teaches nothing actionable. The second encounter requires the

player to destroy a crystal that grants Seath immortality before dealing damage. The crystal mechanic is a solver's uncertainty (Costikyan's type 2) rather than a performative uncertainty (type 1), and once the solution is known, it resolves permanently. Seath's actual combat patterns are among the simplest in the game; large, slow tail sweeps with generous dodge windows. The gradient is steep but short: the crystal puzzle generates one large prediction error that resolves in a single insight, and the combat generates only small, quickly resolved motor errors. Seath is widely considered one of the weakest bosses precisely because the conversion chamber runs out of material.

20.3 Death as the purest error signal

Death in *Dark Souls* is not punishment. It is the purest error signal in game design.

A *Dark Souls* death communicates exactly three things: which attack killed you (the final hit animation), when you made the error (the moment your dodge or block failed), and what the alternative would have been (the correct timing or positioning that would have avoided the hit). Every death teaches something specific and actionable. The player's model of the encounter improves with every death, and the improvement is perceivable; each subsequent attempt goes slightly further, dodges slightly more attacks, deals slightly more damage.

This is why *Dark Souls* players describe the game as "fair" despite its punishing difficulty. Hidetaka Miyazaki articulated the principle: "It's not a matter of simply cranking up the difficulty; it's doing so fairly. When players are killed and they can understand why they were killed, and it feels justified, that makes sense." In prediction-error terms, this is a commitment to ensuring that all difficulty arises from learnable pattern complexity rather than irreducible stochasticity.

Jesper Juul's *The Art of Failure* (MIT Press, 2013) provides the theoretical frame: games exploit a "paradox of failure" in which humans have a basic desire to succeed yet voluntarily engage in activities where they are nearly certain to fail. The resolution is that "the feeling of escaping failure; often by improving skills; is a central enjoyment of games." *Dark Souls* makes this mechanism explicit: every death is a failure, and the escape from that failure through improved play is the entire point. Juul notes that when you fail in a game, "you; not a character; are in some way inadequate," and the drive to escape that inadequacy through skill improvement is the game's motivational engine.

The souls mechanic amplifies the error signal. When the player dies, they drop all accumulated souls (the game's currency for levelling and purchasing). They have one chance to recover them by returning to the spot of their death. If they die again before recovering, the souls are lost permanently. This creates a meta-level prediction error: the player must decide whether to play cautiously (preserving souls) or aggressively (risking loss for faster progress). The possibility of permanent loss makes each death more salient, increasing the attention paid to the error signal. But the loss is never mechanically devastating; the player can always earn more souls. The system creates emotional weight without punitive consequence, sharpening the learning gradient without corrupting it.

Vella (2015, *Game Studies*) theorised this as the "ludic sublime": the aesthetic moment when mastery confronts irreducible system mystery. The player knows they can learn the boss's patterns; the question is whether they can execute under pressure. This tension between knowledge and execution sustains the learning gradient long after the cognitive model is complete, because motor automaticity takes much longer to develop than cognitive understanding. A player who can verbally describe every attack in the Nameless King's repertoire may still die to him repeatedly because the motor-level conversion from "I know what to do" to "I can do it automatically" is incomplete.

20.4 The spatial learning gradient

Dark Souls generates a secondary learning gradient through its interconnected world design. The game's world is a single continuous structure where areas loop back on themselves, shortcuts create connections between distant locations, and spatial memory replaces the need for a map.

The Undead Burg, the game's effective opening area, illustrates the spatial gradient at its clearest. The player begins at a bonfire (checkpoint) and must navigate forward through a linear sequence of narrow streets populated by low-level enemies. The route is winding and opaque on first traversal; the player does not know where they are going, what is around each corner, or how the spaces connect. As they progress, they discover a shortcut: a locked door that opens from the far side, creating a direct connection back to the starting bonfire. The spatial prediction error (how does this area connect to where I started?) resolves in a single satisfying moment, and all subsequent traversals use the shortcut rather than the original route.

This pattern repeats throughout the game at increasing scales. The elevator from the Undead Parish to Firelink Shrine reveals that two areas the player thought were distant are actually vertically adjacent. The hidden bonfire in Blighttown transforms a punishing descent into a manageable expedition. The door from the Darkroot Garden to the Valley of Drakes opens a cross-world shortcut that connects three distinct regions. Each discovery generates a spatial prediction error that, once resolved, permanently improves the player's mental map. The spatial learning gradient operates on a different timescale from the combat gradient; it resolves over hours rather than minutes; but it contributes to the total engagement by providing a second simultaneous source of model improvement.

The interconnected world design also serves a narrative function within the learning gradient framework. Unlike games that separate

story from gameplay (cutscenes interrupting combat), *Dark Souls* embeds its narrative in the spatial structure. The player learns the world's history by discovering where things are relative to each other: the graveyard leads to the catacombs which leads to the tomb of the giants, tracing a descent into deeper and darker regions of the underworld. The environmental storytelling generates narrative prediction errors (what happened here? why is this ruin connected to that kingdom?) that layer on top of the spatial and combat gradients without interrupting them.

20.5 The estus flask and resource management as a learning layer

The estus flask system adds a third gradient on top of combat and spatial learning: **resource management under pressure**.

The player carries a limited number of healing charges (estus flasks) that refill only at bonfires. Each use of a flask requires a long animation during which the player is vulnerable. This creates a persistent resource management prediction error: should I heal now (risking punishment during the animation) or wait (risking death from the next attack)? The optimal timing depends on the specific enemy's attack patterns, the distance to the next bonfire, and the player's remaining flask count.

This system transforms healing from a binary decision (do I have health? then I'm fine) into a continuous strategic calculation that generates prediction errors at every moment of low health. A player who heals at the wrong moment and is punished during the animation learns to wait for a safe window. A player who hoards flasks too conservatively and dies with three remaining learns to use them more aggressively. Each death where flask management played a role generates a prediction error about resource allocation that is distinct from the prediction errors about combat execution and spatial navigation.

The bonfire placement intensifies this effect. Bonfires are spaced so that the distance between checkpoints is approximately 5-15 minutes of gameplay. This means the player must manage their flask supply across multiple encounters, creating a macro-level resource gradient that operates on the same timescale as the spatial gradient. Running out of flasks far from a bonfire is a specific kind of failure that teaches something specific: the player was too aggressive with flask use early in the sequence, or they took too much avoidable damage, or they chose a suboptimal route.

20.6 Build diversity and player-directed difficulty

Dark Souls offers no difficulty settings, but it offers something more nuanced: **build diversity that allows players to implicitly select their own learning gradient.**

A player who invests in heavy armour and a greatshield can block most attacks without timing, reducing the motor-level prediction error density while increasing the strategic error density (managing stamina for blocks, finding counter-attack windows). A player who invests in a fast weapon and light armour faces maximal motor-level errors (dodging must be frame-perfect) but simpler strategic decisions (hit and move). A magic-focused build can engage many bosses at range, fundamentally changing the prediction error profile from "can I dodge this attack?" to "can I maintain distance and resource management?"

Each build creates a different learning gradient through the same content. The game does not adjust its difficulty; the player adjusts their approach to the difficulty, self-selecting into the gradient steepness that matches their skills and preferences. This is player-directed difficulty regulation without explicit difficulty settings; the same design principle that *Halo*'s four tiers implement through menus, *Dark Souls* implements through character building.

20.7 Where the gradient fails: Blighttown and the Bed of Chaos

Not every area in *Dark Souls* sustains the learning gradient successfully, and the failures are diagnostic.

Blighttown is the most criticised area in the original game, and the primary reason is not difficulty but **feedback legibility**. The area is dark, visually cluttered, and plagued by severe frame rate drops in the original console release. Enemies attack from off-screen. Poison-inflicting blowdart snipers fire from positions the player cannot see. Narrow walkways over lethal drops punish any positioning error with instant death. The prediction errors are present but degraded: the player knows they were hit by a dart but cannot identify where it came from; the player knows they fell but cannot determine whether the walkway was shorter than expected or the camera angle was misleading. When feedback is unclear, prediction errors cannot be resolved, and the gradient stalls into frustration.

The **Bed of Chaos** is the franchise's most universally criticised boss, and it fails for a different reason: its difficulty is **platforming-based rather than combat-based**. The player must navigate collapsing floor segments and make precision jumps to reach weak points. The prediction errors are about spatial positioning rather than attack timing, and the spatial system is not calibrated for precision platforming; the controls, designed for combat, lack the responsiveness that platforming demands. The prediction errors are present but irresolvable through the game's normal skill development pathway. A player who has spent 40 hours perfecting their combat timing finds that none of that learning transfers to the Bed of Chaos, which demands a completely different skill set that the game has not trained. The gradient does not collapse; it resets to zero in a domain the player has no tools to navigate.

These failures confirm the model: *Dark Souls* succeeds not because it is difficult but because its difficulty is **learnable through the**

skills the game develops. When difficulty arises from sources the player cannot perceive (Blighttown's visual noise) or cannot address with the skills they have built (the Bed of Chaos's platforming), the learning gradient breaks down regardless of the overall quality of the design.

20.8 Final insight

Dark Souls is the strongest test case for the unified model because it demonstrates the principle in its most extreme form. Difficulty is not the enemy of fun. Unlearnability is. A game can kill the player hundreds of times and sustain engagement for hundreds of hours, provided that every death generates a prediction error the player can resolve through improved play. The learning gradient is sustained not by reducing difficulty but by ensuring that every source of difficulty converts into model improvement at a rate the player can perceive.

The lesson for designers: the question is never "how hard should this be?" It is "how learnable is this difficulty, and how fast can the player convert errors into improvements?"

Chapter 21 - Tetris: The Infinite Simple Gradient

On the simplest possible infinite learning gradient; speed scaling as automatic difficulty adjustment; and why forty years of engagement emerges from the interaction of seven shapes and a single rule.

21.1 Purity of design

Tetris is the purest test case for the unified model because it eliminates every variable except the core prediction-error loop. There are no enemies, no narrative, no exploration, no strategic depth beyond the immediate placement decision. The game is a single system: seven distinct tetromino shapes fall from the top of a 10-wide, 20-tall grid. The player can rotate and translate each piece before it locks into place. Completed rows are cleared. The game ends when the stack reaches the top.

From this minimal ruleset emerges a learning gradient that has sustained engagement across forty years, hundreds of millions of players, and dozens of platform iterations. The unified model explains how: *Tetris*'s design generates prediction errors at every level of the learning gradient simultaneously, and its speed-scaling mechanism ensures that the gradient never collapses to zero.

21.2 The decision space

The simplicity of *Tetris*'s rules disguises the depth of its decision space. Each piece placement involves a combinatorial evaluation: given the current stack topology, the current piece, and (in modern

versions) the preview of upcoming pieces, where should this piece go?

For a novice, this decision is reactive and local. The player places each piece to fill obvious gaps without considering future implications. The prediction errors are spatial: "I thought this piece would fit there, but it didn't" or "I cleared a row I wasn't expecting to clear." These errors resolve quickly because the rules are simple and the feedback is immediate; the piece either fits or it doesn't, and the result is visible instantly.

For an intermediate player, the decision becomes anticipatory. The player maintains a mental model of the stack's topology and plans placements two or three pieces ahead using the preview queue. The prediction errors shift from spatial (will it fit?) to strategic (is this placement optimal given what's coming?). T-spins, a technique where the T-shaped tetromino is rotated into a gap after it would normally lock, introduce a layer of motor-strategic prediction error: the player must execute a precise input sequence within a tight timing window to achieve a higher-scoring line clear.

For an expert, the decision space extends to long-term stack management. Competitive Tetris players maintain specific stack shapes (flat tops, wells for Tetris line clears) that maximise scoring opportunities several pieces in advance. The prediction errors at this level are about strategic planning under uncertainty: the piece sequence is random, and the player must build a stack that accommodates any possible sequence while maintaining the option for high-scoring clears. This is analytic complexity (Costikyan's type 5) operating at the highest level the game supports.

21.3 Speed scaling as automatic gradient management

Tetris's most elegant design feature is that its difficulty adjustment is built into the core loop rather than layered on top of it. As the

player clears rows, the game level increases and the fall speed accelerates. This creates an automatic, implicit DDA system that tracks the player's improving skill: better play produces faster levels, which demand better play.

The mechanism is simple but its implications for the learning gradient are profound. At low speed, the player has ample time to evaluate placements, and the dominant prediction errors are strategic (where should I put this piece?). As speed increases, the time available for evaluation shrinks, and the dominant errors shift from strategic to motor (can I rotate and translate this piece to the correct position before it locks?). The game smoothly transitions from a System 2 puzzle (deliberate evaluation of placement options) to a System 1 challenge (automatic execution of placement under time pressure) as the player improves. The speed curve ensures that the player's cortical-to-subcortical transfer is always slightly behind the game's demands, maintaining a positive learning gradient indefinitely.

The kill screen in classic NES Tetris (level 29, where the fall speed exceeds the standard DAS autorepeat rate, making it physically impossible to move pieces to the sides of the board using standard techniques) created an absolute ceiling that defined competitive play for three decades. In 2024, a 13-year-old player named Willis Gibson became the first person to crash the game by reaching level 157, using a technique called "rolling" that bypasses the DAS limitation by vibrating the fingers across the bottom of the controller to generate faster inputs than the standard thumb method. The rolling technique is itself a learning gradient phenomenon: it required the competitive community to discover, develop, and master an entirely new motor schema that the game's designers never anticipated. The prediction error landscape extended beyond the game's intended design into the physical interface, demonstrating that learning gradients can emerge from any aspect of the player-game interaction, including hardware.

21.4 Neural evidence

Haier et al.'s (1992) PET study remains the most vivid direct evidence for the cortical-to-subcortical transfer that the unified model predicts. Subjects practised Tetris daily for four to eight weeks. Performance improved seven-fold. Cortical glucose metabolic rate **decreased** despite the improvement. The brain became dramatically more efficient at the task while performing it dramatically better. This is the neural signature of automaticity: processing has migrated from metabolically expensive cortical circuits (System 2) to metabolically cheaper subcortical circuits (System 1), freeing cortical resources for other tasks.

The speed-scaling mechanism ensures that this efficiency gain is perpetually challenged. As soon as the player's cortical processing becomes efficient enough to handle the current speed, the speed increases, re-engaging cortical resources. The game is a treadmill that adjusts its speed to match the runner's pace, ensuring that the distance between current capability and current demand never closes completely.

Bavelier and Green's research programme extended this finding across two decades of work. Their overarching synthesis (Bavelier & Green, 2025, *Current Directions in Psychological Science*) proposed that the primary mechanism of game-based cognitive enhancement is not specific skill transfer but **"learning to learn"**: enhanced attentional control that allows more efficient pattern extraction in novel environments. On this account, games like Tetris do not merely train spatial reasoning; they train the brain's prediction error resolution machinery itself. Bediou et al.'s (2018, *Psychological Bulletin*) meta-analysis quantified this: cross-sectional effect $g = 0.55$, intervention effect $g = 0.34$ across 105 cross-sectional and 28 intervention studies.

21.5 Competitive Tetris and the limits of automaticity

Berry (2024, *Sage Journals*) complicated the simple automaticity narrative with an important finding: in competitive Tetris, performance was better when players remained mentally engaged and used focused attention to plan ahead, rather than operating on automatic pilot. The highest level of Tetris mastery requires maintained System 2 strategic engagement (stacking strategy, upcoming piece planning) on top of automatised System 1 perceptual-motor skills.

This finding supports the book's argument that flow is a third cognitive mode rather than pure System 1 dominance. The optimal state is not automatic execution without thought, but automatic execution of motor skills with freed cognitive resources applied to higher-order strategy. The moving boundary between System 1 and System 2 is the flow channel, and Tetris's speed curve keeps the player at that boundary by constantly pushing motor demands to the edge of automaticity while strategic demands remain firmly in System 2.

The competitive Tetris scene also demonstrates how social competition extends the learning gradient beyond what the game's mechanics alone can sustain. Head-to-head play introduces player unpredictability (Costikyan's type 3): the opponent's garbage lines arrive at unpredictable intervals, forcing the player to adapt their stack management strategy in real time. A player who has fully automated their solo play still faces fresh prediction errors in competition because the opponent's actions create a second source of uncertainty that the player's model must accommodate.

21.6 Why Tetris endures

Tetris has sustained engagement for forty years across every platform ever manufactured, from the original Electronika 60 to smartphones, VR headsets, and the Nintendo Switch. The unified model explains this endurance through a single principle: the game's speed-scaling mechanism ensures that the learning gradient

never reaches zero for any human player. There is always a faster level. There is always a more efficient placement. There is always a finer motor calibration to achieve. The seven tetrominoes and the 10×20 grid contain, in principle, an infinite learning gradient compressed into the simplest possible design.

Chapter 22 - Dota 2: The Inexhaustible Gradient

On the inexhaustible learning gradient of human opposition; the role of social prediction errors; and the meta-game as a macro-scale learning gradient that evolves on timescales of weeks and months.

22.1 Why multiplayer sustains engagement indefinitely

Dota 2 has sustained a professional competitive scene and a player base of millions for over a decade. Individual players have logged 10,000+ hours. The game receives no narrative updates and relatively few mechanical changes proportional to its total complexity. Yet engagement persists.

The unified model explains this through a single principle: **human opponents generate an inexhaustible prediction error landscape**. No matter how completely a player models the game's mechanical systems, the behaviour of nine other humans in the match introduces uncertainty that cannot be fully resolved. The learning gradient, in a competitive multiplayer game, is bounded only by the complexity of human behaviour.

22.2 The layered prediction error landscape

Dota 2 generates prediction errors across at least five simultaneous layers, each operating at a different timescale and engaging different neural circuitry.

Mechanical execution (seconds). Last-hitting creeps for gold requires clicking on an enemy unit at the precise moment its health drops below the player's damage value. The prediction error is motor-perceptual: the player must time their click to coincide with the creep's death threshold, accounting for allied creep damage, attack animation wind-up, and projectile travel time. Professional players last-hit at rates exceeding 80% in uncontested lanes; the difference between 60% and 80% represents hundreds of hours of motor-level gradient refinement.

Lane management (minutes). Creep equilibrium; the position of the front line where allied and enemy creeps meet; determines safety, farm accessibility, and gank vulnerability. The player manipulates equilibrium by selectively attacking or not attacking enemy creeps, pulling neutral camps to redirect allied creeps, and blocking creep waves at spawn. The prediction errors are about understanding the mechanical system deeply enough to predict and control creep behaviour. This is a pure model-building exercise: the underlying rules are deterministic, and a player with a perfect model can maintain optimal equilibrium indefinitely.

Tactical decision-making (minutes to tens of minutes). When to rotate between lanes, when to commit to a team fight, when to push objectives, when to take Roshan. Each decision involves predicting what the enemy team will do in response. The prediction errors are strategic: the player's model of the opponent's likely behaviour is tested against their actual behaviour, and the model is updated accordingly.

Drafting (pre-game). The hero selection phase generates strategic prediction errors before gameplay even begins. Each team alternately picks and bans heroes from a pool of 120+. The prediction errors are about team composition synergy, opponent strategy prediction, and counter-picking. Professional drafts involve recursive strategic reasoning: I pick this hero because I predict you will pick that hero, which you will pick because you predict I will pick this other hero. Coricelli and Nagel (2009, *PNAS*)

demonstrated that this kind of recursive strategic reasoning; what they called "strategic IQ"; correlates with mPFC activation during mentalising. The draft is a pure mentalising exercise dressed up as hero selection.

Meta-game evolution (weeks to months). Beyond individual matches, the strategic landscape of hero picks, item builds, and team compositions evolves continuously. Balance patches change hero statistics, adding or removing strategic options. The professional scene discovers and popularises new strategies, which filter down to public matchmaking. Counter-strategies emerge to defeat the new dominant approaches, shifting the landscape again. Each shift generates a wave of prediction errors for players whose models were calibrated to the previous meta.

22.3 Social prediction errors and the mentalising network

Multiplayer games introduce a category of prediction error absent from single-player design: **social prediction errors** about the behaviour of other humans. Gallagher, Jack, Roepstorff, and Frith (2002, *NeuroImage*, PET) demonstrated that playing rock-paper-scissors against a purported human opponent activated the anterior paracingulate cortex; a mentalising region silent during computer opponent play. Kättsyri et al. (2013b, *Cerebral Cortex*) showed that winning against a human opponent produced stronger ventral and dorsal striatal activation than winning against a computer, with ventral striatal activation correlating with self-rated pleasure.

Zhu, Mathewson, and Hsu (2012, *PNAS*) is the most directly relevant study. In a competitive "Patent Race" game, they identified two neurally dissociable learning signals: **reinforcement prediction errors** (tracked by bilateral putamen; "did my action produce a better or worse outcome than expected?") and **belief prediction errors** about opponents' strategies (tracked by

dmPFC/TPJ; "did my opponent behave as I predicted?"). Competitive games like *Dota 2* generate both types continuously, creating a learning gradient with two independent dimensions.

In a *Dota 2* match, social prediction errors include: predicting enemy rotations based on map information visible through ward placement, reading opponent item choices to anticipate their mid-game strategy, inferring which heroes the enemy will prioritise in team fights, and adapting to individual opponents' aggression levels and tendencies across the course of a match. Each of these predictions is tested against the actual behaviour of five human opponents whose strategies are adaptive, context-dependent, and never fully predictable.

22.4 The MMR system as gradient matching

Dota 2's matchmaking rating (MMR) system serves a function analogous to *Halo*'s difficulty tiers: it matches players against opponents whose skill level is close to their own, ensuring that the social prediction error density remains in the optimal range.

If a player is matched against significantly weaker opponents, their predictions are too accurate; they know what the opponent will do, and the social learning gradient is flat. If they are matched against significantly stronger opponents, their predictions are too inaccurate; the opponent's behaviour is incomprehensible, and the gradient tips into overload. The MMR system keeps the match quality in the zone where the player's model of the opponent is good enough to be useful but wrong often enough to generate productive errors.

The Elo-style rating system also creates a macro-level learning gradient through ranking itself. A player whose MMR is rising is perceiving their improvement through a numerical proxy: each rating increase represents an objectively measurable improvement in performance against calibrated opposition. When the MMR plateaus, the player knows they have reached a temporary ceiling

and must develop new skills (or new strategic understanding) to resume climbing. The number becomes a feedback signal for the meta-learning gradient; the rate at which the player's overall model is improving.

22.5 The meta-game as a macro learning gradient

The meta-game operates at the longest timescale of any learning gradient in game design: it refreshes the prediction error landscape for players who have long since exhausted the mechanical and tactical gradients.

A player who has automated their mechanical execution (micro-learning exhausted), learned standard encounter patterns (meso-learning substantially resolved), and developed sophisticated strategic frameworks (macro-learning partially complete) still faces a continuously evolving meta-game that generates novel strategic prediction errors. When a balance patch changes a hero's base damage by 3 points, the ripple effects propagate through drafting strategy, lane matchups, item builds, and team compositions. The player whose model was calibrated to the previous patch discovers that their predictions are slightly wrong in specific contexts, and the process of recalibrating generates fresh engagement.

Professional *Dota 2* history illustrates this at the highest level. The transition from the "4 protect 1" carry-focused meta of 2011-2012 to the fighting-focused meta of 2013-2014, to the "deathball" push strategies of TI4, to the comeback-mechanic-driven late-game metas of 2015-2016, and through countless subsequent evolutions; each transition generated a wave of strategic prediction errors that demanded model revision from every player in the ecosystem. The learning gradient at the meta level is sustained not by the game's designers introducing new content but by the competitive community discovering, developing, and countering strategies within the existing system.

22.6 The onboarding problem

Dota 2 also provides the clearest example of the onboarding failure mode discussed in Chapter 18. A new player faces 120+ heroes (each with four unique abilities), 200+ items (with complex interaction rules), lane mechanics, neutral camps, Roshan timing, rune spawns, ward placement, and the behaviours of nine other humans. The prediction error density is maximal across every layer simultaneously. The result is overload: the player cannot identify which errors to focus on because every aspect of the game is generating errors at maximum rate.

The game has attempted various onboarding solutions (tutorials, limited hero pools, coaching systems), but none fully resolves the fundamental problem: the game's complexity is front-loaded in a way that produces a near-vertical initial gradient for most players. This is the inverse of *Dark Souls*'s onboarding, which is steep but learnable because the prediction errors are predominantly of one type (combat) and arrive one at a time. *Dota 2*'s prediction errors arrive in five simultaneous types from the first second of the first match.

The model predicts that the players who survive *Dota 2*'s onboarding are those whose error resolution rate is high enough to extract patterns from the noise, or whose social context (playing with experienced friends who provide real-time scaffolding) reduces the effective error density to a manageable level. This prediction is consistent with the game's famously steep but survivable learning curve; the players who persist through the initial overload become the most dedicated long-term players in gaming, because the gradient they have accessed is the deepest and longest-lasting available.

22.7 Final insight

Dota 2 demonstrates that the learning gradient in competitive multiplayer is, in principle, inexhaustible. The game's mechanical complexity provides a deep initial gradient. Its combinatorial strategic depth extends the gradient through thousands of hours. And its human opponents ensure that the gradient can never reach zero, because human behaviour; adaptive, context-dependent, and endlessly variable; is a prediction error source that no model can fully resolve.

The lesson for designers: if you want engagement that lasts indefinitely, design systems where the prediction error landscape is co-generated by human opponents rather than authored entirely by the designer. Authored content has a finite learning gradient. Human behaviour does not.

Chapter 23 - Portal: The Architecture of Insight

On puzzle design as a sequence of controlled "aha" moments; the distinction between discrete and continuous learning gradients; and why a single mechanic can sustain an entire game.

23.1 One mechanic, infinite depth

Portal achieves something remarkable: sustained engagement across its entire three-hour runtime using a single core mechanic. The player can place two linked portals on flat surfaces; entering one exits the other, preserving momentum, direction, and velocity. The game introduces no additional mechanics after the first few chambers. Instead, it generates an expanding space of spatial reasoning puzzles by applying the same mechanic in increasingly complex spatial configurations.

Most games sustain their learning gradient by introducing new elements: new enemy types, new abilities, new systems. *Portal* sustains its gradient through **conceptual deepening**: each chamber requires the player to extend their spatial model of how portals interact with the existing physics. The prediction errors are not motor (the execution is simple; click to place portals, walk through them) but cognitive (the spatial reasoning is challenging; where must the portals be placed to reach the exit?). The gradient is maintained because the combinatorial space of portal interactions with gravity, momentum, and multi-room geometry far exceeds what any three-hour game can exhaust.

Shute, Ventura, and Ke (2015, *Computers & Education*; n=77) provided direct empirical support in a randomised controlled experiment comparing Portal 2 and Lumosity: Portal 2 players showed significant advantages on problem solving, spatial skill, and persistence. Lumosity players showed no gains on any measure. The game's puzzle structure genuinely develops cognitive abilities, not just game-specific skills; evidence that the learning gradient produces measurable skill transfer.

23.2 The teaching progression

Portal's onboarding is a masterclass in implicit instruction. Each chamber introduces exactly one concept, allows the player to discover it through interaction, and then requires them to apply it to progress. The teaching progression follows a strict one-insight-per-chamber pacing:

Chambers 00-01: Portals exist. You can see through them. You can walk through them. The player observes a portal in the wall, sees themselves through it, and walks through to the other side. No explanation is provided; the spatial logic is self-evident.

Chambers 02-04: You can create one portal. The game gives the player a portal gun that fires one colour; the other colour is pre-placed. The player learns to choose where their portal goes while the destination is fixed. The prediction errors are about placement: where on the wall will produce a useful connection?

Chambers 05-07: Momentum is preserved through portals. The player must fall from a height into a floor portal and exit a wall portal with the momentum of the fall, launching them across a gap. This is the first concept that violates intuitive physics: falling downward results in moving horizontally. The prediction error is cognitive; the player's model of "what happens when I go through a portal" must be updated to include momentum conservation.

Chambers 08-11: You can create both portals. The full mechanic is now available. The player must evaluate the entire room, identify which two surfaces need to be connected, and place portals accordingly. The prediction error density increases because the solution space has expanded from "where should I place one portal?" to "which two surfaces should be linked?"

Chambers 12-18: Environmental hazards and timing. Turrets, energy balls, and moving platforms introduce elements that must be navigated while solving the portal puzzle. The prediction errors now have a temporal dimension: the player must not only determine the correct portal placement but execute it within timing constraints.

Chamber 19 and the escape sequence: The environment changes from clinical test chambers to behind-the-scenes industrial spaces. The portal mechanic does not change, but the context does: surfaces are irregular, ceilings are high, and the spatial reasoning problems are embedded in naturalistic rather than designed geometry. The prediction error shifts from "what is the puzzle?" to "where is the puzzle?" The player must identify which surfaces in a cluttered environment are portal-compatible and construct their own solutions in spaces not obviously designed as puzzles.

23.3 The discrete learning gradient

Portal's learning gradient has a distinctive shape that differs fundamentally from combat games: it is **discrete rather than continuous**.

In *Halo* or *Dark Souls*, the player improves incrementally. Each attempt is slightly better than the last; the dodge is timed more precisely, the aim is marginally more accurate, the positioning is fractionally improved. The learning gradient is a smooth upward slope.

In *Portal*, improvement is discontinuous. The player stares at a chamber, tries various portal placements, fails, stares more, and

then suddenly sees the solution. The transition from "I have no idea" to "I see it" is a step function, not a ramp. Before the insight, the player's model of the chamber is incomplete and their prediction errors are maximal. After the insight, the model is complete and the prediction errors drop to zero. The "aha" moment is the entire gradient compressed into a single cognitive event.

Van de Cruys's framework explains why this moment produces such intense positive affect. During the stagnation period before the insight, prediction errors are large and apparently irresolvable. The brain predicts a low rate of error resolution; the expectation is that the current confusion will persist. When the insight arrives, the prediction error collapses to zero instantaneously. The rate of error reduction spikes far above the brain's expectation. The gap between expected resolution rate (low) and actual resolution rate (maximal) produces the amplified positive valence that Van de Cruys identifies as the processing signature of both humour and the "aha" experience.

This is why puzzle games feel different from action games at the phenomenological level. The pleasure in combat comes from continuous, incremental improvement; a steady positive gradient producing sustained mild-to-moderate positive affect. The pleasure in puzzles comes from sudden resolution after extended confusion; a spike that punctuates long periods of near-zero gradient, producing brief but intense positive affect. Both are explained by the same mechanism (valence tracks the rate of error reduction) operating at different timescales and with different dynamics.

23.4 GLaDOS and the narrative prediction error layer

Portal layers a narrative learning gradient on top of the puzzle gradient through its AI antagonist, GLaDOS. The narrative operates through a progressive violation of the player's model of the game's context.

Initially, GLaDOS is a helpful but slightly odd testing supervisor, providing instructions and encouragement. The player's model is "I am a test subject completing approved tests." Gradually, GLaDOS's dialogue introduces dissonant information: references to "the android hell" that awaits disobedient test subjects, conspicuous surveillance, suspiciously enthusiastic reassurances about safety, and the famous promise of cake as a reward for completing the tests.

Each piece of dissonant dialogue generates a narrative prediction error: the player's model of "helpful supervisor" is violated by evidence of something more sinister. These errors are small individually but cumulative in effect, building toward the model-shattering reveal that GLaDOS intends to kill the player at the end of the tests. The escape sequence (Chamber 19 onward) is the moment when the narrative prediction errors, accumulated over two hours of play, resolve into a new model: "I am a prisoner of a homicidal AI, and I must escape."

This narrative gradient operates in parallel with the puzzle gradient without interfering with it. The dialogue plays during puzzle-solving, and the narrative information is processed by different cognitive systems (language comprehension, social inference, narrative modelling) than the spatial reasoning required for portal puzzles. The player is simultaneously building a model of the puzzle (spatial-cognitive) and a model of the situation (narrative-social), and the two gradients reinforce each other: solving a puzzle advances the player toward the next narrative revelation, and narrative tension motivates continued puzzle-solving.

23.5 Portal 2: extending the gradient

Portal 2 faced the design challenge of extending a three-hour game into an eight-hour sequel without introducing mechanics that would dilute the core portal concept. Its solution was threefold.

First, three gel types (Repulsion Gel for bouncing, Propulsion Gel for speed, Conversion Gel for creating portal-compatible surfaces) expanded the combinatorial space without replacing the portal mechanic. Each gel interacts with portals in specific ways, generating new prediction errors about how the portal-gel combination behaves. The gels are layered on top of the existing mechanic rather than substituting for it, preserving the knowledge the player built in the first game while adding new sources of uncertainty.

Second, the cooperative campaign introduced social prediction errors. Two players must coordinate portal placement to solve puzzles that require four simultaneous portals. The prediction errors are partly spatial (where should all four portals go?) and partly social (does my partner understand the plan? will they place their portal correctly? when should I execute my part?). The social coordination layer generates prediction errors that are qualitatively different from the solo puzzles and that sustain the gradient through the full cooperative campaign.

Third, the single-player campaign's pacing alternated between puzzle chambers and narrative-exploration sequences (traversing the ruins of Aperture Science, listening to Cave Johnson's prerecorded messages). These exploration sequences reduced the puzzle gradient to zero temporarily, allowing the player's spatial reasoning to rest while the narrative gradient sustained engagement. The oscillation between puzzle-focus (System 2 spatial reasoning, phasic LC-NE mode) and narrative-exploration (curiosity-driven discovery, tonic LC-NE mode) prevents either mode from exhausting the player.

23.6 Final insight

Portal demonstrates that a single mechanic can sustain an entire game's learning gradient if the conceptual space it generates is deep enough. The game's design is a proof of concept for the

principle that learning depth, not content volume, determines engagement duration. Three hours of unique prediction errors, each requiring genuine spatial insight to resolve, produce stronger engagement than sixty hours of repetitive content with a flat gradient.

The lesson for designers: depth is not the same as breadth. One mechanic that generates hundreds of unique spatial configurations is more engaging than twenty mechanics that each generate ten. The learning gradient is sustained by the depth of the interaction space, not by the number of elements in the system.

Chapter 24 - Halo: The Architecture of Combat Flow

24.1 Why Halo matters

Few games demonstrate the principles developed in this book as cleanly as *Halo*. Its combat mechanics are deceptively simple: carry two weapons, throw grenades without switching, punch things that get close, hide behind cover until your shields recharge. Compared to the weapon-wheel arsenals of *Doom* and *Half-Life*, the complex character builds of *Diablo*, or the hundred-hero roster of *Dota 2*, Halo's combat sandbox looks almost minimalist. Yet this simplicity produces sustained engagement across difficulty tiers, supports a multiplayer ecosystem that peaked at over a billion matches in *Halo 3* alone, and generated one of the most culturally significant franchises in gaming history.

The reason, analysed through the framework of this book, is that Halo's design is a nearly optimal implementation of learning gradient management. The game's combat sandbox generates prediction errors at precisely the right density and resolution for the player to reduce them at a sustained, positive rate across multiple timescales simultaneously. Every major design decision, from the two-weapon limit to the shield recharge system to the structure of the Covenant enemy hierarchy, can be understood as a mechanism for regulating the player's learning rate: keeping it high enough to sustain engagement, low enough to prevent overload, and varied enough to prevent the gradient from flattening into grinding.

This chapter traces that architecture through the first four *Halo* games (Combat Evolved, 2, 3, and Reach), in both single-player and

multiplayer, identifying where the gradient is maintained, where it is disrupted, and what the disruptions teach us about the conditions for flow.

24.2 The nested loop architecture

Halo's combat operates at three nested temporal scales, a structure first described by Jaime Griesemer and Chris Butcher at GDC 2002 in their talk "The Illusion of Intelligence." Griesemer later called it "a 3-second loop inside of a 30-second loop inside of a 3-minute loop that is always different, so you get a unique experience every time."

The 3-second loop encompasses individual micro-actions: aim a burst, throw a grenade, dodge incoming plasma fire, close distance for a melee. These decisions are processed at or near the automatic level; an experienced player does not consciously deliberate over whether to melee a Grunt that has closed to within arm's reach. The prediction errors at this timescale are motor-perceptual: the difference between where you aimed and where the target was, the difference between when you threw the grenade and when the Elite dodged. The learning that resolves these errors is the kind tracked by Haier's PET studies; cortical engagement decreasing as the motor mapping from thumbstick input to reticle movement becomes automatic.

The 30-second loop is the single encounter. A player enters a combat space, reads the enemy composition (three Elites, a cluster of Grunts, two Jackals holding a corridor), selects an approach (flank left using the rock formation for cover, open with a plasma grenade on the Elite Major, finish stragglers with the assault rifle), executes it, adapts as enemies react, and resolves the engagement. The prediction errors here are tactical: the difference between the player's model of how the encounter will unfold and how it actually unfolds. The Covenant's territorial AI; enemies that hold positions, take cover, flank, and retreat rather than simply rushing the player; creates readable but non-trivial tactical situations where the

player's predictions are frequently wrong in informative ways.

The 3-minute loop encompasses the full progression through a combat space: the initial encounter, reinforcement waves arriving by Phantom or Spirit dropship, shifting control of terrain as enemies fall and new threats emerge, and the eventual resolution that opens the path to the next space. The prediction errors at this scale are strategic: the player learns the overall shape of the encounter over multiple attempts or through attentive observation on a single pass.

The critical feature of this architecture is that the three loops are **simultaneously active**. At any given moment, the player is reducing motor-perceptual errors (micro), tactical errors (meso), and strategic errors (macro) concurrently. This means the total learning gradient is the sum of three independent gradients, and the probability that all three simultaneously reach zero is very low. The nested structure is what prevents the gradient from flattening; it provides multiple redundant sources of learning gradient operating at different timescales and levels of cognitive abstraction.

24.3 The Golden Triangle: constraint as gradient stabiliser

Halo's core combat framework - internally called **the Golden Triangle** - consists of three actions, each mapped to a dedicated input, each available at all times without interrupting the others: weapons (right trigger), grenades (left trigger), and melee (face button). In 2001, this was radical. In *Doom* and *Quake*, grenades were simply another weapon requiring a full equipment switch. In Halo, the player can throw a grenade while their weapon remains ready and punch an enemy that closes range without switching to a melee weapon. Three combat verbs, always accessible, zero switching cost.

The design effect is combinatorial. Every encounter can be approached through multiple valid sequences: soften a group with a

frag grenade, then finish with rifle fire; overcharge the plasma pistol to strip an Elite's shield, switch to the magnum for a headshot; close range on a Hunter and melee its exposed back. The triangle generates **combinatorial prediction errors**: the player learns not just how each verb works in isolation but how they interact in sequence, which combinations are efficient against which enemy types, and when to shift between them mid-encounter.

The two-weapon limit amplifies this effect. Where *Doom* lets the player carry every weapon simultaneously, Halo forces a binary choice that must be re-evaluated constantly. The weapon sandbox is designed around the **plasma/ballistic dichotomy**: plasma weapons (Covenant origin) strip energy shields efficiently; ballistic weapons (human origin) damage health effectively. A player carrying only ballistic weapons will waste ammunition on shielded Elites. A player carrying only plasma weapons will struggle to finish unshielded targets at range. The constraint stabilises the learning gradient by preventing a dominant strategy from collapsing the decision space.

24.4 The Covenant as a learning system

Halo's Covenant enemy faction is designed as a layered curriculum. Each enemy type teaches a specific pattern; together, they compose encounters of escalating complexity where the player must combine previously learned patterns in novel configurations.

Grunts are the entry-level lesson. They are weak, numerous, and display exaggerated emotional reactions: they cheer when they land hits, scream and flee when their leader is killed, and occasionally suicide-charge with primed plasma grenades while wailing. Their behaviour is **maximally readable**; even a first-time player can extract the pattern within seconds. The prediction errors they generate are small and immediately resolvable, establishing the baseline of the learning gradient.

Jackals introduce a positional puzzle. Their arm-mounted energy shields block frontal attacks, forcing the player to learn flanking or

to target the small gap where the shield does not cover their hand and weapon. The prediction error is spatial: the player's model of "shoot the enemy" is violated by the shield; the updated model ("shoot from the side, or aim for the gap") resolves the error.

Elites are the core lesson. They have recharging energy shields (mirroring the player's own survivability model), they dodge grenades, take cover, and pursue aggressively when they have the advantage. Fighting an Elite teaches the **strip-and-finish** loop that defines Halo's combat rhythm: deplete the shield with plasma fire, then finish with a precision headshot or melee. This two-phase engagement pattern generates persistent prediction errors because Elite behaviour is variable; they dodge in different directions, retreat at different health thresholds, and occasionally surprise the player with a flanking manoeuvre.

Hunters teach positioning and timing. They are armoured everywhere except a small orange patch on their back. Their melee attack is lethal but slow and telegraphed. The prediction error is binary and dramatic: frontal assault fails completely (massive negative feedback), circling to the exposed weak point succeeds rapidly (massive positive feedback).

The power of the Covenant as a learning system lies not in any individual enemy type but in their **combinatorial composition**. An encounter with three Grunts is trivial. An encounter with three Grunts and an Elite is substantially harder because the Grunts occupy attention while the Elite flanks. Add Jackals and the player must also solve the positional puzzle. Each new combination generates novel prediction errors even though the component patterns are familiar. The learning gradient remains positive because the combinatorial space is much larger than the number of enemy types.

Damian Isla, Bungie's AI programmer, framed this through what he called "primal games" at the Develop Conference: the AI plays games of hide and seek, tag, and king of the hill with the player. "It's evolution that taught us these primal games. They're the ones that

are played with our reptilian brains." The prediction errors are ancient, rooted in movement patterns the human brain has been solving since the Pleistocene.

24.5 Shields and the recovery principle

Halo's recharging shield system is perhaps the single most important design decision in the franchise, and its contribution to learning gradient management cannot be overstated.

In pre-Halo shooters, the player had a static health pool depleted by enemy damage and restored only by finding health packs. This creates two flow-breaking problems. First, a player at low health enters the **anxiety zone**: the challenge-skill ratio is catastrophically skewed because any mistake is fatal, and the error signal is noise ("I died because I was at 12% health") rather than information ("I died because I misjudged the Elite's dodge pattern"). Second, the search for health packs introduces **goal displacement**: the player's objective shifts from engaging with combat (where the learning gradient lives) to navigating the environment looking for green boxes (where no prediction errors worth reducing exist).

Halo's shields solve both problems simultaneously. After approximately five seconds without taking damage, shields begin regenerating to full. This ensures that every encounter begins with the player at or near maximum defensive capacity, which means the challenge-skill ratio is reset to its designed value before each engagement. Every death teaches something about the current encounter rather than reflecting residual damage from a previous one. The error signal is clean.

The rhythm this creates; engage, take damage, retreat to cover, wait for shields to recharge, re-engage from a new position; maps directly onto the flow channel. Challenge rises above skill temporarily (shield-down vulnerability creates urgency), then settles back to equilibrium (shield recovery restores full capability), sustaining the prediction error cycle without allowing it to spike into

anxiety or collapse into boredom.

In multiplayer, the relatively long kill time (compared to tactical shooters like *Counter-Strike* where death is near-instantaneous) means that engagements are **extended interactions** rather than reflex tests. Each encounter generates rich, multidimensional error signals across multiple decisions, supporting a deeper and longer-lasting learning gradient.

24.6 Feedback clarity and the legibility of error

A prediction error is only useful for learning if the player can identify what went wrong. This is the principle of **feedback legibility**, and Halo implements it with unusual thoroughness.

Visual feedback operates at multiple layers. The shield indicator shows remaining defensive capacity in real time. The reticle changes colour when aimed at an enemy within effective range. Enemy shields flash and flare when hit. Headshot kills produce a distinctive visual pop. All of these operate at the pre-attentive perceptual level; the player absorbs them automatically through the visual system.

Audio feedback is equally precise. Martin O'Donnell designed the audio system so that the player could, in principle, play effectively with their eyes closed; every combat-relevant state change has a corresponding audio cue. Shields produce a distinct alarm sound when depleted and a rising tone when recharging. Each weapon has a distinctive report. Enemy vocalisations communicate state: Grunts cheer, panic, or issue warnings; Elites growl when aggressive and bark orders to subordinates.

Behavioural feedback is the most important layer. Covenant enemies display their internal state through animation and vocalisation: an Elite whose shields have been stripped staggers and stumbles - a Grunt whose leader has been killed drops its weapon and flees screaming - a Jackal that has been flanked turns in obvious

surprise. These reactions are readable confirmation that the player's strategy is working.

This is why the Flood, introduced in *Halo CE*'s seventh mission, degrade the learning gradient so dramatically. Flood combat forms display no readable state changes: they do not flinch, do not take cover, do not communicate, and do not respond to the player's tactical decisions in ways that provide informative feedback. The prediction error landscape collapses from a rich, multi-dimensional space to a single dimension (can I kill them before they reach me?). The learning gradient flattens accordingly.

24.7 The Silent Cartographer: a case study in gradient composition

Halo CE's fourth mission, The Silent Cartographer, is widely considered the best level in the series. Analysed through the learning gradient framework, its excellence comes from the way it **stacks multiple gradient types** across a continuously evolving play space.

The level begins with a beach assault. The opening encounter is spatially open; the player learns how the Warthog handles on sand, how to coordinate with Marine passengers, and how to read an open battlefield. After clearing the beach, the player drives counterclockwise around the island, encountering progressively harder Covenant forces. The game locks the front door of the main facility, forcing a detour through a secondary facility; a structural decision that extends the exploration gradient and introduces a navigation puzzle layered on top of the combat gradient.

What makes this sequence exceptional is the continuous variation of combat context. The beach assault is a vehicle encounter in open terrain. The approach to the secondary facility is on-foot infantry combat through narrow paths. The interior introduces the first Hunters, demanding the flanking pattern in close quarters. The

return features a Covenant counterattack with reinforcements arriving by dropship. The descent to the map room shifts to vertical combat in enclosed, multi-level Forerunner architecture. The return to the surface introduces stealth Elites with active camouflage. Each segment shifts the dominant prediction error type - vehicular → infantry → close-quarters boss → defensive → vertical → stealth - so the gradient never flattens even though the underlying combat mechanics remain constant.

24.8 The Library: a case study in gradient collapse

The Library, *Halo CE*'s eighth mission, is the most widely criticised level in the series, and it functions as a nearly perfect negative case study. Every principle that The Silent Cartographer exemplifies, The Library violates.

The level consists of four virtually identical floors of narrow, repetitive corridors. The environmental prediction errors are zero: every room looks like every other room. The enemy composition is exclusively Flood for the entire approximately 30-minute duration. The tactical learning gradient collapses because there is no variation in enemy behaviour. The optimal strategy (backpedal, fire shotgun, repeat) is apparent within the first minute and does not evolve. The weapon sandbox narrows correspondingly: the shotgun is overwhelmingly dominant, reducing weapon selection prediction errors to zero.

The Library demonstrates that **the learning gradient is the primary determinant of engagement**, independent of difficulty. The level is challenging, sometimes extremely so on Heroic and Legendary. But the challenge is not learnable in the sense that matters: it does not offer a positive gradient of improving player models. It is difficult in the way that endurance is difficult; it tests stamina rather than skill acquisition.

24.9 Multiplayer: the learning gradient that never ends

Halo's multiplayer extends the learning gradient into a domain where the prediction error landscape is, in principle, inexhaustible: competition against other human beings.

Equal starts (every player spawns with the same weapons in the same condition) ensure the learning gradient is determined entirely by skill rather than by character selection. **Power weapon placement** introduces a spatial learning gradient layered on top of the combat gradient: learning where weapons spawn, when they respawn, and how to time rotations creates strategic prediction errors that persist for dozens of hours. **Map design** generates spatial prediction errors through the balance between navigability and complexity.

The **skill gap** in Halo multiplayer has been analysed across all four Bungie-era titles. In *Halo CE*, the M6D pistol's three-shot-headshot kill time created a massive gulf between players who could consistently land headshots and those who could not. The prediction error at the top of this gradient is extraordinarily fine-grained: the difference between a professional and a merely good player comes down to fractions of a second in target acquisition. This is the motor-level learning gradient at its most extended.

Tracing the multiplayer learning gradient across four titles reveals a consistent pattern: the gradient deepens when design decisions increase the combinatorial richness of encounters, and it shallows when decisions reduce encounter variance to knowledge checks. *Halo 3* achieved the best balance, with the equipment system adding combinatorial depth, the weapon sandbox at its most complete (26 weapons, 11 vehicles, 11 equipment items), and the Forge/Theatre ecosystem extending the gradient into creative and analytical domains. The game held the top of Xbox Live's most-played list for years and accumulated over one billion matches.

24.10 Design lessons from the franchise arc

Halo's design history across four games provides a practical set of principles derivable from the learning gradient framework.

Simplicity enables depth. The Golden Triangle's three verbs generate more tactical depth than a complex ability tree because their interactions are discoverable, learnable, and never fully exhausted. Design for combinatorial richness through the interaction of simple elements rather than for surface complexity through the enumeration of many elements.

Constraints improve learning. The two-weapon limit forces continuous re-evaluation of loadout, preventing a dominant strategy from collapsing the decision space.

Feedback must be immediate, graded, and legible. The Covenant's readable behaviour transforms every combat interaction into an informative prediction error. The Flood's unreadable rush demonstrates that without legible feedback, the learning gradient collapses regardless of difficulty.

Recovery mechanics stabilise the gradient. Shield regeneration resets the challenge-skill ratio between encounters, ensuring clean error signals.

The nested loop structure is the key to gradient longevity. By generating prediction errors simultaneously at motor, tactical, and strategic timescales, the probability that all layers reach zero simultaneously is kept very low.

Vehicle transitions are pacing devices. Each shift from on-foot to vehicular combat resets the motor-layer gradient while maintaining the tactical-layer gradient.

Music manages arousal subliminally. O'Donnell's dynamic audio system regulates physiological arousal without becoming a competing source of prediction errors.

Player-directed difficulty selection preserves agency. Halo's four-tier system lets the player choose their learning gradient steepness without hidden manipulation.

24.11 Final insight

Halo is not great because it is exciting. Excitement is a by-product. Halo is great because its combat sandbox generates prediction errors at the right density, resolution, and variety across multiple simultaneous timescales, and its feedback systems ensure that every error the player encounters is legible, informative, and resolvable through improved play.

The series' decline under 343 Industries can be traced through the same framework. Promethean enemies reduced feedback legibility. Sprint disrupted the temporal calibration of encounter pacing. Loadouts shifted multiplayer prediction errors from skill-based to knowledge-based. The open world in *Halo Infinite* broke the back-to-back encounter stacking that sustained the 3-minute loop.

Halo's combat was never about making the player feel powerful. It was about making the player feel like they were **getting better**, continuously, at exactly the right rate. The lesson is the same one that echoes through every chapter of this book: **fun is not a property of the game. It is a property of the rate at which the player's brain is reducing uncertainty about the game.** Halo understood this before the theory existed to explain it.

Chapter 25 - The Legend of Zelda: Curiosity, Exploration, and Player-Driven Learning

On curiosity as a primary engagement mechanism; how environmental legibility replaces explicit direction; and why Breath of the Wild represents the purest implementation of the player-directed learning gradient.

25.1 A different kind of game

Where *Halo* is about combat flow and *Dark Souls* is about the conversion chamber, *The Legend of Zelda* - particularly *Breath of the Wild* and *Tears of the Kingdom* - is about **curiosity-driven learning**. There are fewer tightly controlled encounters, fewer explicit instructions, fewer enforced sequences. And yet, these games produce some of the strongest and most sustained engagement in modern game design.

This raises a question the unified model must answer: how do you maintain a learning gradient in a system that refuses to guide the player directly? The answer reveals that the learning gradient is not something the designer imposes; it is something the player co-constructs, and the designer's role is to create a world where every direction the player turns, something worth learning is waiting.

25.2 The core loop: curiosity, discovery, understanding

Zelda replaces the traditional action-game loop (encounter → fight → reward → next encounter) with a different structure:

1. The player notices something unusual in the environment
2. They investigate
3. They experiment with the game's systems
4. They discover a rule or solve a puzzle
5. That understanding applies elsewhere in the world

This loop is not imposed by the game. It is **generated by the player's curiosity**. The brain's mesolimbic curiosity circuits (Gruber, Gelman, & Ranganath, 2014, *Neuron*) transform information gaps into intrinsic rewards; the player approaches the unusual landmark or interacts with the unfamiliar object because the anticipation of discovery is itself rewarding. Gruber et al. demonstrated that high-curiosity states enhance memory not only for the target information but for entirely incidental material encountered during the curiosity period. A *Breath of the Wild* player in a state of curiosity - wondering what is behind the next ridge, what a new Sheikah structure does, how two physics objects interact - is in a neurochemically enhanced state for learning of all kinds.

25.3 The Great Plateau: onboarding through freedom

The Great Plateau, *Breath of the Wild*'s opening area, is a masterclass in open-world onboarding that solves the overload problem without resorting to tutorials.

The player wakes in a sealed room. They pick up basic clothes and a tablet-like device (the Sheikah Slate). They exit the cave onto a plateau overlooking the entire game world. From this vantage point, every biome is visible: mountains, forests, deserts, volcanic regions, a distant castle at the centre. The visual composition is not arbitrary; Hidemaro Fujibayashi and Satoru Takizawa described at

CEDEC 2017 how the team used the "**triangle rule**" as a fundamental design principle: triangular shapes (peaks, roofs, towers) are visible from great distances and naturally draw the eye, creating points of interest that guide the player without explicit markers.

The Plateau contains four shrines, each teaching one of the Sheikah Slate's rune abilities: Magnesis (manipulating metal objects), Stasis (freezing objects and storing kinetic energy), Cryonis (creating ice pillars on water), and Remote Bombs. Each shrine isolates a single concept, provides a controlled environment for experimentation, and rewards completion with a Spirit Orb. The teaching is entirely environmental; no tooltip explains that Magnesis can be used to pull metal treasure chests from underwater, or that Stasis can freeze a boulder and accumulate kinetic energy for a catapult effect. The player discovers these applications through experimentation, and each discovery is a prediction error resolved: "I didn't know I could do that, but now I understand the rule."

The Plateau is bounded by a cliff that prevents exploration of the wider world until the four shrines are completed. This is the game's only linear constraint, and it serves a precise learning gradient function: it ensures that every player has the four core rune abilities before entering the open world, guaranteeing a minimum model complexity that prevents the open-world phase from producing overload. Once the player paraglides off the Plateau, they are free to go anywhere, in any order, including directly to the final boss. The game trusts that the Great Plateau has provided sufficient foundational learning for the player to construct their own gradient from that point forward.

25.4 The physics chemistry system

Breath of the Wild's most important innovation is not its open world but its **physics chemistry system**: a set of interacting physical rules that produce emergent behaviour.

Fire spreads to grass and wood. Metal conducts electricity. Ice melts near heat. Wind carries sound and fire. Temperature affects the player's survival. Rain makes surfaces slippery. Every material in the world has physical properties that interact according to consistent rules, and these rules produce combinatorial outcomes that no designer explicitly programmed.

The learning gradient generated by this system is enormous because the player discovers rules through experimentation rather than instruction. Consider the fire system alone:

- The player discovers that striking flint near wood creates fire
- They discover that fire spreads to grass, creating updrafts
- They discover that the paraglider catches updrafts, providing vertical mobility
- They discover that fire arrows shot into grass create the same effect at range
- They discover that setting grass on fire near enemies damages them
- They discover that the updraft from burning grass can be used to reach otherwise inaccessible locations

Each of these discoveries is a prediction error resolved: the player's model of "what fire does in this world" expands with each interaction. And because the rules are consistent (fire always spreads to flammable materials, always creates updrafts, always damages), the learning transfers: a rule discovered in one location applies everywhere. This is the opposite of hand-crafted puzzle design, where each puzzle has its own rules that apply nowhere else. The physics chemistry system generates a learning gradient that compounds; each new rule multiplies with all previously learned rules to expand the space of possible interactions.

The system also supports multiple valid solutions to any given problem. A camp of enemies on a wooden platform can be approached by: sneaking past (stealth), fighting directly (combat), setting the platform on fire (environmental), rolling boulders

downhill (physics), dropping items from above (aerial), or simply ignoring them (exploration). Each approach uses different combinations of learned rules, and the player's choice of approach is itself a learning gradient decision: which of my learned rules do I want to test in this context?

25.5 Shrine design as controlled learning

The 120 shrines in *Breath of the Wild* (152 in *Tears of the Kingdom*) function as **controlled learning environments** embedded within the open world. Each shrine isolates a concept, provides a safe space for experimentation, and allows the player to test their understanding before returning to the open world where that understanding applies.

Shrines fall into several categories that generate different prediction error profiles:

Combat shrines (Test of Strength) pit the player against a Guardian Scout at one of three difficulty tiers. These are pure conversion chambers in the *Dark Souls* sense: the enemy's attack patterns are learnable through repetition, and the player's model improves with each attempt. The increasing difficulty tiers (Minor, Modest, Major) create a cross-shrine combat gradient.

Physics puzzle shrines require the player to use rune abilities and physics interactions to navigate an obstacle course. These generate discrete "aha" moments like *Portal*'s chambers: the player studies the layout, experiments with different rune applications, and eventually sees the solution. The prediction error collapses in a single cognitive event.

Apparatus shrines use motion controls to manipulate platforms and mazes, generating motor-perceptual prediction errors about how the physical controller maps to the in-game object. These are the most controversial shrines because the motion control system introduces a control uncertainty (Costikyan's type 10) that some

players find frustrating.

Blessing shrines are rewards for solving an overworld puzzle (finding the shrine's entrance was the puzzle). These generate zero prediction errors within the shrine itself; the gradient was entirely in the discovery.

The shrine system creates a rhythm of exploration and focus that maps onto the LC-NE phasic/tonic oscillation described in Chapter 16. Traversal between shrines is exploratory (tonic mode): the player scans the environment, notices landmarks, and investigates. Shrine puzzles are focused (phasic mode): attention narrows to a contained challenge with immediate feedback. The oscillation prevents either mode from exhausting the player.

25.6 Environmental legibility replaces markers

Breath of the Wild's most consequential design decision is its refusal to use traditional open-world navigation markers. There are no quest icons on the map, no minimap waypoints, no percentage completion counters for regions, and no quest log directing the player to specific locations.

Instead, the game relies on **environmental legibility**: the world itself communicates where interesting things are. The triangle rule ensures that peaks and towers are visible from great distances. Unusual terrain features (a ring of mushrooms, a suspiciously arranged set of rocks, a lone tree on a hilltop) signal Korok seed puzzles. Columns of smoke indicate enemy camps. Shrine pedestals glow orange against the landscape. The Sheikah Tower network provides elevated vantage points from which the player can survey the surrounding terrain and identify points of interest visually.

This design exploits the same perceptual-attentional system that the Bavelier lab has shown is enhanced by game play. The player's attentional system does the "quest marker" work: scanning the environment for anomalies, filtering relevant signals from irrelevant

noise, and directing movement toward points of expected information gain. The exploration process itself engages active attention rather than passive waypoint-following. In the terms of the undermining effect (Deci, Koestner, & Ryan, 1999; meta-analysis $k = 128$, tangible-reward effect $d = -0.40$), the absence of extrinsic markers preserves the intrinsic curiosity that drives engagement.

The contrast with Ubisoft's open-world formula is instructive. *Assassin's Creed*, *Far Cry*, and similar titles use map towers that reveal every collectible, quest, and point of interest in a region. This reduces uncertainty immediately and completely: the player knows exactly where everything is. The remaining engagement is task-completion (go to the marked location, do the marked thing). *Breath of the Wild* keeps uncertainty high by never telling the player where things are, ensuring that exploration continuously generates prediction errors about what lies beyond the next ridge. Deterding, Andersen, Kiverstein, and Miller (2022, *Frontiers in Psychology*) tested this directly across three large game datasets totalling approximately 1.8 million player votes: people preferred levels of intermediate difficulty and were motivated by success, consistent with the predictive processing account of play.

25.7 Weapon durability as forced prediction error

The weapon durability system, the game's most controversial mechanic, serves a specific learning gradient function: it **prevents the dominant-strategy collapse that would flatten the gradient**.

Weapons break after a limited number of uses. The player cannot hoard a single powerful weapon and rely on it for the entire game. Instead, they must continuously scavenge, adapt, and make do with whatever is available. This forces three types of prediction error that would otherwise resolve to zero:

Resource management errors: Should I use my best sword on these enemies, or save it for a harder fight? The answer depends on

predictions about what the player will encounter next, which are never certain in an open world.

Combat adaptation errors: When a weapon breaks mid-fight, the player must switch to a different weapon type with different properties (range, speed, damage), generating fresh motor-level prediction errors about timing and spacing.

Exploration incentive errors: The player must continuously explore to find replacement weapons, ensuring that the exploration gradient (curiosity about what is in the next chest or enemy camp) is never allowed to decouple from the combat gradient.

Without weapon durability, a player who finds a powerful sword early would have no reason to engage with the weapon variety system, no reason to explore for replacements, and no forced adaptation in combat. The gradient across all three dimensions would flatten. Weapon durability is a constraint that prevents gradient collapse, functioning analogously to *Halo's* two-weapon limit: it removes the possibility of a dominant strategy, ensuring that the player faces meaningful prediction errors about resource management throughout the entire game.

25.8 Tears of the Kingdom: extending the gradient

Tears of the Kingdom (2023) faced the sequel problem of extending a game whose physics chemistry system had been thoroughly explored by millions of players. Its solution was to add a new category of prediction error: **construction**.

The Ultrahand ability lets the player grab, move, rotate, and attach any physical object to any other physical object. Fuse lets the player attach materials to weapons and shields, modifying their properties. Recall reverses an object's trajectory. Ascend moves the player vertically through ceilings.

These abilities transform the game's prediction error landscape from physics-discovery (how do the rules work?) to to

physics-application (what can I build with these rules?). The construction system generates prediction errors about engineering: will this bridge support my weight? Will this vehicle move? Will this flying machine generate enough lift? The combinatorial space is vastly larger than the original game's because the player is no longer limited to interacting with designer-placed objects; they can create their own objects from components.

The construction gradient is sustained by the fact that the system supports genuine engineering creativity. Players have built functioning computers, aircraft carriers, combat mechs, and vehicles that the designers could not have anticipated. Each construction project generates prediction errors about whether the design will work as imagined, and the immediate physical simulation provides feedback that is both precise (the vehicle tips over because the weight distribution is wrong) and interpretable (the player can see which component caused the failure).

25.9 Micro, meso, and macro learning

Zelda maintains a learning gradient at all three timescales simultaneously:

Micro (seconds): Climbing requires stamina management. Gliding requires reading wind currents and thermals. Combat requires timing parries and dodge-flurries. Each of these generates motor-level prediction errors that resolve through repetition.

Meso (minutes): Shrine puzzles require spatial reasoning and rule application. Enemy camps require tactical assessment. Environmental puzzles (Korok seeds) require pattern recognition. Each generates cognitive-level prediction errors that resolve through insight.

Macro (hours): Understanding the full physics chemistry system, building a mental map of Hyrule, discovering the interconnections between systems (fire creates updrafts; updrafts enable paragliding;

paragliding enables access to elevated shrines; elevated shrines contain runes that enable new physics interactions). The macro gradient is sustained by the combinatorial depth of the system interactions.

25.10 Discovery as reward

Zelda replaces traditional progression rewards (XP, loot, level-ups) with **understanding itself** as the primary reward. The reward for completing a shrine is a Spirit Orb, which contributes to health or stamina upgrades. But the actual reward; the thing that produces positive affect; is the discovery: "I figured out how this works." The Spirit Orb is a token; the insight is the dopamine.

This aligns perfectly with the unified model. Fun is the reduction of uncertainty, and Zelda makes uncertainty reduction the primary reward rather than a means to earning secondary tokens. Bromberg-Martin and Hikosaka (2009, *Neuron*) demonstrated that the brain treats information as intrinsically rewarding; monkeys will sacrifice actual juice to gain advance information about upcoming rewards. Zelda's design exploits this directly: the player explores not for what they will receive but for what they will learn.

25.11 Final insight

Breath of the Wild succeeds because it transforms the player into an active learner navigating a world where every direction contains something worth understanding. The learning gradient is not imposed by the designer through paced content delivery; it is co-constructed by the player's curiosity interacting with a world whose rules are consistent, discoverable, and endlessly combinatorial.

The lesson for designers: the most durable learning gradient is one the player builds themselves, from a world that rewards curiosity

with understanding. Do not tell the player where to go. Make everywhere worth going.

Chapter 26 - Resident Evil: Tension, Scarcity, and Controlled Overload

On fear as an arousal amplifier of the learning gradient; resource scarcity as a mechanism for making prediction errors consequential; and why horror games demonstrate that the gradient operates at any arousal level.

26.1 A different kind of engagement

Where *Halo* creates flow through action, *Zelda* through curiosity, and *Dark Souls* through conversion chambers, *Resident Evil* creates engagement through **tension and constraint**. It is designed to make the player feel vulnerable, uncertain, and cautious. By any hedonic measure, this should be aversive. Yet the franchise has sustained engagement across three decades and multiple reboots, and the survival horror genre it codified remains one of gaming's most distinctive.

The unified model must account for this. If fun is the subjective experience of reducing uncertainty at an optimal rate, why does a game designed to maximise the player's feeling of uncertainty sustain engagement?

The answer is that fear does not oppose the learning gradient. It **amplifies** it.

26.2 Fear as gradient amplifier

Andersen, Schjoedt, Price, Rosas, Scrivner, and Clasen (2020, *Psychological Science*; n=110) provided the empirical foundation in

a non-game context: enjoyment of a haunted house showed an **inverted-U relationship with fear**, with heart rate data confirming that "just-right" deviations from physiological baseline maximised enjoyment. Too little fear was boring. Too much was aversive. The optimal zone was where the threat was real enough to sharpen attention but controllable enough that the participant could process and respond to it.

This maps directly onto the Yerkes-Dodson curve: moderate arousal enhances performance and learning; excessive arousal degrades it. Survival horror games operate near the upper end of this curve. Fear narrows attention (the player is hyper-focused on threats), deepens error encoding (adrenaline enhances memory consolidation for emotionally salient events), and makes every decision feel consequential (the stakes of each prediction error are amplified by the emotional context).

The core loop of *Resident Evil* is: encounter uncertainty (enemy, sound, unfamiliar room) → experience fear (arousal spike) → make a decision under pressure (fight, flee, conserve) → resolve the uncertainty (the enemy is dead, the room is clear, the path is safe) → experience relief (arousal drops). This loop is the prediction error cycle operating at elevated arousal: the same mechanism as *Halo's* combat loop, but with the norepinephrine dial turned higher.

26.3 The Spencer Mansion as a spatial learning gradient

The original *Resident Evil* (1996) and its 2002 remake use the Spencer Mansion as a spatial puzzle that generates prediction errors through architecture, locked doors, and key items.

The mansion is a single interconnected structure, but the player's access is gated by keys, crests, and puzzle solutions. Initial exploration reveals a small fraction of the total space; each subsequent key opens new rooms and corridors that connect to

previously explored areas. The spatial learning gradient operates through progressive revelation: the player's mental map expands with each key, and the connections between areas generate spatial prediction errors (this corridor connects to the main hall? the underground connects to the guardhouse?).

This structure serves two gradient functions simultaneously. First, it sustains curiosity: the player knows there are locked doors they have not yet opened, creating information gaps that drive exploration. Second, it creates backtracking that tests the player's spatial model: returning to a previously visited area with new knowledge (a key, a puzzle solution) requires the player to navigate from memory, testing their internal map against the actual layout.

The zombie placement intensifies both functions. Enemies do not respawn in most *Resident Evil* games, but they persist in rooms the player has visited. A zombie left alive in a corridor becomes a persistent threat that the player must manage on every subsequent traversal. The decision to kill or avoid each zombie is a resource management prediction error (is this zombie worth the ammunition?), and the consequences of that decision persist across the entire game.

26.4 Resource scarcity as consequential prediction error

The defining mechanic of *Resident Evil* is **resource scarcity**: limited ammunition, limited healing herbs, limited inventory space (managed through the famous item box system). Every bullet and every herb is a finite resource whose expenditure cannot be reversed.

Scarcity transforms prediction errors from informational to **consequential**. In *Halo*, a suboptimal weapon choice costs a few seconds of reduced effectiveness; the player can switch weapons immediately. In *Resident Evil*, wasting ammunition on a

non-threatening zombie costs a resource that may be needed for a future encounter the player cannot yet anticipate. The prediction error "should I fight or avoid this enemy?" carries real stakes because the answer depends on predictions about future resource requirements that are uncertain.

This consequentiality deepens the learning gradient in three ways:

Attention is sharpened. When every bullet matters, the player attends more carefully to each encounter. Enemy behaviour is observed more closely because the cost of a missed shot is higher. This increased attention produces richer error encoding and faster model building.

Strategic depth is forced. The player must plan across encounters rather than optimising each encounter independently. Inventory management, route planning, and enemy triage (which enemies to kill, which to avoid, which to incapacitate) create a macro-level strategic gradient that operates on a timescale of hours rather than minutes.

Tension is sustained. A player with abundant resources feels safe; their arousal drops to baseline, and the fear-enhanced learning gradient collapses. A player with scarce resources feels vulnerable; their arousal remains elevated, and every encounter generates heightened prediction errors. Scarcity is the mechanism that keeps the player in the optimal zone of the Yerkes-Dodson curve throughout the game.

26.5 Mr. X and the adaptive threat gradient

The *Resident Evil 2* Remake (2019) introduced Mr. X (the Tyrant) as a persistent, unkillable threat that patrols the Raccoon City Police Department. Mr. X is the purest implementation of what Miller et al. (2024) called "controlled prediction-error environments" in horror games.

Mr. X cannot be killed. He can only be temporarily staggered. He tracks the player by sound, follows them between rooms, and appears at unpredictable intervals. The player must learn to read his audio cues (heavy footsteps that grow louder as he approaches), identify safe rooms (save rooms where he cannot enter), and plan routes that minimise exposure.

The learning gradient follows a characteristic arc. Initial encounters produce maximal prediction errors: the player does not know Mr. X's rules (Can he be killed? Where does he go? Can he follow me through doors?). Over the next hour, the model develops: the player learns his movement speed, his patrol patterns, his room-entry rules, and his audio cues. By the mid-game, the player can navigate the police station while managing Mr. X as a persistent background threat, planning routes that avoid his patrol path and using safe rooms as staging areas.

This arc; from panic to management to mastery; is a learning gradient compressed into a single mechanic. The prediction errors are not about combat execution (Mr. X cannot be defeated through combat skill) but about spatial prediction and threat management. The player who has learned to predict Mr. X's position from audio cues alone has built a model that is qualitatively different from the model they had during initial panic, and the improvement is perceivable at every stage.

26.6 Where horror gradient management fails

Horror games fail when fear overwhelms learnability. If the threat is truly unpredictable (random instant-kill events with no telegraph), the prediction errors are irresolvable and the gradient produces frustration rather than engagement. If the threat is too predictable (scripted jump scares at fixed locations), the prediction errors resolve after one encounter and the gradient collapses to zero on subsequent playthroughs.

The best horror games calibrate fear so that it is **learnable but not trivially so**. *Resident Evil*'s zombies have consistent behaviour but variable placement. Mr. X has consistent rules but adaptive patrol routes. *Alien: Isolation*'s xenomorph has AI that responds to the player's behaviour, becoming more aggressive if the player uses the same hiding strategy repeatedly. Each of these systems generates prediction errors that reduce with practice but never fully resolve, sustaining the gradient through the tension between growing competence and persistent threat.

26.7 Final insight

Resident Evil demonstrates that the learning gradient operates at any point on the arousal spectrum. Fear does not oppose fun; it amplifies the learning gradient by sharpening attention, deepening error encoding, and making every decision consequential. The model does not require low-stress engagement to produce fun. It requires learnable uncertainty, regardless of the emotional context in which that uncertainty is experienced.

Chapter 27 - Narrative Games: Prediction Error Through Story

On how narrative systems generate prediction errors through expectation violation; why Disco Elysium, Outer Wilds, and The Stanley Parable sustain engagement without motor automaticity; and the unique gradient management challenges of story-driven design.

27.1 Narrative prediction errors

Stories generate prediction errors through the same mechanism as game systems: the audience builds a model (of the characters, the world, the plot trajectory) and the story violates that model through twists, revelations, and subversions. A murder mystery generates prediction errors about the killer's identity. A character drama generates prediction errors about how relationships will evolve. A science fiction narrative generates prediction errors about the rules of its world.

But narrative prediction errors have three properties that distinguish them from mechanical ones:

They are unrepeatable. Once a twist is known, the prediction error it generated cannot be regenerated. The first time the player learns that [spoiler for any narrative game], the prediction error is massive. The second playthrough generates zero narrative prediction error at that moment. This makes narrative gradients **non-renewable** in a way that mechanical gradients are not.

They are not skill-dependent. A player does not need to improve their ability to experience narrative prediction errors; they simply need to continue playing. This means narrative gradients cannot be

calibrated through difficulty adjustment. They are paced entirely through content delivery: the rate at which the writer reveals information.

They engage different neural circuitry. Narrative prediction errors recruit mentalising networks (mPFC, TPJ), language comprehension circuits (left IFG, temporal cortex), and emotional processing systems (amygdala, insula) rather than the motor and spatial circuits that mechanical prediction errors engage. This means narrative engagement can co-exist with mechanical engagement without competing for the same cognitive resources; a player can simultaneously build a model of a boss's attack patterns (motor-spatial prediction errors) and a model of the story's meaning (narrative prediction errors).

27.2 Disco Elysium: System 2 as gameplay

Disco Elysium (2019) is unusual because it remains System 2-dominant throughout. There is no motor automaticity to develop; the game is entirely dialogue, investigation, and decision-making. Yet it produces deep engagement for 30-40 hours; longer than many action games.

The game sustains engagement through three simultaneous learning gradients:

The mystery gradient. The player is investigating a murder. Each conversation, each clue, each environmental detail updates the player's model of who killed the hanged man and why. This is a classic detective narrative gradient: the information space is large, the relevant information is distributed across dozens of characters and locations, and the player must assemble a coherent model from fragments. The gradient is sustained by the sheer density of information and the multiple competing hypotheses the evidence supports.

The world-model gradient. The city of Revachol has a complex political history (a failed communist revolution, a capitalist occupation, labor unrest, international tensions) that is not explained directly but must be inferred from conversations, books, murals, and environmental details. The player builds a model of Revachol's political landscape incrementally, and each new piece of information generates prediction errors about the larger context. This gradient operates at a longer timescale than the mystery; understanding the politics takes the full game, while murder hypotheses are generated and revised throughout.

The psychological gradient. The player character has amnesia. His 24-skill system simulates dual-process cognition within the narrative: skills like Encyclopedia, Authority, and Inland Empire function as literal System 1 voices in the player-character's head; automatic impulses and pattern-recognition outputs that interrupt deliberative dialogue. The player's System 2 must evaluate and sometimes override these System 1 "suggestions."

The skill check system creates a formal interface between processing modes: active checks (dice rolls modified by skill investment) represent the character's automatic competencies, while the player's System 2 decides whether to attempt them. Passive "black checks" occur without player agency; pure System 1 events that the player-character responds to involuntarily. The player is simultaneously building a model of the murder, a model of the world, and a model of their own character's psychology. Three gradients operating in parallel, each at a different timescale, ensure that the total learning rate never drops to zero even though none of them involves motor skill.

27.3 Outer Wilds: the epistemic gradient

Outer Wilds (2019) is the purest implementation of a purely epistemic learning gradient. The game has no combat, no upgrades, no permanent progression. The solar system resets every 22

minutes (a sun-death time loop). The only thing that changes between loops is what the player knows.

Every piece of information in the game is available from the first minute. The "progression" is entirely epistemic: the player discovers clues about the Nomai (an ancient alien civilisation), pieces together the history of the solar system, and eventually understands the mechanism of the time loop and how to resolve it. The learning gradient is maintained solely by the player's growing understanding, and the game achieved near-universal critical recognition despite having none of the conventional reward structures that most games rely on.

The design is remarkable because it demonstrates the learning gradient operating in its purest form: no mechanical skills to develop, no items to collect, no numerical progression. The prediction errors are entirely about the player's model of the world (what does this inscription mean? where does this quantum signal lead? why does this planet fragment at the 12-minute mark?), and the reward for resolving each error is understanding rather than any in-game token. Bromberg-Martin and Hikosaka (2009) demonstrated that the brain treats information as intrinsically rewarding; *Outer Wilds* is a 20-hour game built entirely on this principle.

27.4 The Stanley Parable: meta-prediction errors

The Stanley Parable (2013, expanded 2022) generates prediction errors at the meta-level: it violates the player's expectations about how **games themselves** work.

The narrator describes what the player "will" do. The player can comply or defy. Each choice leads to a different narrative branch, and the branches comment on the player's choice, on game design conventions, and on the nature of player agency. The prediction errors are not about the game's mechanical systems (which are trivially simple: walk, press buttons, open doors) but about the

player's model of what a game is and what it means to "play."

This meta-level gradient is inherently self-limiting: the insight that "this game subverts expectations" is itself a model the player builds quickly. After a few branches, the player expects subversion, and subversion-of-subversion becomes the new prediction. *The Stanley Parable* manages this through **sufficient branching depth** (the 2022 Ultra Deluxe edition contains dozens of endings) and **escalating meta-levels** (early branches subvert game conventions; later branches subvert the concept of subversion; the deepest branches subvert the player's relationship to the act of playing).

The game demonstrates that the learning gradient can operate at any level of abstraction. The prediction errors in *Mario* are about physics. The prediction errors in *Dark Souls* are about combat patterns. The prediction errors in *The Stanley Parable* are about the nature of interactive narrative. The mechanism is identical; only the domain changes.

27.5 The unique challenge of narrative gradient management

Narrative games face a design challenge that action and puzzle games do not: the gradient is **front-loaded and non-renewable**. Once the story is known, narrative prediction errors drop to zero. This limits replay value and creates a structural tension between narrative density (more story = longer gradient) and mechanical engagement (story delivery must not interrupt gameplay flow).

Games manage this tension through several strategies:

Branching narrative (*Disco Elysium*, *Mass Effect*, *Baldur's Gate 3*) creates multiple gradient paths through the same content. Each playthrough generates narrative prediction errors about how different choices affect outcomes. The gradient extends across multiple playthroughs rather than exhausting in one.

Environmental narrative (*Dark Souls*, *Elden Ring*, *BioShock*) embeds story in the environment rather than delivering it through cutscenes. Players discover narrative fragments through exploration, which means the narrative gradient is co-constructed with the exploration gradient. A player who investigates every item description in *Dark Souls* experiences a different (and deeper) narrative gradient than a player who ignores them.

Emergent narrative (*Dwarf Fortress*, *RimWorld*, *Crusader Kings III*) generates stories procedurally from system interactions. The narrative prediction errors arise from the player's own experience rather than from authored content, which makes them renewable: each playthrough generates a unique story. The gradient is sustained by the system's capacity to produce surprising narrative outcomes from rule interactions.

Integrated narrative (*The Last of Us*, *Hades*, *Portal*) weaves story into mechanical progression so that narrative and mechanical gradients advance simultaneously. Story beats arrive at transition points between mechanical challenges, and the emotional stakes of the narrative amplify the mechanical prediction errors. *Hades* is the strongest example: death (a negative mechanical prediction error) advances the narrative (a positive narrative prediction error), converting mechanical failure into narrative progress.

27.6 Final insight

Narrative games demonstrate that the learning gradient is not limited to mechanical skills. The brain builds predictive models of stories, characters, worlds, and meanings, and the process of refining those models generates the same kind of engagement that refining a motor skill or solving a puzzle does. The gradient mechanism is domain-general: it operates wherever the brain is building and improving a model, regardless of whether that model is about physics, combat, spatial reasoning, or human nature.

Chapter 28 - Idle and Mobile Games: The Ethics of the Gradient

On how micro-transactions interact with the intrinsic learning gradient - the neuroscience of "wanting" without "liking" - and the distinction between engagement and exploitation.

28.1 Engagement without learning

Idle games (*Cookie Clicker*, *Adventure Capitalist*) and many mobile games represent a challenge for the unified model: they sustain engagement without any obvious learning gradient. The player clicks (or waits) and numbers go up. There are no learnable patterns, no skill development, no prediction errors to resolve. Yet millions of players engage with these games for hundreds of hours.

The model's explanation is that these games exploit the dopamine system's responsiveness to **variable-ratio reinforcement** and **number escalation** rather than genuine prediction error reduction. The engagement they produce is "wanting" without "liking" (Berridge & Robinson, 2016): the dopamine system sustains approach behaviour (the player keeps checking the game, keeps clicking) without the hedonic satisfaction that comes from actual uncertainty reduction.

Deterding, Andersen, Kiverstein, and Miller (2022, *Frontiers in Psychology*) showed that idle games do generate micro-prediction errors through accumulation mechanics; each click or time interval produces a small, positive uncertainty about the exact outcome (how much currency will this produce? will I unlock the next threshold?). Under the prediction-error-rate framework, idle games operate at

the lowest possible prediction error magnitude but the highest possible resolution frequency; tiny errors resolved continuously. This is the opposite end of the spectrum from *Dark Souls* (massive errors resolved slowly) but mechanistically identical.

28.2 Micro-transactions and the corrupted gradient

The ethical concern with mobile and free-to-play game design is that micro-transactions can corrupt the learning gradient by substituting **purchased progress** for earned progress. When a player can buy their way past a challenge, the prediction error that the challenge would have generated is eliminated without being resolved. The player's model does not improve; they have simply removed the obstacle that would have required improvement.

Drummond and Sauer (2018, *Nature Human Behaviour*) evaluated loot boxes in 22 games against five psychological criteria for gambling and found 45% met all five. Zendle and Cairns (2018, *PLoS ONE*; n=7,422) found a significant link ($\eta^2 = 0.054$) between loot box spending and problem gambling severity, replicated in a second study (n=1,172; $\eta^2 = 0.051$). Larche et al. (2021, *Journal of Gambling Studies*; n=48/40) demonstrated that rarer loot box items triggered larger skin conductance responses and greater urge to open more boxes; physiological responses paralleling gambling reward reactivity.

The wanting/liking dissociation is the mechanism: variable-ratio reinforcement schedules in loot boxes and gacha systems maintain dopaminergic "wanting" (the compulsion to pull, to open, to check) while "liking" (actual enjoyment of the game's systems) may remain flat or decline. Singer et al. (2012) and Mascia et al. (2018, *Neuropsychopharmacology*) demonstrated that chronic variable-ratio reinforcement produces dopamine system sensitisation; the "wanting" escalates with exposure.

28.3 The ethical distinction

The unified model provides a clear ethical distinction between engagement and exploitation:

Genuine engagement occurs when the game sustains a positive learning gradient through prediction errors that the player resolves through skill development. The reward is uncertainty reduction; the player is getting better, and the improvement is intrinsically satisfying. The "wanting" and "liking" are aligned: the player desires to play because playing feels good.

Exploitation occurs when the game sustains engagement through variable-ratio reinforcement schedules that maintain "wanting" without providing genuine learning. The player is not getting better; they are being maintained in a state of compulsive approach behaviour through the manipulation of the dopamine system's uncertainty-sensitivity. The "wanting" and "liking" are dissociated: the player feels compelled to play but does not enjoy it.

This distinction is not always clean; many games contain elements of both. But the unified model provides a principled basis for evaluating where a design falls on the spectrum: does this mechanic generate prediction errors that the player resolves through improved skill, or does it generate variable rewards that sustain engagement without learning?

28.4 Design implications

The ethical path for mobile and free-to-play design is to monetise around the learning gradient rather than against it:

- Sell **cosmetics** (which do not affect the learning gradient)
- Sell **content** (new levels, new challenges, new prediction error sources)
- Sell **convenience** (quality-of-life features that do not replace learning)

- Do not sell **progress** (bypassing challenges removes the prediction errors that sustain genuine engagement)
- Do not sell **power** (purchased advantages corrupt the feedback that makes the learning gradient legible)

Games that follow these principles (*Fortnite*'s cosmetic-only model, *Path of Exile*'s stash tabs and cosmetics, *Hades*'s paid DLC) demonstrate that free-to-play monetisation is compatible with a healthy learning gradient. Games that violate them (pay-to-win mobile games, loot-box-driven progression systems) demonstrate that monetisation can corrupt the gradient into a compulsive loop that serves neither the player nor the long-term health of the product.

CLOSING CHAPTER - The Nature of Fun

The argument in full

This book began with a simple observation: every known human culture plays games. It traced that observation through biology (play is a primary emotional drive, subcortical in origin, older than the neocortex), through neuroscience (the brain is a prediction machine that generates specific chemical signals when predictions fail, that treats information as intrinsically rewarding, and that physically reorganises its circuitry as skills are acquired), through psychology (flow is a third cognitive mode in which task-relevant executive control operates through well-trained procedural pathways while metacognitive overhead is suppressed), and through design (the learning gradient - the rate at which prediction errors are generated and resolved - is the hidden variable governing the entire experiential landscape of play).

The central claim can now be stated in its most general form:

Games are engineered environments for regulating the rate at which the brain reduces uncertainty.

From this claim, the corollaries follow:

- **Fun** is the subjective experience of reducing uncertainty at a rate that matches or exceeds the brain's expected rate of progress
- **Flow** is the cognitive state that emerges when this rate is sustained and stable over time
- **Engagement collapses** when the rate reaches zero (boredom), becomes negative or incoherent (frustration), or

exceeds processing capacity (overload)

The four novel theoretical contributions of this book are:

1. **Flow is a third cognitive mode**, distinct from both System 1 automatic processing and System 2 deliberate reasoning. It is characterised by task-relevant executive control operating through well-trained procedural pathways, with metacognitive overhead suppressed. The evidence converges from Dietrich's transient hypofrontality, Weber and Huskey's synchronisation theory, Harris et al.'s finding that objective effort peaks during flow while subjective effort is minimal, and the LC-NE phasic mode as the shared neurochemical mechanism.

2. **Games and language share frontal-basal ganglia circuits for hierarchical sequential structure.** The procedural memory system that computes grammatical rules also underpins game rule learning, strategic chunking, and expert intuition. This is supported by Ullman's declarative/procedural model, Wan et al.'s demonstration of caudate activation in shogi experts, Thibault et al.'s finding of common basal ganglia substrates for tool use and syntax, and formal computational parallels between game description languages and context-free grammars.

3. **Game quality correlates with the quality of the System 2-to-System 1 conversion pipeline.** Every game is a machine for converting deliberate processing into automatic execution. The most acclaimed games introduce challenges that engage System 2 at the right rate, provide feedback that supports pattern extraction, and scale difficulty to match automaticity acquisition. The cortical-to-subcortical transfer documented by Poldrack, Lehericy, Haier, and Wan provides the neural mechanism.

4. **The learning gradient is the primary design metric.** The rate at which a player reduces prediction error relative to their expected rate of progress is the hidden variable governing engagement. Wilson et al.'s 85% rule for optimal learning, Van

de Cruys's affective error dynamics, Schmidhuber's compression progress, and Andersen et al.'s predictive processing account of play all converge on the same quantity as the hedonic signal: the first derivative of model accuracy over time.

The convergence

The strongest evidence for this framework is not any individual finding but the convergence across disciplines. Anthropology establishes that play is universal. Evolutionary biology establishes that it is functional. Affective neuroscience establishes that it is driven by a dedicated subcortical system (Panksepp's PLAY circuit). Reward neuroscience establishes that engagement tracks prediction error (Schultz) and that uncertainty itself is rewarding (Fiorillo). Curiosity research establishes that the brain treats information as intrinsically valuable (Bromberg-Martin & Hikosaka; Gruber et al.). Skill acquisition research establishes that mastery involves a measurable cortical-to-subcortical transfer (Poldrack; Lehericy; Haier; Wan et al.). Flow research establishes that optimal experience involves a distinctive neural configuration that is neither System 1 nor System 2 (Dietrich; Weber & Huskey; Harris et al.). Predictive processing theory establishes that valence tracks the rate of model improvement (Van de Cruys; Schmidhuber). And game design practice, from Koster to Griesemer to Miyazaki to Thorson, demonstrates that the designers who produce the most acclaimed games are those who intuitively manage the learning gradient; even when they lack the theoretical vocabulary to describe what they are doing.

If anthropology, biology, neuroscience, psychology, computational theory, and design practice all point to the same structure, the explanation is unlikely to be accidental.

The designer's role

The unified model changes what it means to be a game designer. You are not creating content, crafting narratives, or engineering spectacles. You are designing **a system that regulates the player's rate of learning**. Every design decision - from millisecond input responsiveness to hundred-hour content pacing - should be evaluated against a single criterion: does this sustain, enhance, or disrupt the learning gradient?

The five design laws provide the practical framework:

1. **Maintain the learning gradient:** the player must always be learning something
2. **Hide the learning:** players should feel like they are playing, not studying
3. **Player-regulated challenge:** the player must be able to control their own difficulty
4. **Immediate, legible feedback:** learning requires clear and immediate error signals
5. **Never fully solve the system:** the game must stay slightly ahead of the player

These laws are not rules to follow blindly. They are tools for answering a single question: does this system support the player's learning process? If the answer is yes, flow will follow.

The nature of fun

Games are often treated as trivial. They are not.

They are one of the clearest windows we have into how the brain learns, predicts, and engages with the world. The 50-kHz ultrasonic vocalisations of tickled rats, the place cells firing in hippocampi of virtual taxi drivers, the caudate nucleus activating in shogi experts generating intuitive moves, the seven-fold performance improvement with decreased cortical metabolism in Tetris players,

the sustained dopamine ramp at maximum uncertainty, the convergent sweet spot at 85% accuracy where gradient-descent learners learn fastest; these are not isolated curiosities. They are facets of a single phenomenon: the brain is built to reduce uncertainty, it finds the process intrinsically rewarding, and games are the most precisely engineered environments we have for facilitating that process.

To understand games is to understand something fundamental about the human mind. To design games well is to engineer environments that serve that understanding with precision, craft, and respect for the extraordinary machinery that evolution built to navigate an uncertain world.

Fun is not a property of the game. It is not a property of the player. It is a property of the dynamic relationship between the game's pattern generation and the player's pattern absorption; a relationship that must be actively maintained through design decisions at every scale.

The designer who understands this relationship; who sees their work not as creating content but as engineering a learning gradient; has the most powerful lens available for predicting, diagnosing, and improving player engagement.

This is the nature of fun.